LARGE BIPARTITE GRAPHS OF GIVEN DEGREE AND DIAMETER FROM GENERALIZED POLYGONS

GABRIELA ARAUJO-PARDO, NACHO LÓPEZ, AND DOĞA ÖZDEN

ABSTRACT. Let $n_b(\Delta, D)$ be the largest order of a bipartite graph of maximum degree at most Δ and diameter at most D. We give constructions from generalized polygons that improve the lower bounds in the 2013 table of Miller and Širáň at diameters 3, 4, 6 and 7, and Haar graphs that improve it at diameters 5, 8, 9 and 10. At diameter 3 we add a fourth hat to the three hats of Araujo-Pardo and López and obtain $n_b(q+2,3) \geq 2(q^2+q+6)$ with regular graphs. At diameter 4 we copy points and lines of a generalized quadrangle with weights and stand-ins; a weighted 2-ovoid of the Ahrens–Szekeres quadrangle built from two elliptic quadrics gives $n_b(q+2,4) \geq 2q^2(q+2)$ for odd q, and Singer 2-ovoids of $W(3,q)$ give $n_b(q+3,4) \geq 2(q^2+1)(q+3)$, both with regular graphs. At diameter 6 an explicit set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$ ("whiskers") added to Delorme's doubling gives $n_b(q+m,6) \geq 2(q+1)(q^4+q^2+1)+(m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big)$, and a regular version. At diameter 7 we join $2r+1$ copies of the incidence graph of $H(q)$ by matchings, delete matching edges at suitable sets and attach new vertices; we prove which sets are allowed. All constructions are given in projective coordinates with the exact neighbourhood of every new vertex. We also show that the graphs of Gómez behind six diameter-8 entries of the table have diameter at least 9 as defined, and that relabelling their copies by automorphisms does not help.

POINTS FOR THE CO-AUTHORS

- Title: a placeholder. Choose the final title. 3
- Authors, order, affiliations, e-mail addresses. The byline of [AL20] is "Gabriela Araujo, Nacho López"; use whichever form you prefer here and in the references. 3
- AI disclosure. The constructions, proofs and checks in this draft were produced in a research repository run by Claude Code agents coordinated by D. Özden, and most explicit ingredients (sets, connection sets, hub graphs) were found by computer search. Following the guidance of the 20 October meeting this draft keeps search and AI out of the mathematics and mentions computation only as "computer-checked". Decide how to acknowledge the AI use (most journals now ask for a statement) and whether to cite the repository `https://github.com/dogaozden/degree-diameter`, which holds every graph, builder and check log. 3
- Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16,3)$ 396; $(4,5)$ 162, $(5,5)$ 354, $(8,5)$ 2 048, $(9,5)$ 2 758, $(10,5)$ 4 800, $(11,5)$ 7 020, $(12,5)$ 9 840, $(13,5)$ 11 700, $(14,5)$ 17 568, $(15,5)$ 20 736, $(16,5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4,7)$ 864, $(5,7)$ 3 120, $(6,7)$ 8 880, $(8,7)$ 43 344; $(3,10)$ 810; and later Haar graphs at $(9,8)$ 546 720, $(5,9)$ 29 756, $(7,10)$ 2 298 240, $(8,10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper. 5

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- The referee of the diameter-7 note recalls that this form of the quadric and of Tits' equations is [40, §2.4.13]. We could not consult [VM98] to confirm the place; please check and cite it. . . . 6
- Gabriela, Nacho: please check Remark 3.5 against the journal version of [AL20] (we have only seen arXiv v1), and decide whether the remark should stay in the paper. . . 12
- Example 4.6: the optimality of 21 comes from an integer-programming solver (HiGHS). Keep the sentence, or drop it as computer-dependent? . . . 14
- Proposition 4.10: do you know whether the 2-ovoids $C_a \cup C_{q-a}$ are those of [BLP09], [CCEM08] or [FWX19]? If so, cite and shorten. . . . 17
- Corollary 5.15, $q = 2$: the 12 whiskers are a computer-found set (its exactness is also proved in Lean). List them in an appendix, or drop the $q = 2$ case? . . . 23
- Theorem 5.18: the completing anti-flags are only shown to exist (as in the Lean proof). If you want every neighbourhood written out, we need an explicit completion (for instance a formula on the deficient elements); this is open work. . . . 24
- Corollary 6.23 was suggested by the referee of the diameter-7 note (who built its $q = 5$, $r = 1$ and $r = 2$ cases and checked them by breadth-first search from every vertex); the note itself left it out because it added no new statement. It is the only way the $(15, 7)$ cell is improved without computer-found sets. Keep it? . . . 32
- Corollary 6.24: the $W(E_7)$ description of R_0 ($q = 7, 11$) is from the lab notes; a precise statement (which lattice, which embedding into $Q(6, q)$) should be written out if we keep it. For $q = 5, 8$ the sets are only data. . . . 33
- Section 6.10: you may know [17] and how the $D = 7$ table values were obtained. The literature search behind this paragraph did not query Google Scholar or zbMATH. . . 34
- Section 7 questions six entries of the 2013 table credited to Delorme, Gómez and Quisquater. Decide how to present it (in this paper, as a separate note, or after contacting J. Gómez), and how much of the relabelling part (Section 7.4) to keep. . . 34
- Fact 7.14 is a finite enumeration over block patterns and wiring words; it does not depend on q. Should we replace it by a hand count, or include the enumeration as an appendix table? . . . 41
- The thesis value ΩH_5 at $(8, 8)$ (195 300) is below the 2013 value 201 480, so it is not a table entry; $(8, 8)$ is improved by a Haar graph in Section 8. Our own graphs at $(11, 8)$ and $(15, 8)$ (Table 8) beat the table values whatever their status; at $(10, 8)$, $(12, 8)$, $(14, 8)$, $(16, 8)$ we have no graph above the table values. . . . 44
- $(9, 5)$: the repository now has a larger Haar graph, of $\mathbb{Z}_{197} \rtimes \mathbb{Z}_7$ with 2758 vertices (computer-checked and proved in Lean), and $(8, 5)$ has a 2048-vertex lift that is not a Haar graph. Neither is in the refereed notes. Include? . . . 46
- Values at $D = 8$–10 raised after the refereed Haar note (computer-checked by the coordinator and proved in Lean on the lean-all branch): $(9, 8)$ 546 720 ($\mathrm{SL}(2, 16) \times \mathbb{Z}_{67}$), $(5, 9)$ 29 756 ($\mathbb{Z}_{346} \rtimes \mathbb{Z}_{43}$), $(7, 10)$ 2 298 240 ($\mathrm{SL}(2, 7) \times \mathrm{PSL}(2, 19)$), $(8, 10)$ 8 043 840 ($\mathrm{PSL}(2, 7) \times \mathrm{PSL}(2, 19) \times \mathbb{Z}_7$); also $(3, 10)$ 810 (a lift over $\mathbb{Z}_3 \times \mathrm{He}_3$). Use these in Table 8? . . . 47
- Table 9: (i) how to show the four ΩH_q cells we do not improve ($(10, 8)$, $(12, 8)$, $(14, 8)$, $(16, 8)$): keep the 2013 value with a mark, or leave the cell empty? (ii) The 2013 table has 112 cells with $\Delta \le 16$, $D \le 10$; should the paper print the whole table (with the cells we do not touch and the Moore cells)? (iii) The CombinatoricsWiki was last read for these cells in October 2026; re-check it before submission. . . . 49
- Decide how much of this appendix to keep, and whether to cite the Lean files by name or only describe what is covered. . . . 50
- Before submission: have the L^* proofs re-checked independently, or drop the column. . . 51

Should this appendix name the programs and log files (for instance `research/lifts/coordinator_check/apsp_mt.c`, `research/frontier/build_haar2.py`, `research/coordinator_check/check_2026-10-25.txt`)? The data files that define the $D = 7$ sets and hub graphs and the Haar connection sets are in `research/frontier/sols/`. 51

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1. Introduction

A bipartite graph of maximum degree $\Delta \geq 2$ and diameter $D \geq 2$ has at most

$$M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big)$$

vertices (the bipartite Moore bound). Let $n_b(\Delta, D)$ be the largest order of such a graph. For $D \geq 3$ and $\Delta \geq 3$ the bound is attained only by the incidence graphs of generalized polygons: projective planes ($D = 3$), generalized quadrangles ($D = 4$) and generalized hexagons ($D = 6$) of order $(\Delta - 1, \Delta - 1)$ [23, 31]. These are known when $\Delta - 1$ is a prime power. For the other cells only lower bounds are known. They are collected in Table 4 of the survey of Miller and Širáň [31] (version of 2013), which the CombinatoricsWiki [11] reproduces; since 2013 the wiki has changed only at $(13, 3)$ and $(15, 3)$, by the work of Araujo-Pardo and López [3]. We call this table "the 2013 table".

This paper gives constructions that improve the 2013 table in the cells listed in Table 1. Most of them are built from generalized polygons in their classical coordinates, by three kinds of operations: copying points and lines (with Delorme's doubling with levels [15] as the basic tool), changing a bounded number of incidences, and joining several copies of an incidence graph by matchings and new vertices. The remaining cells are Haar graphs (bi-Cayley graphs) of explicit groups.

Results.

- *Diameter 3* (Section 3). Araujo-Pardo and López [3] added three "hats" to $\mathrm{PG}(2, q)$ and obtained $n_b(q + 2, 3) \geq 2(q^2 + q + 5)$. We add a fourth hat (Theorem 3.2): $n_b(q + 2, 3) \geq 2(q^2+q+6)$ for every prime power $q \geq 4$, attained by a $(q+2)$-regular graph; the construction works in every projective plane of order $q \geq 3$.
- *Diameter 4* (Section 4). Theorem 4.2: in any generalized quadrangle, clones of points with arbitrary weights and copies of lines with "stand-ins" keep diameter 4. With a weighted 2-ovoid of the Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ made of two elliptic quadrics (Theorem 4.7) this gives $n_b(q + 2, 4) \geq 2q^2(q + 2)$ for odd q; with the Singer 2-ovoids of $W(3, q)$ (Proposition 4.10) it gives $n_b(q + 3, 4) \geq 2(q^2 + 1)(q + 3)$ for odd q; both with regular graphs. For every prime power $q \leq \Delta - 1$, $n_b(\Delta, 4) \geq 2q^2\Delta$ (Corollary 4.13).

- *Diameter 6* (Section 5). A set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$, given by an explicit rule (Theorem 5.5), can be added at level 0 to Delorme's doubling of $H(q)$ along a Hermitian spread. This gives Theorem 5.8, and deleting repeated lines and completing gives regular graphs (Theorem 5.18).
- *Diameter 7* (Section 6). The 2013 values at $(7,7)$ and $(9,7)$–$(16,7)$ are the orders $(2r+1)\cdot 2|H(q)|$ of $2r+1$ copies of the incidence graph of $H(q)$ joined by matchings. We delete matching edges at suitable sets of elements and attach new vertices to the freed slots. Theorem 6.10 reduces the diameter to one condition per pair of copies; Theorems 6.12 and 6.15 give two sufficient conditions (segment sets at automorphic transitions; local dualities), the first also necessary. With classical sets this improves all nine cells (Corollaries 6.20–6.23); explicit larger sets give the values of Table 1.
- *Diameter 8* (Section 7). Six entries of the 2013 table at $D=8$ are the orders of the compound graphs $\Omega\mathrm{H}_q$ of Gómez [27], for which the thesis states diameter at most 8 without proof. As defined there, these graphs have diameter at least 9 (Theorems 7.3 and 7.5), and relabelling the copies by automorphisms of the hexagon does not give diameter 8 (Theorems 7.15 and 7.21).
- *Haar graphs* (Section 8). We state the covering criterion that decides the diameter of a Haar graph from two breadth-first searches, and list Haar graphs that improve two cells at $D=5$ and 33 cells at $D=8,9,10$ (with one further lift at $(3,8)$). These are single explicit graphs, checked by computer.

Problem 4 of [3] asks to generalize the hat construction to regular bipartite graphs of diameters 4 and 6. The hat lemma (Lemma 2.7) says in what sense it generalizes, and Corollaries 4.8 and 4.12 and Theorem 5.18 give regular graphs at $D=4$ and $D=6$ that beat the 2013 table.

What is proved and what is computed. Every theorem in Sections 2–7 has a complete proof here, with one kind of exception, always marked: a statement about one explicit finite object (a set of points, a connection set, a hub graph, a count of route words) is *computer-checked*, and we name the check. "Computer-checked" for a graph means: built from its definition, and found to be bipartite, of the stated maximum degree and of the stated diameter by breadth-first search from every vertex (or from one vertex of every orbit of an automorphism group that was itself checked), by at least two programs written separately. Appendix C lists the checks. Many of the statements are also proved in Lean 4 with Mathlib; Appendix B says exactly which.

Table 1: The cells improved in this paper. M_b: bipartite Moore bound; 2013: [31, Table 4]; "reg.": the largest regular graph we give, when the record graph is not regular. Values at $D=8,9,10$ are in Table 8.

(Δ, D)	M_b	2013	this paper	reg.	construction
(11,3)	222	190	**192**		fourth hat, $q=9$ (Thm. 3.2)
(13,3)	314	270 [a]	**276**		fourth hat, $q=11$
(15,3)	422	370 [a]	**376**		fourth hat, $q=13$
(7,4)	518	346	**359**	350	clones and shadows on $Q(4,5)$ (Ex. 4.6)
(11,4)	2 222	1 734	**1 782**		AS(9), two quadrics (Cor. 4.8)
(13,4)	3 770	3 064	**3 146**		AS(11)
(15,4)	5 910	4 946	**5 070**		AS(13)
(16,4)	7 232	5 134	**5 440**		Singer 2-ovoid of $W(3,13)$ (Cor. 4.12)
(7,5)	3 110	1 134	**1 152**		Haar graph (Sec. 8)
(9,5)	9 362	2 496	**2 662**		Haar graph
(7,6)	18 662	8 992	**9 592**	9 348	whiskers in $H(5)$ (Thm. 5.8, 5.18)

(Δ, D)	M_b	2013	this paper	reg.	construction
(11,6)	222 222	142 464	**148 944**	147 500	$H(9)$
(13,6)	542 906	374 452	**388 972**	386 328	$H(11)$
(15,6)	1 158 390	842 048	**870 440**	866 068	$H(13)$
(16,6)	1 627 232	884 062	**940 846**	927 724	$H(13)$, $m = 3$
(7,7)	111 974	23 436	**23 598**		3 levels of $H(5)$, local duality (Sec. 6)
(9,7)	599 186	117 648	**118 238**		3 levels of $H(7)$
(10,7)	1 195 742	224 694	**225 152**		3 levels of $H(8)$
(11,7)	2 222 222	398 580	**413 180**		3 levels of $H(9)$, duality
(12,7)	3 897 434	664 300	**693 500**		5 levels of $H(9)$, duality
(13,7)	6 514 874	1 062 936	**1 064 472**		3 levels of $H(11)$
(14,7)	10 458 086	1 771 560	**1 789 588**		5 typed levels of $H(11)$
(15,7)	16 217 462	2 480 184	**2 501 491**		7 typed levels of $H(11)$
(16,7)	24 408 482	4 022 340	**4 052 108**		5 typed levels of $H(13)$

[a] The current values of [11] are 274 and 374 [3].

All graphs in Table 1 except the two Haar graphs at $D = 5$ come from constructions proved here for every prime power q (or every odd q). At $D = 3, 4, 6$ the ingredients are classical or given by explicit formulas (at $(7, 4)$ an explicit list of 21 lines, Appendix A). At $D = 7$ the values in the table use, in addition, explicit sets and hub graphs whose properties are computer-checked; Section 6 also gives values beating the 2013 table in all nine cells from classical ingredients only (Table 6).

Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16, 3)$ 396; $(4, 5)$ 162, $(5, 5)$ 354, $(8, 5)$ 2 048, $(9, 5)$ 2 758, $(10, 5)$ 4 800, $(11, 5)$ 7 020, $(12, 5)$ 9 840, $(13, 5)$ 11 700, $(14, 5)$ 17 568, $(15, 5)$ 20 736, $(16, 5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4, 7)$ 864, $(5, 7)$ 3 120, $(6, 7)$ 8 880, $(8, 7)$ 43 344; $(3, 10)$ 810; and later Haar graphs at $(9, 8)$ 546 720, $(5, 9)$ 29 756, $(7, 10)$ 2 298 240, $(8, 10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper.

Organisation. Section 2 fixes notation, coordinates for the classical polygons, and Delorme's doubling with levels, with a proof and the hat lemma. Sections 3–6 treat diameters 3, 4, 6 and 7, Section 7 the Gómez entries at diameter 8, and Section 8 Haar graphs. Section 9 collects the updated values. Appendix A writes out the $(7, 4)$ example, Appendix B lists what is checked in Lean, and Appendix C the computer checks.

2. Preliminaries

2.1. **Incidence structures.** An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ has points, lines and a symmetric incidence relation; repeated points and repeated lines (with the same incidences) are allowed. Its *incidence graph* $\Gamma(\mathcal{S})$ has vertex set $\mathcal{P} \cup \mathcal{L}$ and the incident pairs (*flags*) as edges; it is bipartite with the points on one side. A non-incident point–line pair is an *anti-flag*. Every bipartite graph with a chosen 2-colouring is of this form, so

$$n_b(\Delta, D) = \max\{|\mathcal{P}| + |\mathcal{L}| : \text{ every element has at most } \Delta \text{ neighbours, } \operatorname{diam}\Gamma(\mathcal{S}) \le D\}.$$

Distances $d(x, y)$ are taken in $\Gamma(\mathcal{S})$, and $\operatorname{diam}\mathcal{S}$ means $\operatorname{diam}\Gamma(\mathcal{S})$. Two distinct points on a common line are *collinear*; two distinct lines on a common point are *concurrent*. $\mathbb{F} = \mathrm{GF}(q)$ is the field of order q, and for q odd χ is its quadratic character, with $\chi(0) = 0$.

Lemma 2.1 (diameter in geometric terms)**.** *Let $\mathcal{S}$ have no isolated element and let $D \ge 2$.*

(i) If D is even, $\operatorname{diam}\mathcal{S} \le D$ *if and only if* $d(p,L) \le D-1$ *for every anti-flag* (p,L).
(ii) If D is odd, $\operatorname{diam}\mathcal{S} \le D$ *if and only if any two points, and any two lines, are at distance at most* $D-1$.

In particular, $\operatorname{diam}\mathcal{S} \le 3$ *if and only if any two points are collinear and any two lines are concurrent, and* $\operatorname{diam}\mathcal{S} \le 4$ *if and only if for every anti-flag* (p,L) *some point of* L *is collinear with* p.

Proof. (i) Let $p \neq p'$ be points and L a line on p'. If $p \mathrel{I} L$ then $d(p,p') = 2$; otherwise $d(p,p') \le d(p,L)+1 \le D$. Lines likewise. (ii) For an anti-flag (p,L) take $p' \mathrel{I} L$; then $d(p,L) \le d(p,p')+1 \le D$. The converses hold by parity. □

Lemma 2.2 (adding flags, deleting twins)**.** *(a) Adding flags to an incidence structure keeps its incidence graph bipartite and does not increase any distance. (b) If two lines have the same set of points, deleting one of them does not change any distance between the remaining elements.*

Proof. (a) is clear. (b) A shortest path through the deleted line can use its twin instead; it does not already contain the twin, since then it could be shortened. □

2.2. **Generalized polygons.** A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; it has *order* (s,t) if every line has $s+1$ points and every point is on $t+1$ lines. It is *thick* if $s,t \ge 2$. A generalized 3-gon of order (q,q) is a projective plane of order q, with q^2+q+1 points and as many lines. A generalized quadrangle (GQ) of order (s,t) has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines, and its axiom is: for every anti-flag (p,L) exactly one point of L is collinear with p. A generalized hexagon of order (q,q) has $(q+1)(q^4+q^2+1)$ points and as many lines. By Feit and Higman [23], thick generalized n-gons exist only for $n \in \{2,3,4,6,8\}$, and a generalized octagon of order (s,s) does not exist; so $M_b(\Delta,D)$ is attained, for $\Delta \ge 3$, only at $D \in \{2,3,4,6\}$.

The projective plane. Points of $\mathrm{PG}(2,q)$ are written $(x:y:z)$ and lines $[u:v:w]$, with $(x:y:z) \mathrel{I} [u:v:w]$ if and only if $ux+vy+wz=0$.

Quadrangles. $W(3,q)$ is the symplectic quadrangle: the points of $\mathrm{PG}(3,q)$ and the lines of $\mathrm{PG}(3,q)$ that are totally isotropic for a non-degenerate alternating form; it is a GQ of order (q,q) [33]. $Q(4,q)$ is the parabolic quadric of $\mathrm{PG}(4,q)$ with the lines it contains; it is a GQ of order (q,q), dual to $W(3,q)$ [33, 3.2.1]. The Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ (q odd) is introduced in coordinates in Section 4.2.

The split Cayley hexagon. Let $V = \mathbb{F}^7$ with coordinates $(X_0,\dots,X_6)$,

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2, \qquad b(X,Y) = Q(X+Y) - Q(X) - Q(Y),$$

and let $Q(6,q)$ be the parabolic quadric $\{\langle x\rangle : Q(x)=0\}$ of $\mathrm{PG}(6,q)$. Points $\langle x\rangle, \langle y\rangle$ of $Q(6,q)$ are *perpendicular*, $x \perp y$, if $b(x,y)=0$, i.e. if the line xy lies on $Q(6,q)$. The *split Cayley hexagon* $H(q)$ [39], [40, §2.4] has as points all $N = N(q) = (q^6-1)/(q-1) = (q+1)(q^4+q^2+1)$ points of $Q(6,q)$, and as lines the lines of $Q(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12}=p_{34}, \quad p_{54}=p_{32}, \quad p_{20}=p_{35}, \quad p_{65}=p_{30}, \quad p_{01}=p_{36}, \quad p_{46}=p_{31}. \tag{2.1}$$

For example, the $q+1$ lines through $e_0 = (1,0,\dots,0)$ are $\langle e_0, \lambda e_5 + \mu e_6\rangle$, $(\lambda:\mu) \in \mathrm{PG}(1,q)$; they lie in the totally singular plane $\langle e_0,e_5,e_6\rangle$. The same equations define $H(q)$ for q even, with the quadric written $X_0X_4 + X_1X_5 + X_2X_6 + X_3^2$ (Section 5.5).

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Fact 2.3 ([39], [40, §2.4])**.** $H(q)$ is a generalized hexagon of order (q,q). The lines of $H(q)$ through a point x are the lines through x of a totally singular plane $\pi_x \subset x^\perp$. The collineation group contains $G_2(q)$; for $A \in \mathrm{SL}(3,q)$ the map $(v,a,w) \mapsto (Av, a, A^{-T}w)$, with $x=(v,a,w)$, $v=(x_0,x_1,x_2)$, $a=x_3$, $w=(x_4,x_5,x_6)$, is a collineation. $H(q)$ has a duality (an incidence-preserving bijection exchanging points and lines) if and only if q is a power of 3.

Let $I = I(q)$ be the incidence graph of $H(q)$, with $2N$ vertices, and $\Gamma_i(x) = \{y : d(x,y) = i\}$. Two elements are *opposite* if they are at distance 6, and $\mathrm{opp}(x) = \Gamma_6(x)$. The following facts are standard; we include the short proofs because Sections 6 and 7 use them repeatedly.

Lemma 2.4. *Let x, y be vertices of I.*

(a) If $d(x,y) \le 5$ there is a unique geodesic from x to y. If $1 \le d(x,y) = k \le 5$, exactly one neighbour of y is at distance $k-1$ from x and the other q at distance $k+1$; if $k=6$ all $q+1$ neighbours of y are at distance 5. Hence $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \le i \le 5$ and $|\Gamma_6(x)| = q^5$.

(b) (covering) *There is $z \in \Gamma_2(x)$ with $d(z,y) \le 3$ if x, y have different types, and with $d(z,y) \le 4$ if they have the same type.*

(c) (orthogonality) *For points x, y: $d(x,y) \le 4$ if and only if $x \perp y$. So $\mathrm{opp}(x) \cap \mathcal{P} = \mathcal{P} \setminus x^\perp$.*

Proof. (a) Two geodesics would close a cycle of length at most $10 < 12$. Neighbours of y are at distance $k \pm 1$ from x; the predecessor on the geodesic is at $k-1$, and a second one would give a second geodesic; at $k=6$ the diameter forces 5. The counts follow by induction. (b) If $2 \le d(x,y) \le 6$ take the vertex at distance 2 from x on a geodesic to y; if $d(x,y)=1$ take $z \in \Gamma_1(y) \setminus \{x\}$; if $x=y$ any $z \in \Gamma_2(x)$. The largest value of $d(z,y)$ is $6-2=4$ (same type) or $5-2=3$ (different types). (c) If $d(x,y) \le 4$ there is a point z with x, y on lines of $H(q)$ through z ($z=x$ if $d(x,y) \le 2$); these lines lie in π_z, so $b(x,y)=0$. Conversely $x^\perp \cap Q(6,q)$ is a cone with vertex x over a $Q(4,q)$ and has $1+q(q^3+q^2+q+1)$ points, which by (a) is the number $1+(q+1)q+(q+1)q^3$ of points at distance at most 4 from x. □

2.3. **Doubling with levels, and the hat lemma.** A *duplicate* of a vertex (a new vertex with the same neighbours) does not increase the diameter (see [16], which we have not seen, and [15, §7]). Delorme [15, §7] refined this into a doubling with levels, stated there without proof, and used it for the 2013 values at $D=4$ and $D=6$ [15, §8(a),(b)]. We state it for incidence structures and give a proof.

Let $\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$ be a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, and $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Fix $m \ge 2$. Two words over $[m] = \{1, \dots, m\}$ are *comparable* if one is a prefix of the other.

Definition 2.5 (Delorme)**.** The m-fold doubling $\mathcal{S}^{(m)}_{\mathcal{A}}$ has the *versions* x^w, for x an element of $\mathcal{S}$ and $w \in [m]^{\lambda(x)+1}$ (x^w is a point if x is a point and a line if x is a line), and

$$x^w \mathrel{\mathrm{I}} y^u \iff x \mathrel{\mathrm{I}} y \text{ and } w, u \text{ are comparable.}$$

So an element of level -1 has one version (identified with x), adjacent to every version of each of its neighbours; and a version x^w of level $k \ge 0$ is adjacent, for each neighbour y of x, to the single version $y^{w_1 \cdots w_{j+1}}$ if $\lambda(y) = j \le k$, and to the m^{j-k} versions y^{wu} ($|u| = j-k$) if $\lambda(y) = j > k$. This is the exact neighbourhood of every version. The map $x^w \mapsto x$ preserves incidence, so $d(x^w, y^u) \ge d(x,y)$.

The conditions on $\mathcal{A}$ are (with $D \ge 2$):

(L0) every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$;

(L1) for every $i \ge 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$;

(C_D) for any two elements x, y (possibly $x=y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is $z \notin \mathcal{A}_0$ with $d(x,z) + d(z,y) \le D$.

Theorem 2.6 (Delorme [15, §7], (a)–(c))**.** *Let $\mathcal{S}$ have diameter at most $D \geq 2$ and let $m \geq 2$.*

(a) If (C_D) holds, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.
(b) If (L0) and (L1) hold, every version satisfies $\deg x^w \leq \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x) + 1$.
(c) $\mathcal{S}_{\mathcal{A}}^{(m)}$ has $|\mathcal{P}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{P}|$ points and $|\mathcal{L}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{L}|$ lines.
(d) Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, the conclusion of (C_D) holds for every pair $x, y \in \mathcal{A}_0$.

Delorme states (a)–(c) without proof ("je ne donne pas la preuve"); (d) is not in [15].

Proof. (c) An element of level k has $m^{k+1} = 1 + (m-1)(1 + m + \cdots + m^k)$ versions.

(b) Let $\lambda(x) = k$. By the description after Definition 2.5, only neighbours of level $> k$ contribute more than one version. If $k = -1$, (L0) gives at most one neighbour in $\mathcal{A}_0$; it has level exactly 0 (a neighbour in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1)), and contributes m versions. If $k \geq 0$, (L1) with $i = k$ gives at most one neighbour in $\mathcal{A}_{k+1}$, and (L1) with $i = k + 1$ gives none in $\mathcal{A}_{k+2}$; that neighbour contributes m versions. So $\deg x^w \leq \deg x - 1 + m$.

(a) For an infinite word ω the *sheet* Σ_ω is the set of versions x^w with w a prefix of ω. It contains one version of each element, and $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions with comparable words lie in a common sheet, hence are at distance at most D. Let x^w, y^u have incomparable words; both are non-empty, so $x, y \in \mathcal{A}_0$. If x, y are in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, \ldots, z_k = y$ ($k \leq D$) passes through some $z_i \notin \mathcal{A}_0$, whose unique version lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet through x^w and $z_i, \ldots, z_k$ into a sheet through y^u: this is a path of length k. If x, y are in the same component (including $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \leq D$, and the same lifting gives a path of length at most D.

(d) Take versions x^w, y^u with w starting with the letter 1 and u with the letter 2, and a path of length at most D between them. Consecutive versions on it have comparable words, and comparable non-empty words start with the same letter; so some version on the path has the empty word, that is, it is an element $z \notin \mathcal{A}_0$. Projecting the path gives a walk $x \to z \to y$ of length at most D. □

Lemma 2.7 (hat lemma)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (s, t), $s, t \geq 1$, and $\mathcal{A}_0$ a set of elements. Then (C_n) holds if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.*

Proof. Suppose a component B contains a path $x = z_0, \ldots, z_{n-1} = y$ of length $n - 1$ (it does if B has diameter at least $n - 1$, and if B contains a cycle, since cycles have length at least $2n$). By the girth this is the only path of length at most $n - 1$ from x to y, so $d(x, y) = n - 1$. If $d(x, z) + d(z, y) \leq n$ then by parity $d(x, z) + d(z, y) = n - 1$, so z lies on a shortest x–y path, i.e. $z = z_i \in \mathcal{A}_0$. So ($C_n$) fails.

Conversely let B be a tree of diameter at most $n - 2$, $x, y \in B$, and P the path from x to y in B, of length d. The incidence graph is connected and has cycles, so some element of B has a neighbour outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and z a neighbour of u outside $\mathcal{A}_0$. Then $d(x, z) + d(z, y) \leq d + 2k + 2$. If $k = 0$ this is at most n. If $k \geq 1$, every element of B at distance less than k from P has all its neighbours in B, and every element has at least two neighbours. So we can walk k steps away from P starting at x, and k steps away from P starting at y (along two different neighbours if $x = y$); in the tree B this gives a path of length $d + 2k$. Hence $d + 2k \leq n - 2$ and $d(x, z) + d(z, y) \leq n$. □

Corollary 2.8. *In a generalized n-gon of order (s, t), $s, t \geq 1$, for any filtration and $m \geq 2$, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n if and only if every component of the substructure on $\mathcal{A}_0$ is a tree*

of diameter at most $n-2$. For $n=3$ the allowed components are single elements and flags; for $n=4$ also stars *(a point with some lines on it, or a line with some of its points); for $n=6$ trees of diameter at most 4.*

Proposition 2.9 (diameter 3 needs patches)**.** *In a projective plane, if $\mathcal{A}$ satisfies (L0), (L1) and (C_3), then $\mathcal{A}_0$ has at most one point and at most one line, and $\mathcal{A}_1=\emptyset$.*

Proof. If $x\neq y$ are points of $\mathcal{A}_0$, the line xy is either outside $\mathcal{A}_0$ and incident with two elements of $\mathcal{A}_0$ (against (L0)), or in $\mathcal{A}_0$, and then x, xy, y is a path of length $2>n-2$ in one component (against Lemma 2.7). Lines likewise. An element of $\mathcal{A}_1$ would have its $q+1\geq 3$ neighbours in $\mathcal{A}_0$ by (L1). □

So pure doubling gains at most $2(m-1)$ elements in a projective plane; the hats of Section 3 double flags and then change a bounded number of incidences.

3. Diameter 3: the fourth hat

Araujo-Pardo and López [3] add three *hats* to $\mathrm{PG}(2,q)$ and obtain $n_b(q+2,3)\geq 2(q^2+q+5)$. We add a fourth. By Proposition 2.9, every hat must change some incidences of the plane; the fourth hat costs one rotation of three incidences.

3.1. **Frame and notation.** Let $\ell=[1:0:0]$ (the line $x=0$) and $A=(1:0:0)\notin\ell$. The points of ℓ and the lines on A are

$$X_t=(0:t:1),\quad P=(0:1:0);\qquad \lambda_t=[0:1:-t],\quad \lambda_\infty=[0:0:1]\qquad (t\in\mathbb{F}).$$

Every point is listed once among A, X_t $(t\in\mathbb{F})$, P, $(x:y:1)$ $(x\neq 0)$, $(x:1:0)$ $(x\neq 0)$; every line once among ℓ, λ_t, λ_∞, $[u:1:w]$ $(u\neq 0)$, $[1:0:w]$ $(w\neq 0)$. We use these facts (one-line computations):

(F1) $\lambda_t\cap\ell=X_t$, $\lambda_\infty\cap\ell=P$; the points of λ_t are A and $(x:t:1)$, $x\in\mathbb{F}$ (X_t at $x=0$); those of λ_∞ are A and $(x:1:0)$, $x\in\mathbb{F}$.
(F2) The lines on X_t are ℓ and $[w:1:-t]$, $w\in\mathbb{F}$ (λ_t at $w=0$); the lines on P are ℓ, λ_∞ and $[1:0:w]$, $w\neq 0$.
(F3) If $u=(x:y:z)$ with $x\neq 0$, then u and X_s lie on $[(sz-y)/x:1:-s]\neq\ell$.
(F4) A line $[u:v:w]$ with $u\neq 0$ meets λ_s in $(-(vs+w)/u:s:1)\neq A$; $[u:1:w]$ meets ℓ in X_{-w}.

Let $q\geq 4$, let $a,c\in\mathbb{F}\setminus\{0,1\}$ be distinct, and put

$$S=\{0,1,a,c\},\qquad T=\{0,1,c\},\qquad \sigma=(0\ c\ 1):\ 0\mapsto c\mapsto 1\mapsto 0.$$

3.2. **The construction.**

Definition 3.1. $\mathcal{S}(q;a,c)$ has the points of $\mathrm{PG}(2,q)$ and five new points Q, H_0, H_1, H_a, H_c, the lines of $\mathrm{PG}(2,q)$ and five new lines M, K_0, K_1, K_a, K_c, and exactly these incidences:

(C1) every incidence of $\mathrm{PG}(2,q)$ except $X_s\ \mathrm{I}\ \lambda_s$ $(s\in S)$, and the *rotation* $X_0\ \mathrm{I}\ \lambda_c$, $X_c\ \mathrm{I}\ \lambda_1$, $X_1\ \mathrm{I}\ \lambda_0$ (that is, $X_t\ \mathrm{I}\ \lambda_{\sigma(t)}$ for $t\in T$);
(C2) H_s lies on $[w:1:-s]$ for every $w\in\mathbb{F}$ $(s\in S)$: the lines on X_s other than ℓ;
(C3) K_s contains $(x:s:1)$ for every $x\in\mathbb{F}$ $(s\in S)$: the points of λ_s other than A;
(C4) M contains P, X_t $(t\in\mathbb{F}\setminus T)$, and H_0, H_1, H_a, H_c;
(C5) Q lies on λ_∞, λ_t $(t\in\mathbb{F}\setminus T)$, and K_0, K_1, K_a, K_c;
(C6) the *links* $A\ \mathrm{I}\ K_a$, $H_a\ \mathrm{I}\ \ell$, $H_0\ \mathrm{I}\ K_c$, $H_c\ \mathrm{I}\ K_1$, $H_1\ \mathrm{I}\ K_0$ (that is, $H_t\ \mathrm{I}\ K_{\sigma(t)}$ for $t\in T$).

$G(q;a,c)$ is its incidence graph. $R(q;a,c)$ is $G(q;a,c)$ together with the *filler* edges

$$(x:y:1)\sim[x:1:-y]\quad (x\neq 0,\ y\notin S),\qquad (x:1:0)\sim[1:0:-1/x]\quad (x\neq 0).$$

The hat pair (H_s, K_s) is a copy of the flag (X_s, λ_s) with the flag "crossed", and (Q, M) is a copy of the anti-flag (A, ℓ).

Theorem 3.2. *For every prime power $q \geq 4$ and distinct $a, c \in \mathbb{F} \setminus \{0, 1\}$, the graphs $G(q; a, c)$ and $R(q; a, c)$ are bipartite, have $2(q^2 + q + 6)$ vertices, maximum degree $q + 2$ and diameter 3, and $R(q; a, c)$ is $(q + 2)$-regular. Hence*

$$n_b(q + 2, 3) \geq 2(q^2 + q + 6),$$

attained by a regular graph.

The construction and the proof use only the axioms of a projective plane together with an anti-flag (A, ℓ) and four points X_s of ℓ, so a coordinate-free version holds in every projective plane of order $q \geq 3$ (in coordinates the four points are labelled by $0, 1, a, c$, which needs $q \geq 4$). It also has a regular completion there. At $q = 3$ the four points X_s are all of ℓ, there is no deficient element, and the structure is already 5-regular. For $q \geq 4$ put $N_0 = (q - 1)(q - 3)$, the number of points of degree $q + 1$ and of lines of degree $q + 1$ (see the table below). A deficient point p lies on exactly $q - 4$ deficient lines: the lines on p other than Ap meet ℓ in the q points other than $Ap \cap \ell$, four of which are the X_s. Dually, a deficient line has exactly $q - 4$ deficient points. So the bipartite graph of non-incident pairs between deficient points and deficient lines is regular of degree $N_0 - (q - 4) = q^2 - 5q + 7 \geq 1$, and a regular bipartite graph of positive degree has a perfect matching (Kőnig); adding it as flags completes the structure. In coordinates, the filler of Definition 3.1 is such a matching.

3.3. **Every neighbourhood.** Table 2 lists the neighbourhood in $G(q; a, c)$ of every vertex, new or old. Here $\sigma^{-1}(0) = 1$, $\sigma^{-1}(1) = c$, $\sigma^{-1}(c) = 0$.

Table 2: The neighbourhood of every vertex of $G(q; a, c)$.

vertex	neighbours	degree
Q	λ_∞, λ_t $(t \notin T)$, K_0, K_1, K_a, K_c	$q + 2$
H_a	$[w : 1 : -a]$ $(w \in \mathbb{F})$, M, ℓ	$q + 2$
H_t, $t \in T$	$[w : 1 : -t]$ $(w \in \mathbb{F})$, M, $K_{\sigma(t)}$	$q + 2$
M	P, X_t $(t \notin T)$, H_0, H_1, H_a, H_c	$q + 2$
K_a	$(x : a : 1)$ $(x \in \mathbb{F})$, Q, A	$q + 2$
K_t, $t \in T$	$(x : t : 1)$ $(x \in \mathbb{F})$, Q, $H_{\sigma^{-1}(t)}$	$q + 2$
A	λ_t $(t \in \mathbb{F})$, λ_∞, K_a	$q + 2$
P	ℓ, λ_∞, $[1 : 0 : w]$ $(w \neq 0)$, M	$q + 2$
X_t, $t \notin S$	ℓ, $[w : 1 : -t]$ $(w \in \mathbb{F})$, M	$q + 2$
X_a	ℓ, $[w : 1 : -a]$ $(w \neq 0)$, M, K_a	$q + 2$
X_t, $t \in T$	ℓ, $[w : 1 : -t]$ $(w \neq 0)$, $\lambda_{\sigma(t)}$, K_t	$q + 2$
$(x : y : 1)$, $x \neq 0$	$[u : 1 : -(ux + y)]$ $(u \in \mathbb{F})$, $[1 : 0 : -x]$, and K_y if $y \in S$	$q + 2$ if $y \in S$, else $q + 1$
$(x : 1 : 0)$, $x \neq 0$	λ_∞, $[1 : -x : w]$ $(w \in \mathbb{F})$	$q + 1$
ℓ	P, X_t $(t \in \mathbb{F})$, H_a	$q + 2$
λ_∞	A, $(x : 1 : 0)$ $(x \in \mathbb{F})$, Q	$q + 2$
λ_t, $t \notin S$	A, $(x : t : 1)$ $(x \in \mathbb{F})$, Q	$q + 2$
λ_a	A, $(x : a : 1)$ $(x \neq 0)$, Q, H_a	$q + 2$
λ_t, $t \in T$	A, $(x : t : 1)$ $(x \neq 0)$, $X_{\sigma^{-1}(t)}$, H_t	$q + 2$
$[u : 1 : w]$, $u \neq 0$	$(x : -ux - w : 1)$ $(x \in \mathbb{F})$, $(-1/u : 1 : 0)$, and H_{-w} if $-w \in S$	$q + 2$ if $-w \in S$, else $q + 1$

vertex	neighbours	degree
$[1:0:w]$, $w \neq 0$	$(-w:y:1)$ $(y \in \mathbb{F})$, P	$q+1$

The vertices of degree $q+1$ are the points $(x:y:1)$ $(x \neq 0,\ y \notin S)$ and $(x:1:0)$ $(x \neq 0)$, and the lines $[u:1:w]$ $(u \neq 0,\ -w \notin S)$ and $[1:0:w]$ $(w \neq 0)$: $(q-1)(q-3)$ of each kind. In $R(q;a,c)$ each of them gains exactly one neighbour, its filler partner.

3.4. **Proof of Theorem 3.2.** *Degrees.* Each row of Table 2 is (F1) or (F2) corrected by (C1)–(C6), and within a row the listed neighbours are distinct. For instance: A keeps its $q+1$ lines and gains only K_a (A is not of the form $(x:s:1)$); X_a loses λ_a and gains M (as $a \notin T$) and K_a; X_t $(t \in T)$ loses λ_t, gains $\lambda_{\sigma(t)}$ and K_t, and is not on M; a point $(x:y:1)$ with $x \neq 0$ is not on ℓ, so it loses nothing, and it gains K_s if and only if $y = s \in S$; Q is on $1+(q-3)+4 = q+2$ lines; H_s is on q lines by (C2), on M, and on exactly one link line. The line rows are the dual count: ℓ gains only H_a; λ_a loses X_a and gains Q, H_a; λ_t $(t \in T)$ loses X_t and gains $X_{\sigma^{-1}(t)}$ and H_t; $[u:1:w]$ $(u \neq 0)$ gains H_s if and only if $s = -w \in S$; M has $1+(q-3)+4$ points; K_s has q points, Q, and one link point.

The filler map $\delta : (x:y:z) \mapsto [x:z:-y]$ is a correlation of $\mathrm{PG}(2,q)$ (since $ux + wz + (-v)(-y) = ux + vy + wz$), a point lies on its image if and only if $x^2 = 0$, i.e. if and only if it is on ℓ, and δ maps $(x:y:1) \mapsto [x:1:-y]$ and $(x:1:0) \mapsto [x:0:-1] = [1:0:-1/x]$. So the filler is a bijection from the deficient points to the deficient lines, and its pairs are anti-flags. Hence $G(q;a,c)$ has maximum degree $q+2$ and $R(q;a,c)$ is $(q+2)$-regular.

Reduction. By Lemma 2.1(ii) it suffices to show that any two points are collinear and any two lines concurrent. In the next two lemmas "incident" means incident in $\mathcal{S}(q;a,c)$. The only incidences of $\mathrm{PG}(2,q)$ that are missing are X_s I λ_s, $s \in S$; so an incidence u I L of the plane survives whenever $u \notin \ell$ or L is not on A.

Lemma 3.3. *Any two points $u \neq v$ of $\mathcal{S}(q;a,c)$ are collinear.*

Proof. (P1) Both in $\mathrm{PG}(2,q)$; let $L = uv$. If both incidences survive, use L. Otherwise, say $u = X_s$ and $L = \lambda_s$ with $s \in S$, and $v = A$ or $v = (x:s:1)$ with $x \neq 0$. If $v = (x:s:1)$: use K_s. If $v = A$ and $s = a$: K_a (by (C3), (C6)). If $v = A$ and $s \in T$: $\lambda_{\sigma(s)}$ (rotation).

(P2) Q and u. If $u \in \{A, P\}$ or $u = (x:1:0)$: λ_∞. If $u = X_t$, $t \notin S$: λ_t. If $u = X_s$, $s \in S$: K_s. If $u = (x:y:1)$, $x \neq 0$: λ_y if $y \notin T$, and K_y if $y \in T$.

(P3) H_s and u. If $u \notin \ell$ (including $u = A$): the line of (F3), which contains H_s by (C2). If $u = X_s$: $[1:1:-s]$. If $u = P$, or $u = X_t$ with $t \notin T$, $t \neq s$: M. If $u = X_t$ with $t \in T \setminus \{s\}$ and $s = a$: ℓ. If $u = X_t$ with $t \in T \setminus \{s\}$ and $s \in T$: as σ is a 3-cycle, $t = \sigma(s)$ or $t = \sigma^{-1}(s)$; use $K_{\sigma(s)}$ (it contains $X_{\sigma(s)}$ and H_s) or λ_s (it contains $X_{\sigma^{-1}(s)}$ by the rotation, and H_s by (C2) with $w = 0$).

(P4) Two new points. $H_s, H_{s'}$: M. Q, H_a: λ_a. Q, H_t $(t \in T)$: $K_{\sigma(t)}$. □

Lemma 3.4. *Any two lines $L \neq N$ of $\mathcal{S}(q;a,c)$ are concurrent.*

Proof. (N1) Both in $\mathrm{PG}(2,q)$; let $p = L \cap N$. If both incidences survive, use p. Otherwise, say $p = X_s$ and $L = \lambda_s$, and $N = \ell$ or $N = [w:1:-s]$ with $w \neq 0$. If $N = [w:1:-s]$: H_s. If $N = \ell$ and $s = a$: H_a. If $N = \ell$ and $s \in T$: $X_{\sigma^{-1}(s)}$ (rotation).

(N2) M and L. If L is on P ($L = \ell$, λ_∞ or $[1:0:w]$): P. If $L = \lambda_t$, $t \notin S$: X_t. If $L = \lambda_s$, $s \in S$: H_s. If $L = [u:1:w]$, $u \neq 0$: it meets ℓ in X_r, $r = -w$ (F4); use X_r if $r \notin T$, and H_r if $r \in T$ (H_r is on $[u:1:-r]$ and on M).

(N3) K_s and L. If L is not on A ($L = \ell$ or $L = [u:v:w]$ with $u \neq 0$): $L \cap \lambda_s = (x:s:1)$ (F1, F4), which is on K_s, and its incidence with L survives. If $L = \lambda_s$: $(1:s:1)$. If $L = \lambda_\infty$ or

$L = \lambda_t$, $t \notin T$: Q. If $L = \lambda_t$ with $t \in T \setminus \{s\}$ and $s = a$: A. If $L = \lambda_t$ with $t \in T \setminus \{s\}$ and $s \in T$: if $t = \sigma(s)$, use X_s (on $\lambda_{\sigma(s)}$ and K_s); if $t = \sigma^{-1}(s)$, use H_t (on λ_t and on $K_{\sigma(t)} = K_s$).

(N4) Two new lines. $K_s, K_{s'}$: Q. M, K_a: X_a. M, K_t $(t \in T)$: $H_{\sigma^{-1}(t)}$. □

Proof of Theorem 3.2. By Lemmas 3.3 and 3.4 and Lemma 2.1(ii), $G(q; a, c)$ has diameter at most 3, and adding the filler does not increase it (Lemma 2.2). The graphs are bipartite and not complete bipartite ($A \notin \ell$), so the diameter is exactly 3. The order is $2(q^2 + q + 1) + 10$. □

The proof uses only that $\mathbb{F}$ is a field and that $0, 1, a, c$ are distinct. The correlation δ, extended by $H_s \leftrightarrow K_s$ and $Q \leftrightarrow M$, maps $\mathcal{S}(q; a, c)$ built with σ onto the dual of the structure built with σ^{-1}, so Lemma 3.4 is also Lemma 3.3 for σ^{-1}; we wrote both out so that every case can be checked directly. Both the rotation and the link H_1 I K_0 are needed: without either, the diameter is 4 in every case we checked by computer (every prime power $q \leq 13$ and every admissible (a, c)).

Remark 3.5 (the three hats of [3] in these coordinates). The structure of [3] has hats $\{0, 1, a\}$, no rotation, only the flag (X_a, λ_a) removed, links A I K_a, H_a I ℓ, H_0 I K_1, H_1 I K_0; M contains P and X_t $(t \neq 0, 1)$, and Q lies on λ_∞ and λ_t $(t \neq 0, 1)$. The dictionary with the affine coordinates of [3] is $(x, y) = (x : y : 1)$, $[m, b] = [m : -1 : b]$, $L_c = [1 : 0 : -c]$, $L = [0 : 0 : 1]$, $P_m = (1 : m : 0)$; so $L_0 = \ell$, $P_0 = A$, $(0, s) = X_s$, $[0, s] = \lambda_s$ and $L = \lambda_\infty$. In the completing matching of [3, arXiv v1, p. 8] the index set of $P_i \sim L_i$ should be $\mathbb{F}^*$; with $\mathbb{F} \setminus \{0, 1, a\}$ the four vertices P_1, P_a, L_1, L_a keep degree $q + 1$. Neither the order nor the diameter is affected.

Gabriela, Nacho: please check Remark 3.5 against the journal version of [AL20] (we have only seen arXiv v1), and decide whether the remark should stay in the paper.

3.5. **Cells.** For $q = 9, 11, 13$, Theorem 3.2 gives $(11, 3)$: 192 (2013 table: 190), $(13, 3)$: 276 (2013: 270; current value 274 [3]) and $(15, 3)$: 376 (2013: 370; current value 374 [3]), all with regular graphs. (The 2013 value 190 at $(11, 3)$ is due to Feria-Purón, Miller and Pineda-Villavicencio [25] and, independently, to [3].) At $q = 5$ it gives $72 < 80 = n_b(7, 3)$, a value known to be exact [25].

4. Diameter 4: clones and shadow lines in generalized quadrangles

4.1. **Clones and shadow lines.** Copies of points and lines of a GQ go back to Delorme [15, §§7–8]: his doubling may copy any induced substructure, points and lines together, and his $D = 4$ construction (Example 4.4) doubles a spread. Araujo-Pardo, Dalfó, Fiol and López [2, Sec. 4] double every line of a GQ to get biregular graphs. The construction below is not a doubling: a clone is joined to every old line through its point, so no tree condition on the copied set is needed, and it allows weights larger than one, repeated lines, and "stand-ins". (A level-0 line all of whose points are at level -1 is a shadow line with trivial stand-ins.)

Definition 4.1 (clones and shadow lines)**.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L})$ be a GQ of order (s, t), let $\omega : \mathcal{P} \to \mathbb{Z}_{\geq 0}$ be a *weight*, and $\mathcal{M}$ a finite multiset of lines. For each entry μ of $\mathcal{M}$, with underlying line L_μ, choose for every point p I L_μ a *stand-in* $\theta_\mu(p) \in \{p, p^1, \ldots, p^{\omega(p)}\}$. The incidence structure $\mathcal{S}[\omega, \mathcal{M}, \theta]$ has

- points: $\mathcal{P}$ and the *clones* $p^1, \ldots, p^{\omega(p)}$ of each $p \in \mathcal{P}$;
- lines: $\mathcal{L}$ and one *shadow line* L^μ for each entry μ of $\mathcal{M}$;
- incidences: those of $\mathcal{S}$; p^i I L whenever p I $L \in \mathcal{L}$; and $\theta_\mu(p)$ I L^μ for p I L_μ.

When all stand-ins are old points ($\theta = \mathrm{id}$), a shadow line is adjacent exactly to the old points of L_μ, and clones are adjacent only to old lines.

Theorem 4.2 (clones and shadow lines)**.** *Let $\mathcal{S}$ be a GQ of order (s, t), $s, t \geq 1$, and $\mathcal{S}' = \mathcal{S}[\omega, \mathcal{M}, \theta]$.*

TABLE 3. Neighbourhoods in $\mathcal{S}[\omega, \mathcal{M}, \theta]$.

vertex	neighbours	degree
clone p^i	the $t+1$ old lines on p; the shadow lines L^μ with $\theta_\mu(p) = p^i$	$t+1+\#\{\mu : \theta_\mu(p) = p^i\}$
shadow line L^μ	$\theta_\mu(p)$ for the $s+1$ points p of L_μ	$s+1$
old point p	its $t+1$ lines; the shadow lines L^μ with $\theta_\mu(p) = p$	$t+1+\#\{\mu : \theta_\mu(p) = p\}$
old line L	its $s+1$ points and all their clones	$s+1+\sum_{p \mathrm{I} L} \omega(p)$

(a) $\mathcal{S}'$ has diameter 4.
(b) $\mathcal{S}'$ has maximum degree at most Δ if and only if $\Delta \geq \max(s,t)+1$, $\sum_{p \mathrm{I} L} \omega(p) \leq \Delta - s - 1$ for every line L of $\mathcal{S}$, and every point of $\mathcal{S}'$ (old point or clone) is the stand-in for at most $\Delta - t - 1$ entries of $\mathcal{M}$.
(c) $\mathcal{S}'$ has $(s+1)(st+1) + \sum_p \omega(p)$ points and $(t+1)(st+1) + |\mathcal{M}|$ lines.

Proof. (b) and (c) are read off Table 3. (a) Write π for the map sending a clone or a shadow line to its underlying point or line. Every point x of $\mathcal{S}'$ is incident with every old line through $\pi(x)$. We use Lemma 2.1(i); no element is isolated. Let (x, Y) be an anti-flag of $\mathcal{S}'$.

- Y old. Then $\pi(x) \notin Y$. Let r be the point of Y collinear with $\pi(x)$, on the line N. Then $x \mathrm{I} N \mathrm{I} r \mathrm{I} Y$.
- $Y = L^\mu$ and $\pi(x) \notin L_\mu$. Let $r \mathrm{I} L_\mu$ be collinear with $\pi(x)$, on the line N. Then $x \mathrm{I} N \mathrm{I} \theta_\mu(r) \mathrm{I} L^\mu$.
- $Y = L^\mu$ and $\pi(x) \mathrm{I} L_\mu$. Pick $r \mathrm{I} L_\mu$, $r \neq \pi(x)$ ($s \geq 1$). Then $x \mathrm{I} L_\mu \mathrm{I} \theta_\mu(r) \mathrm{I} L^\mu$.

So $d(x, Y) \leq 3$ for every anti-flag, and the diameter is at most 4. Two non-collinear old points exist (a point is collinear with $s(t+1) < (s+1)(st+1) - 1$ others) and stay non-collinear: they share no old line, a shadow line L^μ contains only points over points of L_μ, and an old point lies over itself only. So the diameter is 4. □

Corollary 4.3 (exact weights, regular completion)**.** *Let $\Delta \geq \max(s,t)+1$, let ω be a weighted $(\Delta - s - 1)$-ovoid (that is, $\sum_{p \mathrm{I} L} \omega(p) = \Delta - s - 1$ for every line L) and $\mathcal{M}$ a weighted $(\Delta - t - 1)$-cover (every point is on exactly $\Delta - t - 1$ entries of $\mathcal{M}$). Then $\mathcal{S}[\omega, \mathcal{M}, \mathrm{id}]$ has diameter 4, maximum degree Δ and $2\Delta(st+1)$ elements. It becomes Δ-regular, with the same order and diameter 4, by the following explicit flags: number the clones $c_0, c_1, \ldots$ and the shadow lines $y_0, \ldots, y_{b-1}$ ($b = |\mathcal{M}|$), put $k = \Delta - t - 1$, and join c_i to y_j for $j \equiv ik, ik+1, \ldots, ik+k-1 \pmod b$. (If $b = 0$ or $\sum_p \omega(p) = 0$, the structure is already Δ-regular.)*

Proof. Counting flags gives $\sum_p \omega(p) = (\Delta - s - 1)(st+1)$ and $|\mathcal{M}| = (\Delta - t - 1)(st+1)$; the total is $(s+t+2+\Delta-s-1+\Delta-t-1)(st+1) = 2\Delta(st+1)$. Old points and lines have degree Δ, clones $t+1$, shadow lines $s+1$. The added pairs are anti-flags (with $\theta = \mathrm{id}$ no clone is on a shadow line); the blocks $\{ik, \ldots, ik+k-1\}$, $i = 0, 1, \ldots$, run through $0, 1, \ldots, k\sum_p \omega(p) - 1 = b(\Delta - s - 1) - 1$, so modulo b every residue is hit exactly $\Delta - s - 1$ times, and within a block the residues are distinct since $k \leq b$. Adding flags does not increase distances (Lemma 2.2), and two non-collinear old points still share no line. □

Example 4.4 (Delorme's $D = 4$ construction [15, §8(a)]; the 2013 values)**.** In $W(3, q)$ take a spread Σ (a set of q^2+1 lines partitioning the points), a line $v \in \Sigma$, and $\mathcal{A}_1 = \{v\}$, $\mathcal{A}_0 = \Sigma \cup \{\text{points of } v\}$. (L1): the elements on v are its points; a point of v meets only v in $\mathcal{A}_1$, and a line of $\Sigma \setminus \{v\}$ meets nothing in $\mathcal{A}_1$. (L0): a point off v is on exactly one line of Σ and on no other line of $\mathcal{A}_0$; a line not in Σ has at most one point on v. The components of the substructure on $\mathcal{A}_0$

are the star formed by v and its points, and single lines, so (C_4) holds (Lemma 2.7). Theorem 2.6 gives maximum degree $q+m$, diameter 4 and $2(q+1)(q^2+1)+(m-1)\big((q^2+1)+(q+1)+m\big)$ elements. For $m=2$ this is $2q^3+3q^2+3q+6$: 346, 1734, 3064, 4946 at $q=5,9,11,13$; for $m=3$, $q=13$ it is $4760+2\cdot 187=5134$. These are the 2013 values at $\Delta=7,11,13,15,16$.

Remark 4.5 (a limit for shadow lines on $Q(4,q)$). Let q be odd, $\mathcal{O}$ an ovoid of $Q(4,q)$ (a set of q^2+1 points meeting every line once), $\omega=\mathbf{1}_{\mathcal{O}}$, and suppose the maximum degree is $q+2$. Then at most q^2-q+2 shadow lines are possible. Indeed every old point and every clone serves at most one shadow line, so a point off $\mathcal{O}$ is on at most one entry of $\mathcal{M}$ and a point of $\mathcal{O}$ on at most two. Suppose some $o\in\mathcal{O}$ is on two entries. They are distinct lines: if both were copies of one line N, each of the q points of N other than o is off $\mathcal{O}$, has no clone, and would be the stand-in of both entries, giving it degree $q+3$. Each other entry L misses o and so has a unique point y collinear with o; $y\notin\mathcal{O}$ (points of an ovoid are pairwise non-collinear); the line oy is not an entry of $\mathcal{M}$ (otherwise $y\notin\mathcal{O}$ would be on two entries), so it is one of the $q-1$ lines on o other than the two entries through o, and y is one of its q points other than o; distinct L give distinct y. So $|\mathcal{M}|\le 2+q(q-1)$. Otherwise every point is on at most one entry, $\mathcal{M}$ is a partial spread, and $|\mathcal{M}|\le q^2-q+1$ (Tallini [36]; see also [10, §1]).

Example 4.6 ($(7,4)\ge 359$)**.** In $Q(4,5)$ take one clone of each point of an ovoid $\mathcal{O}$ that is an elliptic quadric section ($\omega=\mathbf{1}_{\mathcal{O}}$), and 21 shadow lines with stand-ins. Theorem 4.2 gives diameter 4, maximum degree 7, and $2\cdot 6\cdot 26+26+21=359$ elements. Appendix A lists the 21 lines and the stand-ins in coordinates, so that the conditions of Theorem 4.2(b) can be checked by hand. Every ovoid of $Q(4,p)$, p prime, is an elliptic quadric [4], so the choice of ovoid does not matter. By Remark 4.5 at most $q^2-q+2=22$ shadow lines are possible in this setting; 21 is the maximum for this family (an integer program, solved to optimality). This example is not a member of a proved infinite family. The best regular graph we know at $(7,4)$ has 350 vertices (Corollary 4.8 at $q=5$).

Example 4.6: the optimality of 21 comes from an integer-programming solver (HiGHS). Keep the sentence, or drop it as computer-dependent?

4.2. The Ahrens–Szekeres quadrangle and two elliptic quadrics.

Coordinates. Let q be odd. Let $W(q)=W(3,q)$ be the symplectic quadrangle of $\mathrm{PG}(3,q)$ for the alternating form

$$\beta(u,v)=u_0v_1-u_1v_0+u_2v_3-u_3v_2,$$

and let $x=(1:0:0:0)$; then $x^\perp$ is the plane $X_1=0$. Payne's derived quadrangle $\mathrm{AS}(q)=P(W(q),x)$ [32], [33, 3.1.4] has

- *points*: the q^3 points $(a:1:b:c)$, $a,b,c\in\mathbb{F}$ (the points off $x^\perp$); we write (a,w) with $w=(b,c)$;
- *lines of type (i) (fibres)*, q^2 of them: $\{(a_0+\mu:1:w_0):\mu\in\mathbb{F}\}$, the line $\langle x,(a_0:1:w_0)\rangle$ minus x;
- *lines of type (ii)*, $q^2(q+1)$ of them: for a point $(a_0:1:w_0)$ and a direction $d=(d_1:d_2)\in\mathrm{PG}(1,q)$,

$$\ell(a_0,w_0;d)=\{(a_0+\lambda\rho:1:w_0+\lambda d):\lambda\in\mathbb{F}\},\qquad \rho=\det(w_0,d)=b_0d_2-c_0d_1,$$

the totally isotropic line through $(a_0:1:w_0)$ and $(\rho:0:d_1:d_2)$, minus the latter point.

(Check: $\beta((a_0,1,b_0,c_0),(\rho,0,d_1,d_2))=-\rho+b_0d_2-c_0d_1=0$. A point off $x^\perp$ is on $q+1$ totally isotropic lines, none through x, each meeting $x^\perp$ in one point $(\rho:0:d)$, $d\neq 0$; so each point of $\mathrm{AS}(q)$ is on one fibre and on $q+1$ lines of type (ii), and every line has q points.) $\mathrm{AS}(q)$ is a GQ

of order $(q-1, q+1)$ [32], [33, 3.1.4, 3.2.6]; for q odd it is the Ahrens–Szekeres quadrangle [1]. The fibres form a spread.

Fix a non-square $\varepsilon \in \mathbb{F}$ and put $N(b,c) = b^2 - \varepsilon c^2$, an anisotropic form ($N(w) = 0$ only for $w = 0$). For $\kappa \in \mathbb{F}^*$ let $\mathcal{E}_\kappa$ be the quadric $X_0X_1 = \kappa N(X_2, X_3)$. It is elliptic (a hyperbolic pair plus an anisotropic binary form), passes through x with tangent plane $x^\perp$, and $\mathcal{E}_\kappa \setminus \{x\} = \{(\kappa N(w) : 1 : w) : w \in \mathbb{F}^2\}$ is a set of q^2 points of $\mathrm{AS}(q)$, one on each fibre.

Theorem 4.7 (a weighted 2-ovoid of $\mathrm{AS}(q)$)**.** *Let q be odd, ε a non-square and $\kappa_1\kappa_2 = 1/(4\varepsilon)$. Every line of $\mathrm{AS}(q)$ meets $\mathcal{E}_{\kappa_1} \setminus \{x\}$ and $\mathcal{E}_{\kappa_2} \setminus \{x\}$ in exactly two points in total (a point in both counted twice). The two quadrics meet only in x and $e = (0:1:0:0)$. So $\varpi = \mathbf{1}_{\mathcal{E}_{\kappa_1}\setminus x} + \mathbf{1}_{\mathcal{E}_{\kappa_2}\setminus x}$ is a weighted 2-ovoid of $\mathrm{AS}(q)$, with $\varpi(e) = 2$ and $\varpi \in \{0,1\}$ elsewhere.*

The choice $\kappa_1 = \frac{1}{2}$, $\kappa_2 = \frac{1}{2\varepsilon}$ gives the two quadrics

$$\mathcal{E}^{(1)}:\ 2X_0X_1 = X_2^2 - \varepsilon X_3^2, \qquad \mathcal{E}^{(2)}:\ 2\varepsilon X_0X_1 = X_2^2 - \varepsilon X_3^2.$$

Corollary 4.8. *For every odd prime power q, adjoining to $\mathrm{AS}(q)$ one clone of each point of $\mathcal{E}^{(1)} \setminus \{x\}$ and one of each point of $\mathcal{E}^{(2)} \setminus \{x\}$ (so e gets two clones) gives a $(q+2)$-regular incidence structure of diameter 4 with $2q^2(q+2)$ elements. Hence $n_b(q+2, 4) \geq 2q^2(q+2)$, attained by a regular graph.*

New vertices. For $P = (\kappa_i N(w) : 1 : w) \in \mathcal{E}_{\kappa_i} \setminus \{x\}$ ($i = 1, 2$, $w \in \mathbb{F}^2$), the clone $P^{(i)}$ is adjacent exactly to the $q+2$ lines of $\mathrm{AS}(q)$ through P: the fibre $\{(\kappa_i N(w) + \mu : 1 : w) : \mu \in \mathbb{F}\}$ and, for each $d \in \mathrm{PG}(1,q)$, the line $\ell(\kappa_i N(w), w; d)$. The two clones of e are adjacent to the same $q+2$ lines. Changed old neighbourhoods: the fibre over w_0 gains the clones of $(\kappa_1 N(w_0) : 1 : w_0)$ and $(\kappa_2 N(w_0) : 1 : w_0)$; the line $\ell(a_0, w_0; d)$ gains the clones of its points on $\mathcal{E}_{\kappa_1}$ and on $\mathcal{E}_{\kappa_2}$, which correspond to the roots λ of the quadratic (4.1) below with $\kappa = \kappa_1, \kappa_2$ (two clones in total). Old points are unchanged.

Proof of Theorem 4.7. Fibres. The fibre over w_0 meets $\mathcal{E}_\kappa$ only at $\mu = \kappa N(w_0) - a_0$: one point of each quadric.

Lines of type (ii). Let $B = N(w_0 + d) - N(w_0) - N(d) = 2(b_0d_1 - \varepsilon c_0 d_2)$. The point with parameter λ of $\ell(a_0, w_0; d)$ lies on $\mathcal{E}_\kappa$ if and only if $a_0 + \lambda\rho = \kappa N(w_0 + \lambda d)$, that is,

$$\kappa N(d)\lambda^2 + (\kappa B - \rho)\lambda + \kappa N(w_0) - a_0 = 0. \tag{4.1}$$

As $N(d) \neq 0$ this is a quadratic, with $1 + \chi(\mathrm{disc}_\kappa)$ roots. A direct expansion gives $B^2 - 4N(w_0)N(d) = 4\varepsilon(b_0d_2 - c_0d_1)^2 = 4\varepsilon\rho^2$, hence

$$\mathrm{disc}_\kappa = (\kappa B - \rho)^2 - 4\kappa N(d)(\kappa N(w_0) - a_0) = 4\kappa N(d)\,(a_0 - a_\kappa^*), \qquad a_\kappa^* = -\frac{4\varepsilon\kappa\rho^2 - 2B\rho + \rho^2/\kappa}{4N(d)}.$$

Then $a_{\kappa_1}^* - a_{\kappa_2}^* = -\frac{\rho^2}{4N(d)}(\kappa_1 - \kappa_2)\big(4\varepsilon - \frac{1}{\kappa_1\kappa_2}\big) = 0$; call the common value a^*. The line meets the two quadrics in

$$\big(1 + \chi(\kappa_1)\chi(N(d)(a_0 - a^*))\big) + \big(1 + \chi(\kappa_2)\chi(N(d)(a_0 - a^*))\big) = 2$$

points, because $\chi(\kappa_1)\chi(\kappa_2) = \chi(1/(4\varepsilon)) = \chi(\varepsilon) = -1$, so $\chi(\kappa_1) + \chi(\kappa_2) = 0$. (If $a_0 = a^*$, the line is tangent to both quadrics and meets each in one point.)

Intersection. $\kappa_1 \neq \kappa_2$, since $\kappa_1^2 = 1/(4\varepsilon)$ is impossible for a non-square ε. So $\kappa_1 N(w) = \kappa_2 N(w)$ forces $N(w) = 0$, $w = 0$, and the common points are x and e. □

Proof of Corollary 4.8. Theorem 4.2 and Corollary 4.3 with $(s,t) = (q-1, q+1)$, $\omega = \varpi$ of Theorem 4.7 and $\mathcal{M} = \emptyset$: every line has q points and gains exactly two clones, every old point is on $q+2$ lines, and every clone is on the $q+2$ lines of its point. The order is $q^3 + q^2(q+2) + 2q^2 = 2q^2(q+2)$. □

Remark 4.9. Weights are necessary: AS(3) has no 1- or 2-ovoid and AS(5) has no m-ovoid for $m = 1, \dots, 4$ (computer-checked by integer programming). No choice of clones and shadow lines on AS(q) does better at $\Delta = q + 2$: every point of AS(q) is already on $t + 1 = q + 2$ lines, so no shadow line can be added, and each line has q points and takes at most 2 clones, so $\sum_p \omega(p) \le 2q^2$. Corollary 4.8 beats Delorme's $2q^3 + 3q^2 + 3q + 6$ by $q^2 - 3q - 6 > 0$ for $q \ge 5$.

Cells. $(11, 4)$: 1782 (2013: 1734); $(13, 4)$: 3146 (3064); $(15, 4)$: 5070 (4946). All regular.

4.3. **Singer 2-ovoids of $W(3, q)$.**

The cyclic model. Let q be odd, $\mathbb{F} = \mathrm{GF}(q) \subset K = \mathrm{GF}(q^2) \subset E = \mathrm{GF}(q^4)$, and g a primitive element of E. As E is a 4-dimensional $\mathbb{F}$-space, $\mathrm{PG}(3, q) = E^*/\mathbb{F}^*$ (any $\mathbb{F}$-basis of E gives homogeneous coordinates); the point $\langle g^i \rangle$ has index i modulo $(q + 1)(q^2 + 1)$. Write $\bar{y} = y^{q^2}$ (the involution of E/K), $k' = k^q$ for $k \in K$, $\mathrm{N}(y) = y\bar{y} \in K$ and $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}}$. Let $d = g^{(q^2+1)/2}$; then $\bar{d} = -d$, and $h = d^2 = g^{q^2+1}$ generates K^*. The form

$$b(y, z) = \mathrm{Tr}_{E/\mathbb{F}}(d\, y\bar{z}) = \mathrm{Tr}\big(d(y\bar{z} - \bar{y}z)\big)$$

is alternating and non-degenerate; $W(3, q)$ is its symplectic quadrangle. Multiplication by g^{q+1} multiplies b by $g^{(q+1)(q^2+1)} \in \mathbb{F}^*$, so it is a collineation of $W(3, q)$. Its point orbits are the classes

$$C_a = \{\langle g^i \rangle : \ i \equiv a \pmod{q+1}\}, \qquad |C_a| = q^2 + 1,$$

and $\langle y \rangle \in C_a$ if and only if $\mathrm{N}(y) \in h^a\mathbb{F}^*$ (as $\mathrm{N}(g^i) = h^i$ and $K^*/\mathbb{F}^*$ is cyclic of order $q + 1$, generated by h). These are Ebert's elliptic quadrics partitioning $\mathrm{PG}(3, q)$ [19].

The K-subspaces yK ($y \in E^*$) are lines of $\mathrm{PG}(3, q)$ (the regular spread), with points $\langle g^{i+(q^2+1)j} \rangle$, $j = 0, \dots, q$. They are totally isotropic: for $k_1, k_2 \in K$ we have $\overline{yk_i} = \bar{y}k_i$ (as $\bar{k} = k^{q^2} = k$), so $b(yk_1, yk_2) = \mathrm{Tr}\big(d\,\mathrm{N}(y)(k_1k_2 - k_1k_2)\big) = 0$. So $\Sigma = \{yK\}$ is a spread of $W(3, q)$ with $q^2 + 1$ lines.

Proposition 4.10 (Singer 2-ovoids)**.** *Let q be odd.*

(a) Every line of $\mathrm{PG}(3, q)$ meets each class C_a in at most two points.

(b) Every totally isotropic line of $W(3, q)$ meets $C_a \cup C_{q-a}$ in exactly two points.

So the $(q+1)/2$ sets $C_a \cup C_{q-a}$, $0 \le a \le (q-1)/2$, are 2-ovoids of $W(3, q)$ partitioning its points. (The pairing $a \leftrightarrow q - a$ has no fixed point, as $2a \equiv -1 \pmod{q+1}$ is impossible for $q + 1$ even.)

Proof. Let $L = \langle y, z \rangle_{\mathbb{F}}$ and $u = sy + tz$ ($s, t \in \mathbb{F}$). Then $\mathrm{N}(u) = A_1s^2 + \tau st + A_2t^2$ with $A_1 = \mathrm{N}(y)$, $A_2 = \mathrm{N}(z)$, $\tau = y\bar{z} + \bar{y}z$, all in K. Put $\theta = y\bar{z} - \bar{y}z$. As $\bar{\theta} = -\theta$, $\theta/d \in K$; write $\theta = dk_0$ with $k_0 \in K$. Then

$$\mathfrak{d} := \tau^2 - 4A_1A_2 = \theta^2 = hk_0^2, \qquad b(y, z) = \mathrm{Tr}(hk_0).$$

Fix $\varkappa \in K^*$ with $\varkappa' = -\varkappa$. For $\mu \in K^*$ put $\nu = \varkappa/\mu$; the kernel of $k \mapsto \mathrm{Tr}(\nu k)$ is the $\mathbb{F}$-subspace $\mu\mathbb{F}$. So $\langle u \rangle$ lies in $\widetilde{C}_\mu := \{\langle u \rangle : \mathrm{N}(u) \in \mu\mathbb{F}^*\}$ (thus $C_a = \widetilde{C}_{h^a}$) if and only if $(s : t)$ is a zero of the binary form

$$f_\nu(s, t) = \mathrm{Tr}(\nu A_1)s^2 + \mathrm{Tr}(\nu\tau)st + \mathrm{Tr}(\nu A_2)t^2.$$

(a) Suppose $f_\nu \equiv 0$. Then $A_1, \tau, A_2 \in \mu\mathbb{F}$. If $\mathfrak{d} \ne 0$, then $hk_0^2 = \mathfrak{d} \in \mu^2\mathbb{F}^*$, so $h \in (K^*)^2\mathbb{F}^* = (K^*)^2$ (every element of $\mathbb{F}^*$ is a square in K, since $q - 1$ divides $(q^2 - 1)/2$); but h generates K^*, which has even order. If $\mathfrak{d} = 0$, then $\mathrm{N}(sy + tz) = \mu(\alpha_0 s^2 + \beta_0 st + \gamma_0 t^2)$ with $\alpha_0, \beta_0, \gamma_0 \in \mathbb{F}$, $\beta_0^2 = 4\alpha_0\gamma_0$ and $\alpha_0 \ne 0$ (as $A_1 \ne 0$), which vanishes at $(s : t) = (-\beta_0 : 2\alpha_0)$: impossible, since $\mathrm{N}(u) \ne 0$ for $u \ne 0$. So $f_\nu \not\equiv 0$, and it has $1 + \chi(\mathrm{disc}(\nu)) \le 2$ projective zeros, where $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu\tau)^2 - 4\mathrm{Tr}(\nu A_1)\mathrm{Tr}(\nu A_2)$.

(b) Expanding with $\mathrm{Tr}(\nu k) = \nu k + \nu' k'$,

$$\mathrm{disc}(\nu) = \mathrm{Tr}(\nu^2\mathfrak{d}) + \nu\nu'\Psi, \qquad \Psi = 2\tau\tau' - 4A_1A_2' - 4A_1'A_2 \in \mathbb{F}.$$

If L is totally isotropic, then $\mathrm{Tr}(hk_0) = 0$, so $hk_0 \in \varkappa\mathbb{F}$ and $\eta := h\mathfrak{d} = (hk_0)^2 \in \varkappa^2\mathbb{F} \subseteq \mathbb{F}$. Then $\mathrm{Tr}(h^2\nu'^2\mathfrak{d}) = \eta(h\nu'^2 + h'\nu^2) = hh'\,\mathrm{Tr}(\nu^2\mathfrak{d})$, and since $(h\nu')(h\nu')' = hh'\nu\nu'$,

$$\mathrm{disc}(h\nu') = \mathrm{N}_{K/\mathbb{F}}(h)\,\mathrm{disc}(\nu).$$

$\mathrm{N}_{K/\mathbb{F}}(h) = h^{q+1}$ generates $\mathbb{F}^*$, so it is a non-square and $\chi(\mathrm{disc}(h\nu')) = -\chi(\mathrm{disc}(\nu))$. The class belonging to $h\nu'$ is $\varkappa/(h\nu') = -\mu'/h \equiv (h\mu)^{-1}$ modulo $\mathbb{F}^*$ (using $\mu^{q+1} \in \mathbb{F}^*$); in indices, $a \mapsto -a - 1 \equiv q - a \pmod{q+1}$. Scaling ν by $\mathbb{F}^*$ multiplies disc by a square. So L meets $C_a \cup C_{q-a}$ in $(1 + \chi(\mathrm{disc}(\nu))) + (1 - \chi(\mathrm{disc}(\nu))) = 2$ points. □

Remark 4.11 (literature). For q odd, $W(3,q)$ has m-ovoids for every even m and $(q+1)/2$-ovoids [5], and [24] states, citing [5], that $W(3,p^e)$ (p odd) has a partition into 2-ovoids. Cossidente, Culbert, Ebert and Marino [14] construct m-ovoids of $W(3,q)$ with Singer-type stabilisers. We have not checked whether the 2-ovoids $C_a \cup C_{q-a}$ coincide with those; Proposition 4.10 is very likely known, and we claim only the short proof. The proof uses only that d^2 is a non-square in K: for any δ_0 with $\bar{\delta}_0 = -\delta_0$, $C_a \cup C_b$ is a 2-ovoid of the form $\mathrm{Tr}_{E/\mathbb{F}}(\delta_0 y\bar{z})$ when $h^{a+b}\delta_0^2 \in \mathbb{F}^*$.

Proposition 4.10: do you know whether the 2-ovoids $C_a \cup C_{q-a}$ are those of [BLP09], [CCEM08] or [FWX19]? If so, cite and shorten.

Corollary 4.12. *Let q be odd, $0 \le a \le (q-1)/2$, $\mathcal{O} = C_a \cup C_{q-a}$ and Σ the regular spread $\{yK\}$. Adjoin to $W(3,q)$*

- *one clone P' of each point $P \in \mathcal{O}$, incident with the $q+1$ totally isotropic lines on P (the lines $\langle P, Z\rangle$ with $b(P,Z) = 0$);*
- *two shadow lines L', L'' for each $L \in \Sigma$, each incident with the $q+1$ points of L.*

The result is bipartite, has diameter 4, maximum degree $q+3$ (old points and lines have degree $q+3$, clones and shadow lines $q+1$) and $2(q^2+1)(q+3)$ elements. Joining clone c_i to shadow lines y_{2i}, y_{2i+1} (indices modulo $2(q^2+1)$, any numbering) makes it $(q+3)$-regular with diameter 4. Hence $n_b(q+3,4) \ge 2(q^2+1)(q+3)$, attained by a regular graph.

Proof. $W(3,q)$ is a GQ of order (q,q). By Proposition 4.10(b), $\omega = \mathbf{1}_{\mathcal{O}}$ is a 2-ovoid ($= \Delta - s - 1$ for $\Delta = q+3$), and $\mathcal{M} = 2\Sigma$ is a 2-cover (every point is on exactly one spread line). Apply Corollary 4.3 with $k = 2$. □

This is the point–line dual of clones of an ovoid of $Q(4,q)$ with a dual 2-cover; written in $W(3,q)$ it needs no Klein correspondence. More generally, j copies of the spread and $j/2$ copies of the 2-ovoid give $n_b(q+1+j,4) \ge 2(q^2+1)(q+1+j)$ for even j.

Corollary 4.13 (all degrees)**.** *For every prime power q and every $\Delta \ge q+1$ there is a Δ-regular incidence structure of diameter 4 with $2\Delta(q^2+1)$ elements if q is even or $\Delta - q - 1$ is even, and with $2\Delta q^2$ elements otherwise. In particular $n_b(\Delta,4) \ge 2q^2\Delta$ for every prime power $q \le \Delta - 1$.*

Proof. Put $j = \Delta - q - 1$. In the first two cases we apply Corollary 4.3 in $W(3,q)$ ($s = t = q$, so $\Delta - s - 1 = \Delta - t - 1 = j$). If q is even, $W(3,q)$ has an ovoid $\mathcal{O}_0$ and a spread Σ [33, 3.4.1]; take $\omega = j\mathbf{1}_{\mathcal{O}_0}$ and $\mathcal{M} = j\Sigma$. If q is odd and j even: $\omega = (j/2)\mathbf{1}_{\mathcal{O}}$ with $\mathcal{O}$ a 2-ovoid of Proposition 4.10, and $\mathcal{M} = j\Sigma$. If q is odd and j odd, then $\Delta \ge q+2$, and we apply Corollary 4.3 in $\mathrm{AS}(q)$ ($s = q-1$, $t = q+1$, so $\Delta - s - 1 = \Delta - q$ and $\Delta - t - 1 = \Delta - q - 2$), with $(\Delta - q)/2$ copies of the weighted 2-ovoid of Theorem 4.7 and $\Delta - q - 2$ copies of the fibre spread. □

Cells. $(16,4)$: $q = 13$, $5440 = 2 \cdot 170 \cdot 16$ (2013: 5134), regular. In the range $\Delta \le 16$ this is the only open cell with $\Delta - 3$ an odd prime power. Without stand-ins, 5440 is the most a structure of clones and shadow lines on $W(3,13)$ or $Q(4,13)$ of maximum degree 16 can have (each line takes at most 2 clones and each point lies on at most 2 shadow lines).

5. Diameter 6: whiskers in the split Cayley hexagon

Throughout Sections 5.1–5.4, q is odd; even q is treated in Section 5.5. We use the model of $H(q)$ in Section 2.

5.1. **Elliptic hyperplane sections and Hermitian spreads.** Let $v = \langle e_1, e_6\rangle$. It is a line of $Q(6,q)$ (e_1, e_6 are singular and $b(e_1,e_6)=0$) and a line of $H(q)$ (no equation of (2.1) involves p_{16}). Fix $c \in \mathbb{F}$ with $-c$ a non-square. For $z=(\alpha,\beta)\in\mathbb{F}^2$ let

$$\Pi_z:\ X_0 + cX_4 + \alpha X_2 + \beta X_5 = 0, \qquad \Pi = \Pi_{(0,0)},$$

and let S_z be the set of lines of $H(q)$ contained in Π_z, $S = S_{(0,0)}$. Every Π_z contains v.

Lemma 5.1. *For every $z\in\mathbb{F}^2$:*

(a) $\Pi_z \cap Q(6,q)$ is an elliptic quadric $Q^-(5,q)$.
(b) Every point of $H(q)$ in Π_z is on exactly one line of S_z. So $|S_z| = q^3+1$, the lines of S_z are pairwise opposite, and $v\in S_z$ (S_z is a Hermitian spread [37]).
(c) A line of $H(q)$ not in S_z meets Π_z in exactly one point.
(d) The points of $H(q)$ at distance at most 3 from v are exactly the points of $Q(6,q)$ with $X_2=X_5=0$; among them only the points of v lie in Π_z.
(e) $S_z\cap S_{z'} = \{v\}$ for $z\neq z'$.

Proof. (a) On Π_z substitute $X_0 = -cX_4 - \alpha X_2 - \beta X_5$: $X_0X_4 + X_1X_5 + X_2X_6 - X_3^2 = (X_1 - \beta X_4)X_5 + X_2(X_6 - \alpha X_4) - (X_3^2 + cX_4^2)$, two hyperbolic planes plus the anisotropic form $X_3^2+cX_4^2$. So the form on Π_z is non-degenerate of Witt index 2.

(b) Let $y\in\Pi_z\cap Q(6,q)$. By (a), $\Pi_z\cap Q(6,q)$ contains no totally singular plane, so $\pi_y \not\subseteq \Pi_z$ and $\pi_y\cap\Pi_z$ is one line through y: the only line of $H(q)$ on y inside Π_z. $Q^-(5,q)$ has $(q+1)(q^3+1)$ points, so $|S_z| = q^3+1$. Two lines of S_z are not concurrent (uniqueness). If two were at distance 4, a line N of $H(q)$ would meet both in distinct points, so $N\subseteq\Pi_z$, $N\in S_z$, concurrent with them: a contradiction. So they are opposite.

(c) A line not in the hyperplane Π_z meets it in one point.

(d) Let $y = ae_1 + be_6$ be on v and yu a line of $H(q)$. Then $p_{12}=p_{34}$ gives $au_2=0$, $p_{65}=p_{30}$ gives $bu_5=0$, and $b(y,u)=0$ gives $au_5+bu_2=0$; so $u_2=u_5=0$. Conversely $Q(6,q)\cap\{X_2=X_5=0\}$ is $X_0X_4=X_3^2$, a cone with vertex v over a conic, with $(q+1)+(q+1)q^2$ points: exactly the number of points at distance at most 3 from v ($q+1$ on v, and q more on each of the $q(q+1)$ other lines meeting v). If such a point is in Π_z, then $X_0=-cX_4$ and $X_3^2+cX_4^2=0$, so $X_0=X_3=X_4=0$ and the point is on v.

(e) If $\ell\in S_z\cap S_{z'}$ with $\ell\neq v$, then $(\alpha-\alpha')X_2+(\beta-\beta')X_5=0$ on ℓ, so some point of ℓ has $X_2=X_5=0$ and is at distance at most 3 from v by (d); but ℓ and v are opposite by (b), so every point of ℓ is at distance 5 from v. □

5.2. **Delorme's levels and whiskers.** Let $\mathcal{P}(v)$ be the points of v, $\mathcal{L}_v$ the q^2+q lines other than v that meet v, and $\mathcal{P}_\Pi$ the $(q+1)(q^3+1)$ points of $H(q)$ in Π. A point is *free* if it is not in Π and is at distance 5 from v; by Lemma 5.1(d) there are $(q+1)(q^4+q^2+1)-(q+1)-q^2(q+1)-q^3(q+1) = q^3(q^2-1)$ of them.

Definition 5.2. A *whisker set* is a set W of lines of $H(q)$, each opposite v and not in S, such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W.

A line opposite v and not in S meets Π in one point, not on v (Lemma 5.1(c),(d)), and its other q points are free (every point of a line opposite v is at distance 5 from v). So $|W|\le q^3(q^2-1)/q = q^2(q^2-1)$, with equality if and only if W is exact.

Theorem 5.3 (doubling with whiskers)**.** *Let W be a whisker set and $m \geq 2$. Put*

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \mathcal{P}(v) \cup S, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Pi \cup \mathcal{L}_v \cup W.$$

Then $\mathcal{A}$ satisfies (L0), (L1) and (C_6). Hence $H(q)^{(m)}_{\mathcal{A}}$ is bipartite with maximum degree $q+m$, diameter 6, and

$$2(q+1)(q^4+q^2+1) + (m-1)\big(q^4+2q^3+q^2+2q+2+|W|+m(q^3+q+2)+m^2\big)$$

elements.

Proof. (L1), $i=1$: the elements on v are its points (in $\mathcal{A}_1$); a point of v is on one element of $\mathcal{A}_2$, a line of $S \setminus \{v\}$ on none. (L1), $i=0$: the elements incident with $\mathcal{A}_1$ are the points of v, the lines of $\mathcal{L}_v$ and the points of Π, all in $\mathcal{A}_0$; a line of $\mathcal{L}_v$ meets v in one point and contains no other point of $\mathcal{A}_1$; a point of $\mathcal{P}_\Pi \setminus \mathcal{P}(v)$ is on exactly one line of S (Lemma 5.1(b)) and not on v; a whisker has no point on v. (L0): a point outside $\mathcal{A}_0$ is off Π. If it is at distance 3 from v, it is on exactly one line of $\mathcal{L}_v$ (two would close a circuit of length at most 6) and on no whisker (a whisker through it would be at distance at most 4 from v). If it is free, it is on no line of $\mathcal{L}_v$ and on at most one whisker. A line outside $\mathcal{A}_0$ is not in Π (the lines in Π are those of S), so it meets Π in one point, its only point in $\mathcal{A}_0$. (C_6): a line of $\mathcal{L}_v$ has no point of Π other than its point on v (Lemma 5.1(d)), and a whisker has exactly one point of Π, not on v. So the components of the substructure on $\mathcal{A}_0$ are

$$\{v\} \cup \mathcal{P}(v) \cup \mathcal{L}_v \qquad \text{and} \qquad \{n\} \cup \mathcal{P}(n) \cup \{h \in W : h \text{ meets } n\} \quad (n \in S \setminus \{v\}),$$

trees of diameter 4 (the ball of radius 2 about v; a line, its points and the whiskers hanging from them). Lemma 2.7 gives (C_6), and Theorem 2.6 gives degree at most $q+1+(m-1)$, diameter at most 6 and the count, with $|\mathcal{A}_2| = 1$, $|\mathcal{A}_1| = (q^3+1)+(q+1)$ and $|\mathcal{A}_0| = (q^3+1)+(q^2+q)+|W|+(q+1)(q^3+1)$. The diameter is exactly 6: a free point p (level -1) has q^5 opposite points, and only $(q+1)(q^3+1) < q^5$ points lie in $\mathcal{A}_0$, so some point p' opposite p has level -1, and $d(p,p') \geq 6$ by projection. □

The neighbourhood of every version. Write x^w for the versions ($i,j,k \in [m]$). Every element of $H(q)$ appears once in Table 4. An element of level k has m^{k+1} versions.

Table 4: The neighbourhood of every version in Theorem 5.3.

element (level)	versions	neighbours of the version	degree
v (2)	v^{ijk}	p^{ij} for the $q+1$ points p of v	$q+1$
point p of v (1)	p^{ij}	$v^{ij1},\dots,v^{ijm}$; L^i for the q lines $L \in \mathcal{L}_v$ on p	$q+m$
line $n \in S \setminus \{v\}$ (1)	n^{ij}	p^i for the $q+1$ points p of n	$q+1$
point $p \in \mathcal{P}_\Pi \setminus \mathcal{P}(v)$ (0)	p^i	$n^{i1},\dots,n^{im}$ for the line $n \in S$ on p; h^i for each whisker h on p; every other line on p	$q+m$
line $L \in \mathcal{L}_v$ (0)	L^i	$p^{i1},\dots,p^{im}$ for $p = L \cap v$; the q other points of L	$q+m$
whisker $h \in W$ (0)	h^i	p^i for $p = h \cap \Pi$; the q free points of h	$q+1$
point $r \notin \Pi$ (-1)	r	all m versions of the line of $\mathcal{L}_v \cup W$ on r (if any); every other line on r	$q+m$, or $q+1$ if r is free and on no whisker

element (level)	versions	neighbours of the version	degree
line $N \notin \mathcal{A}_0$ (-1)	N	all m versions of its point in Π (not on v); its q other points	$q+m$

5.3. **An exact whisker set for every odd q.** For a point X of $H(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 - X_4X_5}{cX_2^2 + X_5^2}.$$

The denominator vanishes only if $X_2 = X_5 = 0$, that is (Lemma 5.1(d)) only on v. Let

$$G(a,b) = (c_0a + c_1b)(a^2 + cb^2) - b^3, \qquad F(z) = \frac{G(\alpha,\beta)}{\alpha^2 + c\beta^2} \quad (z = (\alpha,\beta) \neq 0),$$

with $c_0, c_1 \in \mathbb{F}$ chosen so that G has no zero on $\mathbb{F}^2 \setminus \{0\}$.

Lemma 5.4. *Such c_0, c_1 exist for every odd q.*

Proof. $G(a,0) = c_0a^3$, so $c_0 \neq 0$ is needed. For $b \neq 0$ and $t = a/b$, $G(a,b) = b^3\big((c_0t + c_1)(t^2 + c) - 1\big)$, and $t^2 + c \neq 0$. So G has a zero with $b \neq 0$ if and only if (c_0, c_1) lies on one of the q lines $c_0t + c_1 = 1/(t^2 + c)$, $t \in \mathbb{F}$, of the (c_0, c_1)-plane; each has $q - 1$ points with $c_0 \neq 0$. The lines for $t = 0$ and $t = 1$ meet at $c_0 = 1/(1+c) - 1/c = -1/(c(1+c)) \neq 0$ ($c \neq -1$, as $-c$ is a non-square). So these lines cover at most $q(q-1) - 1$ of the $q(q-1)$ points with $c_0 \neq 0$. □

One valid choice for the q of the table ($\mathbb{F}_9 = \mathbb{F}_3[i]$, $i^2 = -1$):

q	3	5	7	9	11	13
c	1	2	1	$1+i$	1	2
(c_0, c_1)	(1,2)	(1,0)	(1,0)	$(i, 2i)$	(1,3)	(1,1)

Theorem 5.5 (exact whiskers)**.** *Let q be odd, $-c$ a non-square, and c_0, c_1 as in Lemma 5.4. Then*

$$W = \big\{\ell \in S_z \setminus \{v\} :\ z \in \mathbb{F}^2 \setminus \{0\},\ \nu(\ell \cap \Pi) = F(z)\big\}$$

is an exact whisker set: every free point lies on exactly one line of W, and $|W| = q^2(q^2-1)$.

Here $\ell \cap \Pi$ is one point not on v: $\ell \subseteq \Pi_z$ and $\ell \neq v$, so $\ell \notin S$ by Lemma 5.1(e), and ℓ is opposite v by Lemma 5.1(b). The proof uses a complex structure on each Π_z. Write a point of Π_z as $x = (x_1, \dots, x_6)$, meaning $(X_0, \dots, X_6) = (-cx_4 - \alpha x_2 - \beta x_5, x_1, \dots, x_6)$; on Π_z the quadratic form is $Q_z(x) = x_1x_5 + x_2x_6 - x_3^2 - cx_4^2 - \alpha x_2x_4 - \beta x_4x_5$, with polar form b_z. Put $\mathrm{num}(y) = y_2y_3 - y_4y_5$ and $\mathrm{den}(y) = cy_2^2 + y_5^2$, and define the linear map

$$J_zx = \big(-\beta x_3 + \alpha x_4 - x_6,\ x_5,\ -cx_4 - \beta x_5,\ x_3 + \beta x_2,\ -cx_2,\ cx_1 + \alpha\beta x_2 + \alpha x_3 + c\beta x_4 + \beta^2 x_5\big);$$

for $z = 0$, $J_0x = (-x_6, x_5, -cx_4, x_3, -cx_2, cx_1)$.

Lemma 5.6 (complex structure)**.** *For every z:*

(a) $J_z^2 = -c$, $Q_z(J_zx) = cQ_z(x)$ and $b_z(x, J_zx) = 0$.
(b) The six differences $p_{12} - p_{34}$, $p_{54} - p_{32}$, $p_{20} - p_{35}$, $p_{65} - p_{30}$, $p_{01} - p_{36}$, $p_{46} - p_{31}$ of the Grassmann coordinates of $x \wedge J_zx$ equal $Q_z(x)$ times $1, 0, 0, -c, \alpha, -\beta$.
(c) For every point x of $Q(6,q) \cap \Pi_z$, the line $\langle x, J_zx\rangle$ is the line of S_z through x.
(d) $\mathrm{num}(sx + tJ_0x) = (s^2 + ct^2)\,\mathrm{num}(x)$ and $\mathrm{den}(sx + tJ_0x) = (s^2 + ct^2)\,\mathrm{den}(x)$. So ν is constant on every line of $S \setminus \{v\}$.

Proof. (a), (b) and (d) are polynomial identities with integer coefficients, verified by direct expansion. For instance, for $z=0$: $Q_0(J_0x) = cx_2x_6 + cx_1x_5 - c^2x_4^2 - cx_3^2 = cQ_0(x)$; $p_{12} = x_1x_5 + x_2x_6$ and $p_{34} = x_3^2 + cx_4^2$, so $p_{12} - p_{34} = Q_0(x)$; in (d), $\mathrm{num}(J_0x) = c\,\mathrm{num}(x)$ and $\mathrm{den}(J_0x) = c\,\mathrm{den}(x)$, and the polar forms $x_2y_3 + x_3y_2 - x_4y_5 - x_5y_4$ and $2cx_2y_2 + 2x_5y_5$ vanish at $y = J_0x$. (c) J_z has no eigenvector ($-c$ is a non-square), so x and J_zx span a line; if $Q_z(x) = 0$, then by (a) the whole line is singular and by (b) its Grassmann coordinates satisfy (2.1), so it is a line of $H(q)$ in Π_z through x, the one of Lemma 5.1(b). (d) By (c), a line $n \in S \setminus \{v\}$ is $\{sx + tJ_0x\}$ for any of its points x, and $s^2 + ct^2 \neq 0$. □

Let p be a free point. By Lemma 5.1(d), $(p_2, p_5) \neq (0,0)$; as $p \notin \Pi$ we normalise $p_0 + cp_4 = 1$. Then $p \in \Pi_z$ if and only if z lies on the affine line

$$A_p = \{(\alpha,\beta) :\ \alpha p_2 + \beta p_5 = -1\} \subseteq \mathbb{F}^2 \setminus \{0\},$$

which has q points and direction $(p_5, -p_2)$. For $z \in A_p$ let $\ell_p(z)$ be the line of S_z on p and $X(z) = \ell_p(z) \cap \Pi$.

Lemma 5.7 (key identity)**.** *For $z = (\alpha,\beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{p_2p_3 - p_4p_5 + \beta p_2^2}{cp_2^2 + p_5^2}.$$

So Φ_p is affine on A_p, with increment $-p_2^3/(cp_2^2 + p_5^2)$ along $(p_5, -p_2)$.

Proof. Put $\Theta = cp_2^2 + p_5^2 \neq 0$ and $\gamma = -\alpha p_5 + c\beta p_2$. On Π_z the linear form $\Lambda(y) = y_0 + cy_4$ equals $-\alpha y_2 - \beta y_5$, and $\Pi = \{\Lambda = 0\}$. We have $\Lambda(p) = 1$ and $\Lambda(J_zp) = -\alpha p_5 + c\beta p_2 = \gamma$. By Lemma 5.6(c), $\ell_p(z) = \langle p, J_zp\rangle$, so $X(z) = \gamma p - J_zp$, with

$$X_2 = \gamma p_2 - p_5, \quad X_3 = \gamma p_3 + cp_4 + \beta p_5, \quad X_4 = \gamma p_4 - p_3 - \beta p_2, \quad X_5 = \gamma p_5 + cp_2.$$

Expanding, $\mathrm{den}(X) = (\gamma^2 + c)\Theta$ and $\mathrm{num}(X) = (\gamma^2 + c)(p_2p_3 - p_4p_5) + \beta(2\gamma p_2p_5 + cp_2^2 - p_5^2)$. Solving $\alpha p_2 + \beta p_5 = -1$, $-\alpha p_5 + c\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = -(cp_2 + \gamma p_5)/\Theta$ and $\beta = (\gamma p_2 - p_5)/\Theta$, hence $\alpha^2 + c\beta^2 = (\gamma^2 + c)/\Theta \neq 0$ and $\beta^3/(\alpha^2 + c\beta^2) = \beta(\gamma p_2 - p_5)^2/((\gamma^2 + c)\Theta)$. The two β-terms add up to $\beta\big(2\gamma p_2p_5 + cp_2^2 - p_5^2 + (\gamma p_2 - p_5)^2\big)/((\gamma^2 + c)\Theta) = \beta p_2^2/\Theta$. Along $(p_5, -p_2)$, β changes by $-p_2$. □

Proof of Theorem 5.5. Let p be free. The lines of $\bigcup_{z\neq 0} S_z \setminus \{v\}$ on p are exactly the q lines $\ell_p(z)$, $z \in A_p$, and they are distinct (a common line would lie in $S_z \cap S_{z'} = \{v\}$); for the same reason each line of W belongs to a single S_z. As $F(z) = c_0\alpha + c_1\beta - \beta^3/(\alpha^2 + c\beta^2)$, we have $\ell_p(z) \in W$ if and only if $\Phi_p(z) - (c_0\alpha + c_1\beta) = 0$. By Lemma 5.7 the left side is affine on A_p with increment

$$-\frac{p_2^3}{\Theta} - (c_0p_5 - c_1p_2) = -\frac{G(p_5, -p_2)}{\Theta} \neq 0$$

along $(p_5, -p_2)$, so it vanishes at exactly one of the q points of A_p: p is on exactly one line of W. The lines of W are opposite v and not in S. Counting incidences between free points and W gives $|W| = q^3(q^2 - 1)/q$. □

5.4. **The bound.**

Theorem 5.8. *For every odd prime power q and every $m \geq 2$, the structure of Theorem 5.3 with the exact whisker set of Theorem 5.5 is bipartite, has maximum degree $q+m$ and diameter 6, and*

$$n_b(q+m, 6) \geq 2(q+1)(q^4 + q^2 + 1) + (m-1)\big(2q^4 + 2q^3 + 2q + 2 + m(q^3 + q + 2) + m^2\big).$$

In particular $n_b(q+2, 6) \geq 2(q+1)(q^4 + q^2 + 1) + 2q^4 + 4q^3 + 4q + 10$ and $n_b(q+3, 6) \geq 2(q+1)(q^4 + q^2 + 1) + 4q^4 + 10q^3 + 10q + 34$.

Proof. Theorems 5.3 and 5.5, with $|W| = q^4 - q^2$. □

With W exact, every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$. For a doubling of a generalized n-gon of order (q,q) with N points, counting incidences gives $q|\mathcal{A}_0|+2k \le 2N$, k the number of components of $\mathcal{A}_0$; here $k=q^3+1$ and $q(2q^4+2q^3+2q+2)+2(q^3+1)=2N$, so the single-level part is as large as this counting bound allows for q^3+1 components.

Remark 5.9 (Delorme's construction; the 2013 values). $W=\emptyset$ is Delorme's doubling of a generalized hexagon [15, §8(b)]: he takes q^3+1 pairwise opposite elements (citing [37]; here the Hermitian spread S), puts one of them in $\mathcal{A}_2$, its neighbours and the others in $\mathcal{A}_1$, and their neighbours in $\mathcal{A}_0$, so $|\mathcal{A}_1|=q^3+q+2$ and $|\mathcal{A}_0|=q^4+2q^3+q^2+2q+2$, exactly as in Theorem 5.3 with $W=\emptyset$; at $q=5$ he states $7812+912+2\cdot 132+4=8992$. At $m=2$ this has $2(q+1)(q^4+q^2+1)+q^4+4q^3+q^2+4q+10$ elements: 8992, 142464, 374452, 842048 at $q=5,9,11,13$; at $q=13$, $m=3$, it has 884062. These are the 2013 values at $\Delta=7,11,13,15,16$. The exact whiskers add $(m-1)q^2(q^2-1)$.

Worked counts. $q=5$, $m=2$: $H(5)$ has $2\cdot 6\cdot 651=7812$ elements, $|\mathcal{A}_0|=1512$ (of which $|W|=600$), $|\mathcal{A}_1|=132$, $|\mathcal{A}_2|=1$, so the order is $7812+1512+2\cdot 132+4=9592$. $q=13$, $m=3$: $H(13)$ has 804468 elements, $|\mathcal{A}_0|=61544$ ($|W|=28392$), $|\mathcal{A}_1|=2212$, so the order is $804468+2(61544+3\cdot 2212+9)=940846$.
Cells. $m=2$: $(7,6)$ 9592 (2013: 8992), $(11,6)$ 148944 (142464), $(13,6)$ 388972 (374452), $(15,6)$ 870440 (842048). $m=3$, $q=13$: $(16,6)$ 940846 (884062). These graphs are not regular; see Section 5.6.

5.5. **Even q.** Let $q=2^e$, Tr the absolute trace of $\mathbb{F}$, and fix $\delta\in\mathbb{F}$ with $\mathrm{Tr}(\delta)=1$, so that $t^2+t+\delta$ is irreducible and $a^2+ab+\delta b^2$ is anisotropic. Now $Q(6,q)$ is $X_0X_4+X_1X_5+X_2X_6+X_3^2=0$, with nucleus e_3, and $H(q)$ is given by the same equations (2.1). Put

$$\Pi_z:\ X_3=X_0+\delta X_4+\alpha X_2+\beta X_5 \qquad (z=(\alpha,\beta)\in\mathbb{F}^2), \qquad \Pi=\Pi_{(0,0)},$$

S_z the lines of $H(q)$ in Π_z, and $v=\langle e_1,e_6\rangle\subseteq\Pi_z$ as before.

Lemma 5.10. *Lemma 5.1 holds verbatim for even q with these Π_z.*

Proof. (a) On Π_z, $X_0X_4+X_1X_5+X_2X_6+X_3^2=(X_0^2+X_0X_4+\delta^2X_4^2)+(X_1+\beta^2X_5)X_5+X_2(X_6+\alpha^2X_2)$. The first term is anisotropic ($\mathrm{Tr}(\delta^2)=\mathrm{Tr}(\delta)=1$) and the polar form is non-degenerate on Π_z (which misses e_3), so $\Pi_z\cap Q(6,q)$ is an elliptic quadric $Q^-(5,q)$. (b), (c), (e) are proved as in Lemma 5.1. (d) The same three equations give $u_2=u_5=0$; on $X_2=X_5=0$ the quadric is the cone $X_0X_4=X_3^2$, and in Π_z this becomes $X_0^2+X_0X_4+\delta^2X_4^2=0$, so $X_0=X_4=X_3=0$, a point of v. □

So Theorem 5.3 holds for even q without change. For $X\in\Pi$ not on v put

$$\nu(X)=\frac{X_2X_3+X_2X_4+X_4X_5}{\delta X_2^2+X_2X_5+X_5^2},$$

$$G(a,b)=(c_0a+c_1b)(a^2+ab+\delta b^2)+b^3, \qquad F(z)=\frac{G(\alpha,\beta)}{\alpha^2+\alpha\beta+\delta\beta^2}.$$

The denominator of ν is anisotropic and vanishes only on v.

Lemma 5.11. *There are $c_0,c_1\in\mathbb{F}$ with G zero-free on $\mathbb{F}^2\setminus\{0\}$ if and only if $q\ge 4$.*

Proof. $G(a,0)=c_0a^3$ needs $c_0\neq 0$. For $b\neq 0$ and $t=a/b$: $G=0$ if and only if $c_0t+c_1=r(t):=1/(t^2+t+\delta)$, a line in the (c_0,c_1)-plane with $q-1$ points having $c_0\neq 0$. If $q\ge 4$ take $\varpi\notin\{0,1\}$; then $r(\varpi)\neq r(0)$ (as $\varpi^2+\varpi\neq 0$), so the lines for $t=0,\varpi$ meet at $c_0=(r(0)+r(\varpi))/\varpi\neq 0$, and the q lines cover fewer than $q(q-1)$ points with $c_0\neq 0$. If $q=2$ then $\delta=1$, $r(0)=r(1)=1$, and the lines $c_1=1$ and $c_0+c_1=1$ contain both points with $c_0=1$. □

On Π_z use coordinates $x = (x_0, x_1, x_2, x_4, x_5, x_6)$, with $X_3 = x_0 + \delta x_4 + \alpha x_2 + \beta x_5$, and

$$J_z x = \big((\delta+1)x_0 + \delta\alpha x_2 + \delta^2 x_4 + (\alpha+\delta\beta)x_5,\ \beta x_0 + x_1 + \alpha\beta x_2 + (\alpha+\delta\beta)x_4 + \beta^2 x_5 + x_6,\ x_2 + x_5,\\ x_0 + (\alpha+\beta)x_2 + \delta x_4 + \beta x_5,\ \delta x_2,\ (\alpha+\beta)x_0 + \delta x_1 + \alpha^2 x_2 + \delta\alpha x_4 + \alpha\beta x_5\big).$$

Lemma 5.12. *With Q the quadratic form and b its polar form: (a) $J_z^2 + J_z + \delta = 0$, $Q(J_z x) = \delta Q(x)$ and $b(x, J_z x) = Q(x)$; (b) the six differences of (2.1) for $x \wedge J_z x$ are $Q(x)$ times $1, 0, 0, \delta, \alpha, \beta$; (c) for every point x of $Q(6,q) \cap \Pi_z$, $\langle x, J_z x\rangle$ is the line of S_z on x; (d) with* $\mathrm{num} = X_2X_3 + X_2X_4 + X_4X_5$ *and* $\mathrm{den} = \delta X_2^2 + X_2X_5 + X_5^2$ *on Π, $\mathrm{num}(sx + tJ_0x) = (s^2 + st + \delta t^2)\mathrm{num}(x)$, and likewise for* den*; so ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b), (d) are polynomial identities over $\mathbb{F}_2[\delta, \alpha, \beta, x]$, verified by expansion. (c) J_z has no eigenvector since $t^2+t+\delta$ is irreducible; if $Q(x) = 0$ then $Q(sx+tJ_zx) = (s^2+st+\delta t^2)Q(x) = 0$ by (a), and by (b) the line satisfies (2.1), so it is the line of Lemma 5.10(b). In (d), $s^2+st+\delta t^2 \neq 0$ for $(s,t) \neq 0$. □

Let p be free; then $(p_2, p_5) \neq (0,0)$ by Lemma 5.10(d), and we normalise $p_0 + p_3 + \delta p_4 = 1$. With $\Lambda(y) = y_0 + y_3 + \delta y_4$ we have $\Pi = \{\Lambda = 0\}$ and $\Lambda = \alpha y_2 + \beta y_5$ on Π_z, so $p \in \Pi_z$ if and only if z lies on the affine line $A_p:\ \alpha p_2 + \beta p_5 = 1$ (direction (p_5, p_2)). For $z \in A_p$ let $\ell_p(z)$ be the line of S_z on p and $X(z) = \ell_p(z) \cap \Pi$.

Lemma 5.13 (key identity, characteristic 2)**.** *For $z = (\alpha, \beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + \alpha\beta + \delta\beta^2} = \frac{p_2p_3 + p_2p_4 + p_4p_5 + \beta p_2^2}{\delta p_2^2 + p_2p_5 + p_5^2},$$

which is affine on A_p with increment p_2^3/Θ along (p_5, p_2), $\Theta = \delta p_2^2 + p_2p_5 + p_5^2 \neq 0$.

Proof. $\Lambda(p) = 1$ and $\gamma := \Lambda(J_zp) = \alpha(p_2+p_5) + \delta\beta p_2$. By Lemma 5.12(c), $X(z) = \gamma p + J_z p$ (then $\Lambda(X) = 2\gamma = 0$); $X_2 = (\gamma+1)p_2 + p_5$ and $X_5 = \delta p_2 + \gamma p_5$. Expanding, $\mathrm{den}(X) = (\gamma^2+\gamma+\delta)\Theta$ and $\mathrm{num}(X) = (\gamma^2+\gamma+\delta)(p_2p_3+p_2p_4+p_4p_5) + \beta\big((\gamma+\delta+1)p_2^2 + p_5^2\big)$, with $\gamma^2+\gamma+\delta \neq 0$. Solving $\alpha p_2 + \beta p_5 = 1$, $\alpha(p_2+p_5) + \delta\beta p_2 = \gamma$ (determinant Θ): $\alpha = (\delta p_2 + \gamma p_5)/\Theta$ and $\beta = ((\gamma+1)p_2 + p_5)/\Theta$, so $\alpha^2 + \alpha\beta + \delta\beta^2 = (\gamma^2+\gamma+\delta)/\Theta$ and $\beta^3/(\alpha^2+\alpha\beta+\delta\beta^2) = \beta((\gamma+1)p_2+p_5)^2/((\gamma^2+\gamma+\delta)\Theta)$. The β-terms add to $\beta\big((\gamma+\delta+1) + (\gamma+1)^2\big)p_2^2/((\gamma^2+\gamma+\delta)\Theta) = \beta p_2^2/\Theta$. Along (p_5, p_2), β changes by p_2. □

Theorem 5.14 (exact whiskers, even $q \geq 4$)**.** *With c_0, c_1 as in Lemma 5.11, $W = \{\ell \in S_z \setminus \{v\} : z \neq 0,\ \nu(\ell \cap \Pi) = F(z)\}$ is an exact whisker set.*

Proof. As for Theorem 5.5: the lines of $\bigcup_{z\neq 0} S_z \setminus \{v\}$ on a free point p are the q distinct lines $\ell_p(z)$, $z \in A_p$, and $\ell_p(z) \in W$ if and only if $\Phi_p(z) + c_0\alpha + c_1\beta = 0$, since $F(z) = c_0\alpha + c_1\beta + \beta^3/(\alpha^2 + \alpha\beta + \delta\beta^2)$. The left side is affine on A_p with increment $p_2^3/\Theta + c_0p_5 + c_1p_2 = G(p_5, p_2)/\Theta \neq 0$, so it vanishes at exactly one point. □

Corollary 5.15. *Theorem 5.8 and Theorem 5.18 hold for every even $q \geq 4$. For $q = 2$ there is also an exact whisker set (of $12 = q^2(q^2-1)$ lines, given explicitly and computer-checked), so they hold for $q = 2$ as well.*

None of this changes a cell with $\Delta \leq 16$, for any m: for $q \in \{2, 4, 8\}$ and $\Delta = q + m \leq 16$, either $\Delta - 1$ is a prime power (a cell where the Moore bound is attained by $H(\Delta-1)$), or $\Delta \in \{7, 11, 13, 15, 16\}$, where Theorem 5.8 with the odd $q' = \Delta - 2$ (or $q' = 13$, $m = 3$ at $\Delta = 16$) gives much more (for instance 9592 against 4468 from $q = 4$, $m = 3$ at $\Delta = 7$).

Corollary 5.15, $q = 2$: the 12 whiskers are a computer-found set (its exactness is also proved in Lean). List them in an appendix, or drop the $q = 2$ case?

5.6. **Regular graphs of diameter 6.** The structure of Theorem 5.8 has equally many points and lines in $\mathcal{A}_0$ (exact whiskers force this), but $\mathcal{A}_1$ has q^3+1 lines and only $q+1$ points, and $\mathcal{A}_2$ is a line. So it has $(m-1)m(q^3-q+m)$ more lines than points. We delete repeated lines and then add anti-flags.

Lemma 5.16 (repeated lines)**.** *In $H(q)^{(m)}_{\mathcal{A}}$, for $n \in S \setminus \{v\}$ and $i \in [m]$, the m lines $n^{i1}, \dots, n^{im}$ all have the point set $\{p^i : p \in n\}$; for $i, j \in [m]$ the lines $v^{ij1}, \dots, v^{ijm}$ all have the point set $\{p^{ij} : p \in v\}$ (Table 4). So there are $mq^3 + m^2$ classes of m repeated lines.*

Lemma 5.17 (completion)**.** *Let a finite incidence structure have maximum degree at most Δ, every element of degree at least $\Delta - r$, and $\sum_{points}(\Delta - \deg) = \sum_{lines}(\Delta - \deg) =: T$. If $T > 2r\Delta$, there is a set J of anti-flags such that adding J makes it Δ-regular.*

Proof. Choose a multiset J of point–line pairs in which each element x occurs $\Delta - \deg x$ times (possible as the sums agree). Call an entry bad if it is a flag or a repeat of another entry, and take J with the fewest bad entries. If (p, L) is bad: L is joined (in I $\cup J$, with multiplicity) to Δ points, each in at most r entries of J; likewise p. So at most $2r\Delta < T$ entries (p', L') have p' joined to L or L' joined to p, and some entry (p', L') has neither (it is not (p, L) itself). Replacing $(p, L), (p', L')$ by $(p, L'), (p', L)$ removes a bad entry and creates none. So the minimum is 0. □

Theorem 5.18. *For every odd prime power q and $m \geq 2$ there is a $(q+m)$-regular incidence structure of diameter at most 6 with*

$$2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$$

elements: $2(q+1)(q^4+q^3+q^2+4)$ for $m=2$ and $2(q+1)(q^4+2q^3+q^2+9)$ for $m=3$. For the five table instances below the order exceeds $M_b(q+m, 5)$, so the diameter is exactly 6.

Proof. Start from Theorem 5.8. Fix a set R of q lines of $S \setminus \{v\}$, and delete the lines v^{ijk} with $k \geq 2$ ($(m-1)m^2$ lines) and n^{ij} with $j \geq 2$ for $n \notin R$ ($(m-1)m(q^3-q)$ lines): in total $(m-1)m(q^3-q+m)$, the excess, so points and lines are now equally many. Each class of Lemma 5.16 keeps at least one line, so by Lemma 2.2(b) no distance between remaining elements increases. After the deletion the degrees are: p^{ij} ($p \in v$): $q+1$; p^i on a line $n \notin R$ of S: $q+1$; p^i on $n \in R$: $q+m$; v^{ij1}, n^{i1} ($n \notin R$), n^{ij} ($n \in R$) and h^i ($h \in W$): $q+1$; all others as in Table 4 ($q+m$). So every deficit is at most $r = m-1$. Points and lines are equally many, so both deficit sums equal $\Delta|\mathcal{P}| - |E| = \Delta|\mathcal{L}| - |E|$ ($\Delta = q+m$, E the set of flags), as Lemma 5.17 requires; and their common value T is at least the deficit of the $m|W|$ whisker versions: $T \geq (m-1)mq^2(q^2-1) > 2(m-1)(q+m)$. Lemma 5.17 completes the structure; the added flags join a point to a line, so the graph stays bipartite, and they do not increase distances. The number of points is $(q+1)(q^4+q^2+1) + (m-1)\big((q+1)(q^3+1)+m(q+1)\big)$. □

Neighbourhoods. Every neighbourhood is that of Table 4, minus the deleted lines listed in the proof, plus the completing anti-flags: exactly $m-1$ of them at each element of degree $q+1$, none elsewhere. The completing anti-flags are given by Lemma 5.17, not by a formula; neither the maximum degree nor the diameter depends on the choice.

Theorem 5.18: the completing anti-flags are only shown to exist (as in the Lean proof). If you want every neighbourhood written out, we need an explicit completion (for instance a formula on the deficient elements); this is open work.

Cells (regular graphs). $(7,6)$ 9348, $(11,6)$ 147500, $(13,6)$ 386328, $(15,6)$ 866068, $(16,6)$ 927724 ($q=13$, $m=3$). Each beats the 2013 value of its cell (8992, 142464, 374452, 842048, 884062), and Delorme's graphs behind those values are not regular (their point side is larger).

6. Diameter 7: level graphs of the split Cayley hexagon, surgery and hubs

In the 2013 table each of the cells $(7,7)$ and $(9,7)$–$(16,7)$ has the value $(2r+1)\cdot 2N(q)$, with $\Delta = q+1+r$, for the (q,r) of Table 5; this is the order of $2r+1$ copies of the incidence graph $I(q)$ of $H(q)$ joined by matchings (Theorem 6.6). The CombinatoricsWiki credits these cells to Delorme, Gómez and Quisquater [17]. We delete some matching edges and attach new vertices ("hubs") to the freed slots. The general theorems (Sections 6.2–6.7) hold for every generalized hexagon of order (q,q), in particular for $H(q)$ and its dual; only Lemma 2.4(c) and Lemma 6.2 are specific to $H(q)$. In this section $\mathcal{P}$ and $\mathcal{L}$ are the point set and the line set of $H(q)$, and $N = |\mathcal{P}| = |\mathcal{L}| = (q+1)(q^4+q^2+1)$.

Table 5. Diameter 7. The 2013 value is $(2r+1)\cdot 2N(q)$.

cell	q, r	2013	this paper	construction
(7,7)	5, 1	23 436	23 598	3 levels, local duality (Cor. 6.24)
(9,7)	7, 1	117 648	118 238	same
(10,7)	8, 1	224 694	225 152	same
(11,7)	9, 1	398 580	413 180	3 levels, duality, spread points freed (Cor. 6.25)
(12,7)	9, 2	664 300	693 500	5 levels, duality, spread points freed (Cor. 6.25)
(13,7)	11, 1	1 062 936	1 064 472	3 levels, local duality (Cor. 6.24)
(14,7)	11, 2	1 771 560	1 789 588	typed levels $HDHDH$ (Cor. 6.26)
(15,7)	11, 3	2 480 184	2 501 491	typed levels $HHDHDHD$ (Cor. 6.26)
(16,7)	13, 2	4 022 340	4 052 108	typed levels $HDHDH$ (Cor. 6.26)

6.1. **Sets of points and lines.**

Definition 6.1. A set of points of $H(q)$ is *non-collinear* if no two of its points are collinear (a partial distance-2 ovoid), and a *partial ovoid* if its points are pairwise opposite; an *ovoid* is a partial ovoid with q^3+1 points. Dually, a set of lines is *non-concurrent*, a *partial spread* (pairwise opposite lines) or a *spread* (q^3+1 pairwise opposite lines).

We follow [40]: an ovoid of a generalized hexagon is a distance-3 ovoid. By Lemma 2.4(c), partial ovoids of $H(q)$ are exactly the sets of pairwise non-perpendicular points of $Q(6,q)$, i.e. partial ovoids of the polar space $Q(6,q)$.

Lemma 6.2. *(a) If Σ is a 3-space of $\mathrm{PG}(6,q)$ with $\Sigma \cap Q(6,q)$ an elliptic quadric $Q^-(3,q)$, its q^2+1 points form a partial ovoid of $H(q)$, for every q.*

(b) (Hermitian spread [37].) If Π is a hyperplane with $\Pi \cap Q(6,q)$ an elliptic quadric $Q^-(5,q)$, the lines of $H(q)$ contained in Π form a spread, for every q.

(c) If S is a spread, every line not in S meets exactly one line of S.

(d) If $q = 3^h$ and φ is a duality, $\varphi(S)$ is an ovoid for every spread S.

Proof. (a) If $x \neq y$ are points of $Q^-(3,q)$ with $b(x,y) = 0$, then $Q(\lambda x + \mu y) = \lambda\mu b(x,y) = 0$, so the line xy lies in $Q^-(3,q)$, which contains no line. So $b(x,y) \neq 0$ and x, y are opposite by Lemma 2.4(c). (b) The proof of Lemma 5.1(b) applies to every hyperplane with an elliptic section (it uses only that $\Pi \cap Q(6,q)$ contains no plane); examples of such hyperplanes for every q are in Lemmas 5.1 and 5.10. (c) A line meeting two lines of S would put them at distance at most 4. So the lines of S together with the $q(q+1)$ lines meeting each of them are $(q^3+1)(q^2+q+1) = N$ distinct lines, that is, all lines. (d) φ preserves distances. □

Ovoids of $H(q)$ (equivalently of $Q(6,q)$) exist for $q = 3^h$ (Lemma 6.2(d), [38]); they do not exist for q even [38] or for q prime, $q \geq 5$ [4].

Definition 6.3 (segment sets)**.** A set F of points is a *segment set* if for every $p \in F$ the points of $F \setminus \{p\}$ collinear with p all lie on one line through p. Equivalently, there are no $p, p', p'' \in F$ with $d(p,p') = d(p',p'') = 2$ and $d(p,p'') = 4$ (a path of two collinearities that turns at p'). Dually, a set G of lines is a *dual segment set* if for every $\ell \in G$ the lines of $G \setminus \{\ell\}$ meeting ℓ all pass through one point of ℓ.

So a segment set is a disjoint union of "segments" (at least two points on a line) and isolated points, such that no two points from different pieces are collinear. Examples: every non-collinear set; every set in which each point is collinear with at most one other point of the set; every subset of the point set of a partial spread; dually, the set of all lines through the points of a partial ovoid.

Proposition 6.4 (size of segment sets)**.** *A segment set F has $|F| \le (q+1)(q^3+1)$, with equality if and only if F is the point set of a spread. Dually, a dual segment set has at most $(q+1)(q^3+1)$ lines, with equality if and only if it is the set of lines through the points of an ovoid.*

Proof. Let F_1 be the points of F collinear with no other point of F, and $C_1, \dots, C_k$ the segments (maximal sets of at least two points of F on a line ℓ_i). A line meeting F in exactly one point is a *private* line of that point. A point of F_1 has $q+1$ private lines and a point of a segment C_i has q (all its lines except ℓ_i, by the segment property). Private lines of different points are different, and different from the ℓ_i. So $(q+1)|F_1| + q\sum_i |C_i| + k \le N$. Since $k \ge \sum_i |C_i|/(q+1)$, the left side is at least $(q + \frac{1}{q+1})|F|$, which gives $|F| \le N(q+1)/(q^2+q+1) = (q+1)(q^3+1)$. Equality forces $F_1 = \emptyset$, $|C_i| = q+1$ and $k = q^3+1$. The lines ℓ_i are then pairwise opposite: if ℓ_i and ℓ_j met in a point z, then $z \in F$ would have other points of F on two lines; if $d(\ell_i, \ell_j) = 4$, a point of ℓ_i would be collinear with a point of ℓ_j and would also have other points of F on ℓ_i. So they form a spread. Conversely the point set of a spread is a segment set of this size. The dual statement is the same proof in the dual hexagon. □

For comparison, a non-collinear set has at most $N/(q+1) = q^4+q^2+1$ points (equality means a distance-2 ovoid; these are known only for $q = 2, 3, 4$ [18], [8]).

6.2. **Level graphs.** Fix $r \ge 1$; there are $2r+1$ *levels*, indexed by $j \in \mathbb{Z}_{2r+1}$. Each level gets a *type* $\tau_j \in \{H, D\}$ (the hexagon or its dual). For a type τ let

$$\mathsf{A}(H) = \mathcal{P}, \quad \mathsf{B}(H) = \mathcal{L}, \qquad \mathsf{A}(D) = \mathcal{L}, \quad \mathsf{B}(D) = \mathcal{P}.$$

A *transition* is a pair $t = (j,k)$ with $j \in \mathbb{Z}_{2r+1}$ and $1 \le k \le r$; its source level is j and its target level $j+k$. It is *friendly* if $\tau_j \ne \tau_{j+k}$; then $\mathsf{A}(\tau_j) = \mathsf{B}(\tau_{j+k})$.

Definition 6.5 (level graph)**.** Given types τ and, for every transition $t = (j,k)$, a bijection $M_t : \mathsf{A}(\tau_j) \to \mathsf{B}(\tau_{j+k})$, the graph $X = X(\tau, M)$ has vertex set $(\mathcal{P} \cup \mathcal{L}) \times \mathbb{Z}_{2r+1}$ and edges

- $(x,j) \sim (y,j)$ whenever x and y are incident (*level edges*);
- $(a,j) \sim (M_{(j,k)}(a), j+k)$ for every j, every $1 \le k \le r$ and every $a \in \mathsf{A}(\tau_j)$ (*matching edges*).

Colour (a,j), $a \in \mathsf{A}(\tau_j)$, with colour A and (b,j), $b \in \mathsf{B}(\tau_j)$, with colour B; every edge joins different colours, so X is bipartite. The neighbourhoods are

$$N_X(a,j) = \Gamma_1(a) \times \{j\} \ \cup \ \{(M_{(j,k)}(a), j+k) : 1 \le k \le r\} \qquad (a \in \mathsf{A}(\tau_j)),$$
$$N_X(b,j) = \Gamma_1(b) \times \{j\} \ \cup \ \{(M^{-1}_{(j-k,k)}(b), j-k) : 1 \le k \le r\} \qquad (b \in \mathsf{B}(\tau_j)),$$

so X is $(q+1+r)$-regular of order $(2r+1) \cdot 2N$. When all levels have type H and all M_t equal one bijection $\mathcal{P} \to \mathcal{L}$, X has the order of the 2013 table value.

Theorem 6.6 (level graphs)**.** *For all types and all bijections M_t, $X(\tau, M)$ has diameter at most 7. More precisely, two vertices of one level are joined by a path of length at most 6 inside that level, and for every transition $t = (j, k)$ every vertex of level j is joined to every vertex of level $j + k$ by a path of length at most 7 that uses only level edges of levels j and $j + k$ and matching edges of t.*

Proof. Inside a level this is the diameter of $I(q)$. Any two distinct levels differ by $\pm k$ for some $1 \le k \le r$, so it is enough to join (u, j) to (v, j'), $j' = j + k$, with $M = M_{(j,k)}$. (1) $u \in \mathsf{A}(\tau_j)$: $(u, j) \to (M(u), j') \rightsquigarrow (v, j')$, of length at most $1+6$. (2) $u \in \mathsf{B}(\tau_j)$, $v \in \mathsf{B}(\tau_{j'})$: with $a = M^{-1}(v)$, $(u, j) \rightsquigarrow (a, j) \to (v, j')$, of length at most $5 + 1$ (elements of different types are at distance at most 5). (3) $u \in \mathsf{B}(\tau_j)$, $v \in \mathsf{A}(\tau_{j'})$: pick $b \in \Gamma_1(v)$ and $a = M^{-1}(b)$; then $(u, j) \rightsquigarrow (a, j) \to (b, j') \to (v, j')$, of length at most $5 + 1 + 1$. □

This is the routine argument for such graphs, and we do not claim it (Section 6.10).

6.3. **Surgery and hubs.**

Definition 6.7 (surgery and hubs)**.** Start from a level graph $X(\tau, M)$ and choose for every transition $t = (j, k)$ a set $R_t \subseteq \mathsf{A}(\tau_j)$ of *freed elements* (possibly empty).

(1) *Deletion.* Delete the matching edges $(a, j) \sim (M_t(a), j + k)$, $a \in R_t$. Call the result X_R. Each deleted edge leaves a *free* A*-slot* (a, t) at the vertex (a, j) and a *free* B*-slot* $(M_t(a), t)$ at the vertex $(M_t(a), j + k)$. Level j has

$$\alpha_j = \sum_{k=1}^{r} |R_{(j,k)}| \text{ free } \mathsf{A}\text{-slots}, \qquad \beta_j = \sum_{k=1}^{r} |R_{(j-k,k)}| \text{ free } \mathsf{B}\text{-slots}.$$

(2) *Hubs.* Add $K_{\mathsf{B}} \le \min_j \alpha_j$ new vertices of colour B (B*-hubs*) and $K_{\mathsf{A}} \le \min_j \beta_j$ new vertices of colour A (A*-hubs*). Each B-hub h takes exactly one free A-slot in every level j and is joined to its vertex, written $(\omega_j(h), j)$; each A-hub g likewise takes one free B-slot in every level, at the vertex $(\omega'_j(g), j)$. Every slot is taken at most once. (The assignment is otherwise arbitrary; a vertex with several free slots may receive several hubs.)

(3) *Hub graph.* Add a bipartite graph Y between the B-hubs and the A-hubs.

Call the result X_R^+.

Exact neighbourhoods. For a level vertex, the neighbourhood is that of Definition 6.5 minus the deleted matching edges plus one hub for each taken free slot. For hubs,

$$N_{X_R^+}(h) = \{(\omega_j(h), j) : j \in \mathbb{Z}_{2r+1}\} \cup N_Y(h), \qquad N_{X_R^+}(g) = \{(\omega'_j(g), j) : j \in \mathbb{Z}_{2r+1}\} \cup N_Y(g).$$

Level vertices have degree at most $q+1+r$ (with equality when all their free slots are used), and hubs have degree $2r+1+\deg_Y$. So X_R^+ is bipartite of maximum degree at most $q+1+r$ as soon as Y has maximum degree at most $q - r$, and $|X_R^+| = (2r + 1) \cdot 2N + K_{\mathsf{A}} + K_{\mathsf{B}}$.

Definition 6.8. A transition $t = (j, k)$ is *repaired* (by R_t) if in X_R every vertex of level j is joined to every vertex of level $j + k$ by a path of length at most 7 that uses only level edges of levels j, $j + k$ and the remaining matching edges of t.

Whether t is repaired depends only on M_t and R_t.

Lemma 6.9 (hubs)**.** *In X_R^+ every hub is within distance 7 of every level vertex, and every* B*-hub is within distance 7 of every* A*-hub.*

Proof. A B-hub h and a level vertex (v, j): $h \to (\omega_j(h), j) \rightsquigarrow (v, j)$ inside level j, of length at most $1 + 6$ (level edges are never deleted). A B-hub h and an A-hub g: in level 0, $\omega_0(h) \in \mathsf{A}(\tau_0)$ and $\omega'_0(g) \in \mathsf{B}(\tau_0)$ have different types, so $h \to (\omega_0(h), 0) \rightsquigarrow (\omega'_0(g), 0) \to g$ has length at most $1 + 5 + 1$. □

Theorem 6.10 (assembly)**.** *Suppose that*

(T) every transition t is repaired by R_t, and

(HG) any two B-hubs, and any two A-hubs, are within distance 6 in X_R^+.

Then X_R^+ has diameter at most 7. If moreover Y has maximum degree at most $q-r$, then

$$n_b(q+1+r,7) \geq (2r+1)\cdot 2N(q) + K_{\mathsf{A}} + K_{\mathsf{B}}.$$

Proof. Pairs of level vertices: same level by the diameter of $I(q)$, different levels by (T), since every pair of levels is the pair of some transition. Pairs with a hub: Lemma 6.9 and (HG). □

(HG) holds for instance when Y alone has all same-colour distances at most 6; it can also use paths through the levels.

6.4. **Transitions that extend to an automorphism.** A transition $t = (j,k)$ is *automorphic* if M_t is the restriction to $\mathsf{A}(\tau_j)$ of an automorphism ψ of $I(q)$ with $\psi(\mathsf{A}(\tau_j)) = \mathsf{B}(\tau_{j+k})$. The two cases used here are: a friendly transition with $M_t = \mathrm{id}$; and an $H \to H$ transition with $M_t = \varphi|_{\mathcal{P}}$ for a duality φ ($q = 3^h$), or a $D \to D$ transition with $M_t = \varphi|_{\mathcal{L}}$.

Lemma 6.11. *Let F be a segment set of points. For all elements u, v of $H(q)$ there is a point $p \notin F$ with $d(u,p) + d(p,v) \leq 6$.*

Proof. If some point p on a geodesic from u to v is not in F, then $d(u,p) + d(p,v) = d(u,v) \leq 6$. Assume that all points on all geodesics from u to v lie in F. A geodesic with three points p, p', p'' in a row ($p - m - p' - m' - p''$ with lines $m \neq m'$) cannot have all three in F (a turning path). A geodesic between two points at distance 4 or 6, a point and a line at distance 5, or two lines at distance 6 contains three points in a row. So only the following cases remain; in each we exhibit $p \notin F$ with $d(u,p) + d(p,v) \leq 6$. We use repeatedly: *if $p_0 \in F$ has a point of F on the line s through it, then no point $p \neq p_0$ collinear with p_0 off s is in F.*

(a) $u = v$ a point of F. Let s be a line through u containing every point of F collinear with u (any line if there is none). Take a line $y \neq s$ through u and $p \in y \setminus \{u\}$: $p \notin F$ and $d(u,p) + d(p,u) = 4$.

(b) u, v collinear points of F on the line s. Take $y \neq s$ through u and $p \in y \setminus \{u\}$: $p \notin F$, $d(u,p) = 2$, $d(p,v) \leq 4$.

(c) $u \in F$ a point and v a line through u. If v has a point $p \neq u$ outside F: $2+1$. Otherwise all points of v are in F; take $y \neq v$ through u and $p \in y \setminus \{u\}$: $p \notin F$, $d(u,p) = 2$, $d(p,v) \leq 3$.

(d) u a point and v a line with $d(u,v) = 3$, geodesic $u - s - p_2 - v$ with $u, p_2 \in F$. A point $p \in v \setminus \{p_2\}$ has $d(u,p) = 4$ and $d(p,v) = 1$. If all of them were in F, p_2 would have points of F on the two lines s and v.

(e) u, v lines meeting in $p_1 \in F$. Not both u and v contain a second point of F (else p_1 turns). Say u does not: take $p \in u \setminus \{p_1\}$; then $p \notin F$, $d(u,p) = 1$, $d(p,v) = 3$.

(f) u, v lines at distance 4, geodesic $u - p_1 - y - p_3 - v$ with $p_1, p_3 \in F$. No other point p of u is in F (else p_1 turns: p on u, p_3 on y), and $d(u,p) = 1$, $d(p,v) \leq d(p,p_1) + d(p_1,v) = 2+3$.

(g) $u = v$ a line. If some point of u is not in F: $1+1$. Otherwise take $p_1 \in u$, a line $y \neq u$ through p_1 and $p \in y \setminus \{p_1\}$: $p \notin F$ and $d(u,p) = 3$, so $3+3=6$.

The cases with u a line and v a point are (c), (d) with u and v exchanged. □

Theorem 6.12 (automorphic transitions)**.** *Let $t = (j,k)$ be an automorphic transition. If $\mathsf{A}(\tau_j) = \mathcal{P}$ and R_t is a segment set of points, or $\mathsf{A}(\tau_j) = \mathcal{L}$ and R_t is a dual segment set of lines, then t is repaired.*

Proof. Let ψ extend $M = M_t$, $j' = j + k$, and take (u,j), (v,j'). For every $a \in \mathsf{A}(\tau_j) \setminus R_t$ the path $(u,j) \rightsquigarrow (a,j) \to (\psi(a), j') \rightsquigarrow (v,j')$ has length $d(u,a)+1+d(\psi(a),v) = d(u,a)+1+d(a,\psi^{-1}(v))$,

since ψ is an isometry of $I(q)$. If $\mathsf{A}(\tau_j) = \mathcal{P}$, Lemma 6.11 applied to u, $\psi^{-1}(v)$ and $F = R_t$ gives a with $d(u,a) + d(a,\psi^{-1}(v)) \le 6$. If $\mathsf{A}(\tau_j) = \mathcal{L}$, apply Lemma 6.11 in the dual hexagon. □

Remark 6.13 (the converse). An automorphic transition is repaired if and only if its freed set is a segment set (a dual segment set for lines). Suppose that $p, p', p'' \in R_t$ turn at p' ($p-m-p'-m'-p''$, $m \ne m'$) and take a line $u \ne m'$ through p''. Then $d(p,u) = 5$, with the unique geodesic $p-m-p'-m'-p''-u$. A path from (u,j) to $(\psi(p), j')$ inside the two levels uses an odd number $2h+1$ of matching edges of t, and (transporting the segments in level j' back by ψ^{-1}) has length at least $2h+1+d(u,p) = 2h+6$. So a path of length at most 7 uses one matching edge, at a point a with $d(u,a) + d(a,p) = 5$, that is $a \in \{p, p', p''\}$; all three edges are deleted, and the distance within the two levels is more than 7.

Corollary 6.14. *The following transitions are repaired:*

(1) a friendly transition with $M_t =$ id *and R_t a non-collinear set of points (if $\tau_j = H$) or a non-concurrent set of lines (if $\tau_j = D$);*
(2) the same with R_t a set in which every element is collinear (concurrent) with at most one other element of R_t;
(3) an $H \to H$ transition with M_t a duality and R_t any subset of the point set of a partial spread (for example all $(q+1)(q^3+1)$ points of the Hermitian spread, $q = 3^h$).

6.5. **Local dualities.** For transitions that are not automorphic (an $H \to H$ or $D \to D$ transition when $q \ne 3^h$, or one whose matching is not the restriction of a duality) we ask M_t to behave like a duality only near the freed elements.

Theorem 6.15 (local dualities)**.** *Let $t = (j,k)$ be any transition, $M = M_t$ and $R = R_t \subseteq \mathsf{A}(\tau_j)$. Suppose*

(H1) no two elements of R are at distance 2;
(H2) no two elements of $M(R)$ are at distance 2;
(H3) $M(\Gamma_2(x)) = \Gamma_2(M(x))$ for every $x \in R$ (M is a local duality *at x when it maps points to lines: it maps the points collinear with x onto the lines concurrent with $M(x)$).*

Then t is repaired.

Proof. Write $j' = j + k$; "A" and "B" refer to $\mathsf{A}(\tau_j), \mathsf{B}(\tau_j)$ in level j and to $\mathsf{A}(\tau_{j'}), \mathsf{B}(\tau_{j'})$ in level j'. Take (u,j) and (v,j').

(1) $u \in \mathsf{A} \setminus R$: $(u,j) \to (M(u),j') \rightsquigarrow (v,j')$, length at most $1+6$.
(2) $u \in \mathsf{B}$, $v \in \mathsf{A}$: the $q+1 \ge 2$ neighbours of v are pairwise at distance 2, so by (H2) at most one is in $M(R)$; take $b \in \Gamma_1(v) \setminus M(R)$ and $a = M^{-1}(b) \notin R$: $(u,j) \rightsquigarrow (a,j) \to (b,j') \to (v,j')$, length at most $5+1+1$.
(3) $u \in \mathsf{B}$, $v \in \mathsf{B} \setminus M(R)$: $(u,j) \rightsquigarrow (M^{-1}(v),j) \to (v,j')$, length at most $5+1$.
(4) $u \in \mathsf{B}$, $v = M(x)$ with $x \in R$: by Lemma 2.4(b) there is $z \in \Gamma_2(x)$ with $d(z,u) \le 3$; $z \notin R$ by (H1), and $M(z) \in \Gamma_2(v)$ by (H3): $(v,j') \rightsquigarrow (M(z),j') \to (z,j) \rightsquigarrow (u,j)$, length at most $2+1+3$.
(5) $u = x \in R$, $v \in \mathsf{A}$: there is $w \in \Gamma_2(M(x))$ with $d(w,v) \le 3$; by (H3) $w = M(z)$ with $z \in \Gamma_2(x)$, and $z \notin R$ by (H1): $(x,j) \rightsquigarrow (z,j) \to (w,j') \rightsquigarrow (v,j')$, length at most $2+1+3$.
(6) $u = x \in R$, $v \in \mathsf{B}$: the same path, now with $d(w,v) \le 4$: length at most $2+1+4$.

These cases cover all pairs. □

For two points x, y at distance 4 let $\mathrm{mid}(x,y)$ be the unique point collinear with both, and likewise for lines.

Proposition 6.16 (what (H1)–(H3) force)**.** *Under (H1)–(H3), M restricted to R preserves distances, and $M(\mathrm{mid}(x,y)) = \mathrm{mid}(M(x),M(y))$ for all $x, y \in R$ at distance 4.*

Proof. Let $x \neq y \in R$; then $d(x,y) \in \{4,6\}$ by (H1). If $d(x,y)=4$, $m = \mathrm{mid}(x,y) \in \Gamma_2(x) \cap \Gamma_2(y)$, so by (H3) $M(m) \in \Gamma_2(M(x)) \cap \Gamma_2(M(y))$; hence $d(M(x),M(y)) \leq 4$, equal to 4 by (H2), and $M(m)$ is the mid-line. If $d(x,y)=6$, $\Gamma_2(x) \cap \Gamma_2(y) = \emptyset$, so $\Gamma_2(M(x)) \cap \Gamma_2(M(y)) = \emptyset$ by (H3), and $d(M(x),M(y))=6$. □

Proposition 6.17 (mirrored sets)**.** *Let R be a non-collinear set of points, R^* a non-concurrent set of lines, and $f : R \to R^*$ a bijection such that*

(S1) every point is collinear with at most two points of R, and every line is concurrent with at most two lines of R^*;
(S2) for $x \neq y \in R$: $d(x,y)=4$ if and only if $d(f(x),f(y))=4$.

Then there is a bijection $M : \mathcal{P} \to \mathcal{L}$ with $M|_R = f$ satisfying (H1)–(H3) for R. One such M is: $M(x) = f(x)$ for $x \in R$; $M(\mathrm{mid}(x,y)) = \mathrm{mid}(f(x),f(y))$ for $x,y \in R$ at distance 4; for each $x \in R$, any bijection from the remaining points of $\Gamma_2(x)$ onto the remaining lines of $\Gamma_2(f(x))$; any bijection on the rest.

Proof. Let $U = \bigcup_{x \in R} \Gamma_2(x)$ and $U^* = \bigcup_{x \in R} \Gamma_2(f(x))$. By (H1) for R and R^*, $R \cap U = \emptyset$ and $R^* \cap U^* = \emptyset$. A point of U lies in $\Gamma_2(x)$ for one or two $x \in R$ (S1); if two, x and y are at distance 4 and the point is $\mathrm{mid}(x,y)$. So the points of U in two sets $\Gamma_2(\cdot)$ correspond to the pairs $\{x,y\} \subseteq R$ at distance 4, and likewise for U^*; by (S2), $\{x,y\} \mapsto \{f(x),f(y)\}$ is a bijection between these pairs. Hence $\mathrm{mid}(x,y) \mapsto \mathrm{mid}(f(x),f(y))$ is a bijection from the shared points of U to the shared lines of U^*, mapping the shared points of $\Gamma_2(x)$ onto the shared lines of $\Gamma_2(f(x))$. The remaining (private) points of $\Gamma_2(x)$ and private lines of $\Gamma_2(f(x))$ are equally many (both $q(q+1)$ minus the number of $y \in R$ at distance 4 from x), and private sets of different x are disjoint. So the pieces define a bijection $R \cup U \to R^* \cup U^*$ with $M(\Gamma_2(x)) = \Gamma_2(f(x))$ for every $x \in R$. As $|\mathcal{P} \setminus (R \cup U)| = |\mathcal{L} \setminus (R^* \cup U^*)|$, it extends to a bijection $\mathcal{P} \to \mathcal{L}$. (H1) and (H2) are the hypotheses on R and R^*. □

Corollary 6.18. *(a) If R is a partial ovoid and R^* a partial spread with $|R| = |R^*|$, every bijection $f : R \to R^*$ satisfies (S1) and (S2) (there are no pairs at distance 4).*
(b) Let R_0 and X be disjoint partial ovoids, $f_0 : R_0 \to S_0$ a bijection onto a partial spread S_0, and $x \mapsto L_x$ an injective map from X onto a partial spread disjoint from S_0. Put $R = R_0 \cup X$, $R^* = S_0 \cup \{L_x : x \in X\}$, $f = f_0$ on R_0 and $f(x) = L_x$ on X, and assume that R is non-collinear and R^* non-concurrent. Then (S1) holds: three points of R collinear with one point would be pairwise at distance 4, but two of them lie in R_0 or two in X, which are pairwise opposite; likewise for lines. And (S2) says: for $r \in R_0$ and $x \in X$, $d(r,x)=4$ if and only if $d(f_0(r),L_x)=4$.

6.6. **Typed level layouts.** By Theorem 6.10 the order is $(2r+1)\cdot 2N + K_{\mathsf{A}} + K_{\mathsf{B}}$ with $K_{\mathsf{B}} \leq \min_j \alpha_j$ and $K_{\mathsf{A}} \leq \min_j \beta_j$.

All levels of type H, one bijection σ, one freed set R on every transition. Every level has $\alpha_j = \beta_j = r|R|$, so one can take $K_{\mathsf{A}} = K_{\mathsf{B}} = r|R|$, and $|X_R^+| = (2r+1)\cdot 2N + 2r|R|$. For $r=1$ the B-hub h_x of $x \in R$ is joined to $(x,0),(x,1),(x,2)$ and the A-hub g_x to $(\sigma(x),0),(\sigma(x),1),(\sigma(x),2)$, plus at most $q-1$ hub-graph edges each.

Types $HDHDH$ ($r=2$). The transitions are $j \to j+1$ and $j \to j+2$. Friendly: $0 \to 1$, $2 \to 3$, $4 \to 1$ ($H \to D$, points freed) and $1 \to 2$, $3 \to 4$, $3 \to 0$ ($D \to H$, lines freed). Not friendly: $0 \to 2$, $2 \to 4$, $4 \to 0$ ($H \to H$) and $1 \to 3$ ($D \to D$). With F_t the freed sets of friendly transitions and R_t those of the others:

level	type	free A-slots α_j	free B-slots β_j
0	H	$F_{0\to1}+R_{0\to2}$	$R_{4\to0}+F_{3\to0}$
1	D	$F_{1\to2}+R_{1\to3}$	$F_{0\to1}+F_{4\to1}$
2	H	$F_{2\to3}+R_{2\to4}$	$F_{1\to2}+R_{0\to2}$
3	D	$F_{3\to4}+F_{3\to0}$	$F_{2\to3}+R_{1\to3}$
4	H	$R_{4\to0}+F_{4\to1}$	$F_{3\to4}+R_{2\to4}$

(sizes of the sets). With all friendly point sets of size f_P, all friendly line sets of size f_L, and m elements freed on each non-friendly transition, $K_{\mathsf{B}} \le \min(f_P+m,\ f_L+m,\ 2f_L)$ and $K_{\mathsf{A}} \le \min(f_L+m,\ 2f_P,\ f_P+m)$. Five levels with two matchings form the regular tournament on 5 vertices, and a split of the types into $3+2$ levels makes 6 of the 10 arcs friendly, the maximum; the split $HDHDH$ also gives every level at least one friendly outgoing and one friendly incoming transition. With three levels some level always has no friendly outgoing transition, so typing does not help for $r=1$. *Seven levels* ($r=3$): the types $HHDHDHD$ make 12 of the 21 transitions friendly, and every level has at least one friendly transition in each direction.

6.7. **Hub graphs.** Write $\mathrm{HG}(K_1,K_2;\delta)$ for a bipartite graph with colour classes of sizes K_1,K_2, maximum degree at most δ, and all same-colour distances at most 6 in the graph itself. If Y is such a graph with $\delta \le q-r$, both conditions on Y in Theorem 6.10 hold.

Lemma 6.19 (plane with pendant trees)**.** *Let s be a prime power and $t,u \ge 0$ integers with $s+1+t \le \delta$ and $u+1 \le \delta$. Then $\mathrm{HG}(K,K;\delta)$ exists for every K with $s^2+s+1 \le K \le (s^2+s+1)(1+t+tu)$.*

Proof. Start from the incidence graph of $\mathrm{PG}(2,s)$ (the *core*: diameter 3, same-colour distance 2). Attach to some core vertices up to t pendant vertices each (depth 1, of the opposite colour), and to some depth-1 vertices up to u pendant vertices each (depth 2, of the colour of the core vertex). Degrees are at most $s+1+t$, $u+1$ and 1. For two vertices x,y of the same colour with core ancestors $c(x),c(y)$, $d(x,y) \le \mathrm{depth}(x)+d(c(x),c(y))+\mathrm{depth}(y)$. The core ancestors have the same colour when the depths have the same parity (core distance at most 2) and different colours otherwise (core distance at most 3); the possible totals are $0+2+0$, $0+3+1$, $0+2+2$, $1+2+1$, $1+3+2$, $2+2+2$, all at most 6. To get K vertices of each colour, give c depth-1 children to the points and c to the lines, and g depth-2 children below each of these two families, with $c=\lceil (K-s^2-s-1)/(1+u)\rceil \le (s^2+s+1)t$ and $g = K-s^2-s-1-c \le uc$. □

With $t=\delta-s-1$ and $u=\delta-1$, Lemma 6.19 gives hub graphs with up to

$$\kappa(\delta) := \max_s (s^2+s+1)\big(1+(\delta-s-1)\delta\big) \approx \tfrac{4}{27}\delta^4$$

vertices of each colour (s over prime powers at most $\delta-1$); $\kappa(3)=7$, $\kappa(4)=35$, $\kappa(7)=315$, $\kappa(8)=527$, $\kappa(9)=868$, $\kappa(10)=1271$, $\kappa(11)=1938$. For $\delta \ge 3$ the intervals for $s=2$ and for the optimal s overlap, so $\mathrm{HG}(K,K;\delta)$ exists for every $7 \le K \le \kappa(\delta)$.

6.8. **Bounds for every q from classical sets.**

Corollary 6.20 (three levels, every $q \ge 4$)**.** *Let R be the q^2+1 points of an elliptic quadric section $Q^-(3,q)$ (Lemma 6.2(a)), R^* any q^2+1 lines of a Hermitian spread, and σ as in Proposition 6.17 (with Corollary 6.18(a)). All three levels of type H, $M_t=\sigma$ and $R_t=R$ for all three transitions, $K=\min(q^2+1,\kappa(q-1))$ hubs of each colour and the hub graph of Lemma 6.19 give*

$$n_b(q+2,7) \ge 6N(q)+2\min\big(q^2+1,\ \kappa(q-1)\big),$$

which is $6N(q)+2(q^2+1)$ for every $q \ge 5$.

Proof. Every transition is repaired by Theorem 6.15; every level has $|R|$ free slots of each colour, and Definition 6.7 allows $K \le |R|$ hubs (the unused slots stay free). Theorem 6.10 with $Y = \mathrm{HG}(K, K; q-1)$. □

At $q = 4$ ($\kappa(3) = 7 < 17$) the bound is $6N(4) + 14 = 8204$, below the table's $(6, 7) = 8310$; the corollary improves the table only for $q \ge 5$.

Corollary 6.21 ($q = 3^h$, a duality)**.** *Let $q = 3^h$, φ a duality of $H(q)$, all types H, $M_t = \varphi|_{\mathcal{P}}$ and $R_t = F$ for every transition, with F the point set of a spread, $|F| = (q+1)(q^3+1)$. For every r with $1 \le r \le q-3$,*

$$n_b(q+1+r, 7) \ge (2r+1) \cdot 2N(q) + 2\kappa(q-r).$$

Proof. Every transition is repaired by Theorem 6.12; every level has $r|F|$ free slots of each colour, and $\kappa(q-r) < q^4 < r(q+1)(q^3+1)$, so $K = \kappa(q-r)$ hubs fit; Theorem 6.10 with the hub graph of Lemma 6.19. □

Corollary 6.22 (typed levels $HDHDH$, every $q \ge 5$)**.** *Take the types $HDHDH$; on the friendly transitions $M_t = \mathrm{id}$, with freed point sets the point set of a Hermitian spread ($(q+1)(q^3+1)$ points, a segment set) and freed line sets the lines through the points of an elliptic quadric section ($(q+1)(q^2+1)$ lines, a dual segment set); on the non-friendly transitions σ (resp. σ^{-1}) and R (resp. $\sigma(R)$) as in Corollary 6.20. The smallest slot counts are $\min_j \alpha_j = \min_j \beta_j = (q+1)(q^2+1) + q^2 + 1 = (q+2)(q^2+1)$, so*

$$n_b(q+3, 7) \ge 10N(q) + 2\min\big((q+2)(q^2+1),\ \kappa(q-2)\big).$$

The minimum is $\kappa(q-2)$ for $q \le 13$ and $(q+2)(q^2+1)$ for $q \ge 16$.

Proof. Theorems 6.12, 6.15 and 6.10, with the slot table of the previous subsection. □

Corollary 6.23 (all levels H, any r)**.** *With R, R^* and σ as in Corollary 6.20, $2r+1$ levels of type H, $M_t = \sigma$ and $R_t = R$ for every transition, for $q \ge r+3$:*

$$n_b(q+1+r, 7) \ge (2r+1) \cdot 2N(q) + 2\min\big(r(q^2+1),\ \kappa(q-r)\big).$$

Proof. Every transition is repaired by Theorem 6.15, each level has $r(q^2+1)$ free slots of each colour, and a vertex with several free slots may receive several hubs; Theorem 6.10 with $Y = \mathrm{HG}(K, K; q-r)$ from Lemma 6.19, $K = \min(r(q^2+1), \kappa(q-r)) \ge 7$. □

Corollary 6.23 was suggested by the referee of the diameter-7 note (who built its $q = 5$, $r = 1$ and $r = 2$ cases and checked them by breadth-first search from every vertex); the note itself left it out because it added no new statement. It is the only way the $(15, 7)$ cell is improved without computer-found sets. Keep it?

6.9. **The values of Table 5.** The graphs behind Table 5 are X_R^+ as in Definition 6.7, with every transition repaired by Theorem 6.12 or 6.15 and (HG) satisfied, so Theorem 6.10 gives the bound. They use explicit sets larger than those of Section 6.8, and explicit hub graphs. These ingredients are given as data (Appendix C); that they satisfy the hypotheses (non-collinearity, (H1)–(H3) at every freed element, the segment property, degrees, (HG)) is computer-checked, and each final graph was also checked by breadth-first search from every vertex.

Corollary 6.24 (three levels, local duality)**.** *Let $r = 1$, all types H, $M_t = \sigma$ and $R_t = R$ for all three transitions, with (R, σ) as in Proposition 6.17, built as in Corollary 6.18(b). If $K \le |R|$ hubs of each colour are attached and the hub graph has maximum degree at most $q-1$ and satisfies (HG), then $n_b(q+2, 7) \ge 6N(q) + 2K$. Values ($K = |R|$): $q = 5$: $|R| = 81$ (R_0: 72 pairwise opposite points; 9 further points; the Hermitian spread has $q^3 + 1 = 126$ lines), order 23 598; $q = 7$: $288 + 7$, order 118 238; $q = 8$: $217 + 12$, order 225 152; $q = 11$: $756 + 12$, order 1 064 472.*

TABLE 6. Diameter 7 from classical ingredients only (elliptic quadric sections, the Hermitian spread, a duality of $H(3^h)$, and the hub graphs of Lemma 6.19).

cell	from	bound	2013
(7,7)	Cor. 6.20, $q=5$	23 488	23 436
(9,7)	Cor. 6.20, $q=7$	117 748	117 648
(10,7)	Cor. 6.20, $q=8$	224 824	224 694
(11,7)	Cor. 6.21, $q=9$, $r=1$	399 634	398 580
(12,7)	Cor. 6.21, $q=9$, $r=2$	664 930	664 300
(13,7)	Cor. 6.20, $q=11$	1 063 180	1 062 936
(14,7)	Cor. 6.22, $q=11$	1 773 296	1 771 560
(15,7)	Cor. 6.23, $q=11$, $r=3$	2 480 916	2 480 184
(16,7)	Cor. 6.22, $q=13$	4 026 216	4 022 340

For $q=7,11$ the partial ovoid R_0 is an orbit of the Weyl group $W(E_7)$ acting on $Q(6,q)$ through the E_7 root lattice modulo q; being pairwise non-perpendicular in $Q(6,q)$, it is a partial ovoid of $H(q)$ by Lemma 2.4(c). The lines L_x are pairwise opposite and not in the Hermitian spread, and f_0 maps R_0 onto Hermitian spread lines.

Corollary 6.24: the $W(E_7)$ description of R_0 ($q=7,11$) is from the lab notes; a precise statement (which lattice, which embedding into $Q(6,q)$) should be written out if we keep it. For $q=5,8$ the sets are only data.

Corollary 6.25 (duality, $q=9$)**.** *In Corollary 6.21 with $q=9$, F the 7300 points of the Hermitian spread and $K=r|F|$ hubs of each colour: for $r=1$, with a hub graph of maximum degree 8 (a Haar graph of $\mathbb{Z}_{73}\rtimes\mathbb{Z}_{100}$ of diameter 7), the order is 413 180; for $r=2$, with 14 600 hubs of each colour and a hub graph of maximum degree 7 (a Haar graph of $(\mathbb{Z}_{73}\rtimes\mathbb{Z}_8)\times\mathbb{Z}_5^2$, for which (HG) holds using paths through the levels), the order is 693 500. By Proposition 6.4 and Remark 6.13, no larger freed set leaves every transition repaired, so within this construction (and as long as (T) is required) $K=r(q+1)(q^3+1)$ is the cap. The duality of $H(9)$ is given explicitly as data and was checked to be a bijection of $\mathcal{P}\cup\mathcal{L}$ exchanging $\mathcal{P}$ and $\mathcal{L}$ and preserving all $N(q+1)$ incidences.*

Corollary 6.26 (typed levels)**.** *Let the types be $HDHDH$ ($r=2$) or $HHDHDHD$ ($r=3$); let every friendly transition carry $M_t=\mathrm{id}$ and a segment set (points) or dual segment set (lines) F_t, and let the non-friendly transitions carry $M_t=\sigma$ ($H\to H$) or σ^{-1} ($D\to D$) with freed sets R (resp. $\sigma(R)$) satisfying Theorem 6.15. If $K_{\mathsf{B}}\le\min\alpha_j$, $K_{\mathsf{A}}\le\min\beta_j$ and the hub graph has maximum degree at most $q-r$ and satisfies (HG), then $n_b(q+1+r,7)\ge(2r+1)\cdot 2N(q)+K_{\mathsf{A}}+K_{\mathsf{B}}$. Values:*

- $(14,7)$, $q=11$, *$HDHDH$: all friendly point sets equal one non-collinear set of 8645 points, all friendly line sets one non-concurrent set of 8246 lines (both unions of orbits of a collineation of order 133), and the 768-point set of Corollary 6.24 on the four non-friendly transitions; $\alpha=(9413,9014,9413,16492,9413)$, $\beta=(9014,17290,9014,9413,9014)$, $K_{\mathsf{A}}=K_{\mathsf{B}}=9014$: order 1 789 588.*
- $(15,7)$, $q=11$, *$HHDHDHD$: twelve friendly sets with 5674–9303 elements each, four of them segment sets that are not non-collinear (5577 collinear or concurrent pairs in all); the 768-point set on the nine non-friendly transitions; $K_{\mathsf{B}}=10\,826$, $K_{\mathsf{A}}=10\,481$: order 2 501 491.*
- $(16,7)$, $q=13$, *$HDHDH$: a non-collinear set of 14 762 points and a non-concurrent set of 14 823 lines (unions of orbits of an element of order 61 of $\mathrm{SL}(3,13)$), and on the*

four non-friendly transitions a partial ovoid of 122 points, with σ as in Corollary 6.18(a); $K_{\mathsf{A}} = K_{\mathsf{B}} = 14\,762 + 122 = 14\,884$*: order 4 052 108.*

Remark 6.27. Theorem 6.12 allows friendly sets larger than those used in Corollary 6.26: up to $(q+1)(q^3+1)$ (Proposition 6.4), for instance the 15 984 points of the Hermitian spread of $H(11)$ against 8645, and the $12 \cdot 756 = 9072$ lines through the 756-point partial ovoid of $H(11)$ against 8246. In the $HDHDH$ layout with these sets and the 768 mirrored elements the slot counts give $K_{\mathsf{A}} = K_{\mathsf{B}} = 9840$, that is 1 791 240 at $(14, 7)$, provided that a hub graph with 9840 vertices of each colour and degree 9 satisfying (HG) exists. We have not built this graph.

6.10. **Prior work.** The level graph of Definition 6.5 is a bipartite analogue of the star product of Bermond, Delorme and Farhi [7, 6] (see [31, §3.3.2]) and of the compound graphs of Gómez and Fiol [26]: copies of a graph indexed by the vertices of a tournament, with a matching for each arc. The cells $(7, 7)$ and $(9, 7)$–$(16, 7)$ of the 2013 table have the form $(2r + 1) \cdot 2N(q)$ of Theorem 6.6 and are credited to [17], which we could not find in print. The values $(6, 7) = 8310$ and $(8, 7) = 40\,586$ equal the order $2q(q^4+q^3+3q^2+q+1)$ ($q = 5, 7$) of Delorme's construction [15, §9], in which vertices are inserted on $2q^3$ edges of the incidence graph of a generalized hexagon and a vertex set is then duplicated. Delorme also builds cyclic multi-level graphs from a duality of a generalized quadrangle or hexagon [15, §3], on the vertex set $\mathbb{Z}/m\mathbb{Z} \times Q^k$ with Q the point set; those graphs are not copies of $I(q)$ joined by matchings. Sets of $q^3 + 1$ pairwise opposite elements of $H(q)$ are used in [15, §8(b)] to raise the degree at diameter 6 (Remark 5.9).

The closest prior work is a 2026 preprint of R. Rajiv [35] (unrefereed). It builds compounds of copies of the incidence graph of a generalized n-gon along a tournament, with point-to-line and type-preserving matchings; its Lemma 5 deletes the type-preserving matching edges at a pairwise non-collinear (or non-concurrent) set and keeps the diameter; its Theorem 3 attaches new vertices, each joined to one point and one line of every copy, connected by a further graph. In our language these are friendly transitions with $M_t = \mathrm{id}$, Corollary 6.14(1), and hubs as in Definition 6.7. The setting differs (general graphs of diameter $n = 4, 6$, not bipartite graphs of diameter 7; hubs with one point slot and one line slot per copy, not hubs of one colour), but the ideas are the same. F. Comellas' table of general graphs [12] records in October 2026 graphs of diameter 6 obtained from $H(q)$ by deleting or subdividing incidence edges and attaching new vertices. Replacing or adding vertices in hexagon-based graphs of diameter 6 is older [13, 28, 34]. In the sources we searched we found nothing on: the bipartite diameter-7 version with hubs of one colour (Definition 6.7, Theorem 6.10); typed levels as the 2-colouring; local dualities at non-automorphic transitions (Theorem 6.15, Propositions 6.16, 6.17); segment sets, Proposition 6.4, Theorem 6.12 beyond non-collinear sets and its converse (Remark 6.13). Proposition 6.4 and Remark 6.13 are elementary and may well be known.

Section 6.10: you may know [17] and how the $D = 7$ table values were obtained. The literature search behind this paragraph did not query Google Scholar or zbMATH.

7. Diameter 8: Gómez's compound graphs $\Omega\mathrm{H}_q$

Section 7 questions six entries of the 2013 table credited to Delorme, Gómez and Quisquater. Decide how to present it (in this paper, as a separate note, or after contacting J. Gómez), and how much of the relabelling part (Section 7.4) to keep.

Six $D = 8$ entries of the 2013 table, credited on the CombinatoricsWiki to Delorme, Gómez and Quisquater [17], are the orders of Gómez's compound graphs $\Omega\mathrm{H}_q$ [27, §2.9.3, values on p. 46]. Here H_q is the incidence graph of $H(q)$, with $N_1 = 2N$ vertices, and $q + 3 = 2\alpha + \beta$ with $\beta \in \{0, 1\}$:

cell	2013 value	graph	α	copies of H_q	order $5(\alpha+1)N_1$
(10,8)	1 176 480	$\Omega\mathrm{H}_7$	5	25	1 176 480
(11,8)	2 246 940	$\Omega\mathrm{H}_8$	5	25	2 246 940
(12,8)	4 650 100	$\Omega\mathrm{H}_9$	6	30	4 650 100
(14,8)	14 172 480	$\Omega\mathrm{H}_{11}$	7	35	14 172 480
(15,8)	14 172 480	$\Omega\mathrm{H}_{11}$ (value of (14,8) reused)	7	35	14 172 480
(16,8)	36 201 060	$\Omega\mathrm{H}_{13}$	8	40	36 201 060

The thesis states $D \le D_1 + 2 = 8$ and gives no proof (its §2.8.2 says that the proofs of this part are left out). The paper [17] was never published, as far as we could find. We show that $\Omega\mathrm{H}_q$, as defined in the thesis, has diameter at least 9 for every reading of its figures (Theorems 7.3 and 7.5), and that giving each copy of the hexagon its own labelling by automorphisms does not help (Theorems 7.15 and 7.21). Whether some other labelling gives diameter 8 is open (Section 7.5).

7.1. **The construction.** [27, §2.9.3, pp. 43–45] takes 5 copies of a bipartite graph B_1 of even diameter D_1 and 10 sets of $N_1/2$ vertices each (copies of K_1); every 10 such vertices, one from each set, are joined to the copies as in Fig. 2.21a of the thesis, so that their degree is 2, and the degree is then raised to 4 by joining them to 5 more copies as in Fig. 2.21b. In general, if $\Delta_1 + 2 = 2\alpha + \beta$, one takes 5α copies of B_1, using either of the two structures, so that the copies of K_1 get degree 2α; the result ΩB_1 has $N = 5(\alpha+1)N_1$, $\Delta = \Delta_1 + 2$ and, as stated, $D \le D_1 + 2$. For $B_1 = H_q$: $D_1 = 6$, $\Delta_1 = q+1$, $N_1 = 2N$.

Labels. In [27, §2.7.1] both colour classes of B_1 are labelled by $\mathbb{Z}/n_1\mathbb{Z}$, $n_1 = N_1/2$, and every adjacency rule of §§2.7–2.9 is written in these labels, the same in every copy; §2.8.9, which §2.9.3 cites for its copies of K_1, joins a copy of K_1 with label t only to vertices labelled t, in either colour class. For $B_1 = H_q$ we write the labelling as two bijections $P_0 : \mathbb{Z}/N\mathbb{Z} \to \mathcal{P}$ and $L_0 : \mathbb{Z}/N\mathbb{Z} \to \mathcal{L}$; the thesis does not fix them, so $\sigma = L_0 P_0^{-1}$ (point $t \mapsto$ line t) is an arbitrary bijection $\mathcal{P} \to \mathcal{L}$.

The figures. Both figures place five copies at the corners of a pentagon, inside a circle and a pentagram. Each copy has an *outer* dot on the circle and an *inner* dot at the corner of the pentagram; each copy of K_1 in the block is a dot on an arc or a pentagram edge, joined to the two dots at its ends. In Fig. 2.21a, the K_1-vertex k lies on the pentagram edge between copies $k-1$ and $k+1$ and $\bar{k}$ on the arc between copies $k+2$ and $k+3$; in Fig. 2.21b, k lies on the arc between copies $k+3$ and $k+4$ and $\bar{k}$ on the pentagram edge between copies $k-1$ and $k+1$ (indices mod 5). The figures do not say which colour class the inner dot is. *Reading 1*: outer dot = points and inner dot = lines in every copy. *Reading 2*: unbarred K_1-vertices always on points and barred ones always on lines. The figures also do not say which blocks are of type (a) and which of type (b), so we allow any pattern; and we allow the inner class to be chosen freely in every copy, which contains both readings and their duals.

Definition 7.1 ($\Omega\mathrm{H}_q$)**.** Data: a prime power q and $\alpha = \lfloor (q+3)/2 \rfloor$; a block pattern $(\tau_0, \dots, \tau_{\alpha-1}) \in \{a, b\}^\alpha$; an inner class $\iota(c) \in \{\mathcal{P}, \mathcal{L}\}$ for every copy c; and a *labelling*, that is, bijections $P_c : \mathbb{Z}/N\mathbb{Z} \to \mathcal{P}$ and $L_c : \mathbb{Z}/N\mathbb{Z} \to \mathcal{L}$ for every copy c.

- *Copies.* $c = 5B + i$ with block $B \in \{0, \dots, \alpha-1\}$ and position $i \in \mathbb{Z}_5$; there are 5α copies. Copy vertices (c, x), $x \in \mathcal{P} \cup \mathcal{L}$, with $(c, x) \sim (c, y)$ when x and y are incident.
- *Hubs* (the copies of K_1). (t, k) and $(t, \bar{k})$ for $t \in \mathbb{Z}/N\mathbb{Z}$ (the *row*) and $k \in \mathbb{Z}_5$; the hub *classes* are $k = \{(t, k)\}$ (unbarred) and $\bar{k} = \{(t, \bar{k})\}$ (barred).
- *Copies met by a class* (positions mod 5): $C(k) = \{5B+k-1, 5B+k+1 : \tau_B = a\} \cup \{5B+k+3, 5B+k+4 : \tau_B = b\}$ and $C(\bar{k}) = \{5B+k+2, 5B+k+3 : \tau_B = a\} \cup \{5B+k-1, 5B+k+1 : \tau_B = b\}$.

- *Where a class sits.* In a copy of a block of type a, unbarred classes sit on the inner class $\iota(c)$ and barred classes on the outer class; in a block of type b, unbarred classes on the outer and barred classes on the inner class.
- *Hub edges.* $(t, Y) \sim (c, P_c(t))$ if Y sits on points in c, and $(t, Y) \sim (c, L_c(t))$ if Y sits on lines in c, for every $c \in C(Y)$.

The *thesis labelling* is $P_c = P_0$, $L_c = L_0$ for all c. Reading 1 is $\iota(c) = \mathcal{L}$ for all c; reading 2 is $\iota(c) = \mathcal{P}$ in a-blocks and $\mathcal{L}$ in b-blocks. The graph $\Omega\mathrm{H}_q$ of the thesis is the thesis labelling with any pattern and either reading.

Exact neighbourhoods. Write $m_c(t)$ for the inner mark of row t in copy c ($P_c(t)$ if $\iota(c) = \mathcal{P}$, $L_c(t)$ if $\iota(c) = \mathcal{L}$) and $\bar{m}_c(t)$ for the outer mark. For $c = 5B + i$, the copy vertex (c, x) is adjacent to (c, y) for the $q + 1$ elements y incident with x, and to two hubs: $(t, i-1), (t, i+1)$ if $\tau_B = a$ and $x = m_c(t)$; $(t, \overline{i+2}), (t, \overline{i+3})$ if $\tau_B = a$ and $x = \bar{m}_c(t)$; $(t, \overline{i-1}), (t, \overline{i+1})$ if $\tau_B = b$ and $x = m_c(t)$; $(t, i+1), (t, i+2)$ if $\tau_B = b$ and $x = \bar{m}_c(t)$. The hub (t, k) is adjacent to $(c, m_c(t))$ for $c = 5B + k \pm 1$ in blocks with $\tau_B = a$, and to $(c, \bar{m}_c(t))$ for $c = 5B + k + 3, 5B + k + 4$ in blocks with $\tau_B = b$; the hub $(t, \bar{k})$ is adjacent to $(c, \bar{m}_c(t))$ for $c = 5B + k + 2, 5B + k + 3$ in blocks with $\tau_B = a$, and to $(c, m_c(t))$ for $c = 5B + k \pm 1$ in blocks with $\tau_B = b$.

Lemma 7.2 (parameters)**.** *For every pattern, inner classes and labelling, $\Omega\mathrm{H}_q$ is a simple bipartite graph on $10(\alpha+1)N = 5(\alpha+1)N_1$ vertices; copy vertices have degree $q+3$ and hubs degree 2α, so $\Delta = q + 3$. Every hub meets 2α of the 5α copies and misses 3α.*

Proof. A copy at position i of an a-block is met by the unbarred classes $i \pm 1$ and the barred classes $i + 2, i + 3$; of a b-block, by the unbarred classes $i + 1, i + 2$ and the barred classes $i \pm 1$. So each copy is met by exactly two classes of each bar, the two of one bar sit on the same type, and each class has one hub per row. Every element of a copy is the mark of exactly one row of its type, so it gets exactly two hubs. A class meets two distinct copies in each block. For the colouring, give unbarred hubs colour 0, in every copy the type on which the unbarred classes sit colour 1 and the other type colour 0, and barred hubs colour 1. □

7.2. **One label: a projection argument.**

Theorem 7.3 (projection)**.** *Let B be a graph and G a graph whose vertices are of two kinds:* copy vertices *v, each with a copy index $c(v)$ and a label $f(v) \in V(B)$, and* new vertices *x, each with a label $f(x) \in V(B)$. Assume*

(h1) if $u \sim w$ are copy vertices, then $c(u) = c(w)$ and $f(u) \sim f(w)$ in B;
(h2) every neighbour w of a new vertex x is a copy vertex with $f(w) = f(x)$;
(h3) no two new vertices are adjacent.

If x is a new vertex with no neighbour in copy j, and v is a copy vertex with $c(v) = j$, then $d_G(x, v) \geq d_B(f(x), f(v)) + 3$.

Proof. Let $(x = w_0, w_1, \ldots, w_l = v)$ be a walk. By (h2), w_1 is a copy vertex with $f(w_1) = f(x)$. Let m be the number of new vertices among $w_1, \ldots, w_{l-1}$. A new vertex w_i $(0 < i < l)$ has copy vertices w_{i-1}, w_{i+1} on both sides, with $f(w_{i-1}) = f(w_i) = f(w_{i+1})$ by (h2); every other edge $w_i w_{i+1}$ $(i \geq 1)$ joins two copy vertices and maps to an edge of B by (h1). Deleting the m new vertices and one of the two equal labels around each, the labels of $w_1, \ldots, w_l$ form a walk in B from $f(x)$ to $f(v)$ of length $l - 1 - 2m$. So $l \geq d_B(f(x), f(v)) + 1 + 2m$. If $m = 0$, then $w_1, \ldots, w_l$ are copy vertices joined by copy edges, so by (h1) $c(w_1) = c(v) = j$, and w_1 is a neighbour of x in copy j, a contradiction. So $m \geq 1$ and $l \geq d_B(f(x), f(v)) + 3$. □

Corollary 7.4. *Take the thesis labelling and inner classes for which every hub class sits on one type in all copies it meets (reading 2; its dual; reading 1 when all blocks have one type). Then for*

every q, every pattern and every pairing σ: every hub x and every copy j that x misses, the q^5 vertices (j, y) with y opposite the mark of x are at distance at least 9 from x. So $\operatorname{diam} \Omega \mathrm{H}_q \geq 9$. *More generally, for any bipartite B_1 with a pair of vertices at distance D_1, ΩB_1 in reading 2 has diameter at least $D_1 + 3$.*

Proof. Label copy vertices by $f(c, y) = y$, $c(c, y) = c$, and a hub by its mark ($P_0(t)$ or $L_0(t)$, the same in all copies by assumption). Copy edges satisfy (h1), hub edges (h2), and there are no hub–hub edges. Every hub misses $3\alpha \geq 1$ copies (Lemma 7.2), and every vertex of $I(q)$ has q^5 opposite vertices (Lemma 2.4(a)). Apply Theorem 7.3. For a general B_1, in reading 2 the unbarred classes sit on all labels of class 0 and the barred classes on all labels of class 1, so whichever class contains the pair, some hub carries one of its labels. □

7.3. **Two labels.**

Theorem 7.5 (two labels)**.** *Let $q \geq 7$. Take $\Omega \mathrm{H}_q$ with the thesis labelling, any block pattern and any inner classes (so both readings and every pairing σ). Let x be a hub and j a copy that x misses. Then at least*

$$q^5 - (A + B + 1) = q^5 - 5q^4 - 8q^3 - 12q^2 - 16q - 10 > 0$$

vertices of copy j are at distance at least 9 from x, where, with $s_0 = 1$ and $s_i = (q+1)q^{i-1}$ for $1 \leq i \leq 5$,

$$A = \sum_{e_1+e_2 \in \{0,2,4\}} s_{e_1} s_{e_2} = 5q^4 + 8q^3 + 6q^2 + 4q + 2, \qquad B = \sum_{e_1+e_2+e_3 \leq 2} s_{e_1} s_{e_2} s_{e_3} = 6q^2 + 12q + 7$$

(sums over $e_i \geq 0$). In particular $\operatorname{diam} \Omega \mathrm{H}_q \geq 9$.

Proof. Let τ be the involution of $\mathcal{P} \cup \mathcal{L}$ with $\tau(P_0(s)) = L_0(s)$ for every row s. Every hub $(s, \cdot)$ is adjacent, in each copy it meets, to $P_0(s)$ or to $L_0(s)$. Let x have row t, and let $T \in \{\mathcal{P}, \mathcal{L}\}$ be the type of the vertices of copy j that have the colour of the neighbours of x (by Lemma 7.2, in each copy one type has that colour). Let $z = P_0(t)$ if $T = \mathcal{P}$ and $z = L_0(t)$ if $T = \mathcal{L}$. We show that at most $A + B + 1$ elements $w \in \operatorname{opp}(z)$ have $d(x, (j, w)) \leq 8$; since $|\operatorname{opp}(z)| = q^5$, the rest are at distance at least 9.

Fix $w \in \operatorname{opp}(z)$ of type T. The graph is bipartite and (j, w) has the colour of the neighbours of x, so $d(x, (j, w))$ is odd; suppose it is at most 7 and take a shortest path. It leaves x to a vertex (c_1, a_0) with $a_0 \in \{z, \tau(z)\}$, then alternates walks inside copies with passes copy $\to$ hub $\to$ copy. A pass through a hub of row s goes from (c, a) to (c', a') with $a, a' \in \{P_0(s), L_0(s)\}$, so $a' = a$ (a *stay*) or $a' = \tau(a)$ (a *switch*). Since x misses j there is at least one pass. With e copy edges and $p \geq 1$ passes the length is $1 + e + 2p \leq 7$, and odd, so e is even, $e \leq 4$ and $p \leq 3 - e/2$. Let $r \leq p$ be the number of switches. Merging the stays, the sequence of labels is a walk in $I(q)$ from a_0 with e edges in total, with τ applied r times in between, ending at w. Every edge of $I(q)$ and every switch changes the type, and w has the type of z, so r is even if $a_0 = z$ and odd if $a_0 = \tau(z)$. The set of ends of walks of length f from a vertex u is $\bigcup \Gamma_g(u)$ over $g \leq f$ with $g \equiv f \pmod 2$, of size at most the sum of the s_g. So:

- $a_0 = z$, $r = 0$: w is the end of a walk of length $e \leq 4$ from z, so $d(z, w) \leq 4$ and $w \notin \operatorname{opp}(z)$.
- $a_0 = z$, $r = 2$: $p \geq 2$, so $e \leq 2$; the label walk is $z \to (f_1 \text{ edges}) \to u_1$, switch, $\to (f_2) \to u_2$, switch, $\to (f_3) \to w$, with $f_1 + f_2 + f_3 = e \leq 2$. At most B such w.
- $a_0 = \tau(z)$, $r = 1$: $\tau(z) \to (f_1) \to u_1$, switch, $\to (f_2) \to w$ with $f_1 + f_2 = e \in \{0, 2, 4\}$. At most A such w.
- $a_0 = \tau(z)$, $r = 3$: $p = 3$, $e = 0$, and $w = \tau^4(z) = z \notin \operatorname{opp}(z)$ (we count it as 1).

So at most $A + B + 1$ elements of $\operatorname{opp}(z)$ are within distance 7 of x. The closed forms are direct expansions (for A: $s_0 s_0 + 2 s_0 s_2 + s_1 s_1 + 2 s_0 s_4 + 2 s_1 s_3 + s_2 s_2$). Finally, for $q \geq 7$, $q^4(q-5) \geq 2q^4 \geq 14q^3$, so $q^5 - 5q^4 - 8q^3 - 12q^2 - 16q - 10 \geq 6q^3 - 12q^2 - 16q - 10 \geq 30q^2 - 16q - 10 > 0$. □

At $q = 7, 8, 9, 11, 13$ the bound is 1348, 7286, 19 286, 75 560 and 208 666 far vertices per hub and missed copy. The proof uses only: the graph is bipartite; all copies carry the labels P_0, L_0; every hub of row s is adjacent, in each copy it meets, only to $P_0(s)$ or $L_0(s)$; there are no hub–hub edges; x misses a copy. For $q \leq 6$ the count is not enough ($A + B + 1 > q^5$).

Corollary 7.6. *As defined in the thesis (every copy carries the same labels, and a copy of K_1 with label t is joined only to vertices labelled t), $\Omega\mathrm{H}_q$ has diameter at least 9 for $q = 7, 8, 9, 11, 13$, under every reading of Figs. 2.21a/b, every pattern of block types and every pairing of point labels with line labels. These are the six table entries.*

Computer checks. The diameter is exactly 9 for $\Omega\mathrm{H}_3$, $\Omega\mathrm{H}_5$ (195 300 vertices, the thesis value at $(8, 8)$), $\Omega\mathrm{H}_7$ and $\Omega\mathrm{H}_9$ in both readings with both block types present, for several pairings σ and patterns (all-pairs breadth-first search for $q = 3, 5$; for $q = 7, 9$ breadth-first search from one vertex in every orbit of an automorphism group that was checked edge by edge). For all six table values, breadth-first search from a hub gives eccentricity 9 in both readings; in reading 2 the vertices at distance 9 from a hub are exactly the points opposite its mark in the 3α copies it misses. With only one block type the diameter is 10.

7.4. **Relabelling the copies.** Diameter at least 9 comes from the shared labels. A labelling that differs from copy to copy is not in the thesis, but it is the obvious repair, and at $q = 2, 3$ it works (Section 7.5). We show that it cannot work at the table's q with labellings that respect the geometry.

A labelling (P_c, L_c) as in Definition 7.1 gives each copy one point labelling and one line labelling, used by every hub class of the corresponding type that meets the copy. For a hub class Y and copies $c, d \in C(Y)$, the *transfer* $T_Y(c \to d)$ maps the mark of row t in c to the mark of row t in d: it is $P_d P_c^{-1}$, $L_d L_c^{-1}$, $L_d P_c^{-1}$ or $P_d L_c^{-1}$, depending on the types Y sits on in c and d.

Definition 7.7. A labelling is *geometric* if every transfer is the restriction (to $\mathcal{P}$ or to $\mathcal{L}$) of an automorphism of $I(q)$, that is, of a collineation of $H(q)$ or, when q is a power of 3, of a duality.

Lemma 7.8 (one automorphism per copy)**.** *Relabelling every copy c of a graph $\Omega\mathrm{H}_q$ by an automorphism g_c of $I(q)$ (replacing every vertex (c, y) by $(c, g_c(y))$) gives an isomorphic graph. In particular, if $P_c = g_c P_0$ and $L_c = g_c L_0$ with collineations g_c (or $P_c = g_c L_0$, $L_c = g_c P_0$ with dualities g_c and the inner class of c exchanged), the graph is isomorphic to the thesis graph it came from, and its diameter is at least 9.*

Lemma 7.9 (reduction)**.** *Take a geometric labelling, with any pattern and inner classes.*

- *(a) If q is not a power of 3, every hub class sits on one type in all copies it meets, all unbarred classes sit on the same type, and all barred classes on the other: the inner classes are those of reading 2 or of its dual.*
- *(b) If q is a power of 3, the graph is isomorphic to one with a geometric labelling in reading 2 (same pattern).*
- *(c) In reading 2 the graph is isomorphic to one with $P_c = P_0$ for all c and $L_c = \lambda_c L_0$ with collineations λ_c (the* normal form*); in the dual of reading 2 likewise, with the unbarred hubs on lines. We call the bar that sits on points the* point bar $\mathfrak{P}$ *and the other the* line bar $\mathfrak{L}$*.*

Proof. (a) A transfer from a copy where Y sits on points to one where it sits on lines maps $\mathcal{P}$ to $\mathcal{L}$; an automorphism of $I(q)$ doing that is a duality, which does not exist (Fact 2.3). In every copy the two unbarred classes sit on the same type. The copies are linked by unbarred classes: inside a block the pairs $\{k \pm 1\}$ (type a) or $\{k + 3, k + 4\}$ (type b), $k \in \mathbb{Z}_5$, form a 5-cycle on the positions, and every class meets every block. So all unbarred classes sit on one type, and barred classes on the other type in every copy. (b) Fix a duality δ and relabel by δ every copy in which the unbarred classes sit on lines (Lemma 7.8). The new transfers are the old ones composed with

$\delta^{\pm 1}$, so they are still automorphisms, and now all unbarred classes sit on points. (c) In reading 2 the transfers of unbarred classes are restrictions of collineations to $\mathcal{P}$, and the unbarred classes link all copies; fixing a copy c_0, every $P_c P_{c_0}^{-1}$ is a product of transfers, so $P_c = a_c P_{c_0}$ with a collineation a_c. In the same way (the barred classes also link all copies) $L_c = b_c L_{c_0}$. Relabel copy c by a_c^{-1}: P_c becomes P_{c_0} and L_c becomes $a_c^{-1} b_c L_{c_0}$. Unbarred rows and barred rows are different vertices, so renaming the unbarred rows by $P_0^{-1} P_{c_0}$ and the barred rows by $L_0^{-1} L_{c_0}$ gives $P_c = P_0$ and $L_c = \lambda_c L_0$ with $\lambda_c = a_c^{-1} b_c$. The dual reading is the same with the bars exchanged. □

From now on we use the normal form, with the rows of the point bar identified with points by P_0 (so the hub (t, X), $X \in \mathfrak{P}$, is adjacent to point t of every copy in $C(X)$), and collineations λ_c acting on $\mathcal{P}$ and on $\mathcal{L}$. Write $B(z, \rho)$ for the ball of radius ρ about z in $I(q)$.

Lemma 7.10 (opposite elements in balls)**.** *Let t, z be vertices of $I(q)$ of the same type, $d = d(t, z)$. Then $|B(z, 4) \cap \mathrm{opp}(t)|$ is 0 if $z = t$ and q^4 if $d \in \{2, 4, 6\}$; and $|B(z, 2) \cap \mathrm{opp}(t)|$ is 0 if $d \in \{0, 2\}$ and q^2 if $d \in \{4, 6\}$.*

Proof. A non-backtracking walk of length $i \le 6$ in $I(q)$ is a geodesic, and for $i \le 5$ each vertex of $\Gamma_i(z)$ is the end of exactly one such walk from z (Lemma 2.4(a)). The vertices of $B(z, 4)$ of the type of t are at even distance $i \in \{0, 2, 4\}$ from z; we follow $d(t, h_j)$ along the walk $z = h_0, h_1, \ldots, h_i$ with Lemma 2.4(a): from h_j at distance $k \in [1, 5]$ from t, if h_{j-1} is the neighbour at distance $k - 1$ all q choices of h_{j+1} go up to $k + 1$; if h_{j-1} is at distance $k + 1$, one choice goes down to $k - 1$ and $q - 1$ go up; from distance 6 all choices go down to 5.

- $d = 0$: $B(t, 4)$ misses $\mathrm{opp}(t)$.
- $d = 2$: from distance 2, four steps reach distance 6 only if every step goes up: $q \cdot q^3$; none of $B(z, 2)$.
- $d = 4$: $i = 2$: both steps up, q^2. $i = 4$: if h_1 goes down (one way), h_2 must go up ($q - 1$ ways) and then h_3, h_4 up (q^2 ways): $(q-1)q^2$. If h_1 goes up (q ways, distance 5), h_2 goes up to 6 (q ways), h_3 goes down to 5 (q ways), and $h_4 \ne h_2$ must go up ($q - 1$ ways): $q^3(q - 1)$. Total $q^2 + (q-1)q^2 + (q-1)q^3 = q^4$; and $B(z, 2)$ gives q^2.
- $d = 6$: $i = 0$ gives 1; $i = 2$: $(q + 1)(q - 1)$; $i = 4$: $(q + 1)(q - 1)q + (q + 1)(q - 1)q(q - 1) = (q + 1)(q - 1)q^2$. Total $1 + (q^2 - 1) + (q^4 - q^2) = q^4$, and $B(z, 2)$ gives q^2.

□

For $X \in \mathfrak{P}$ and $c' \notin C(X)$, let $N(X, c')$ be the set of copies $c_1 \in C(X)$ such that some class of $\mathfrak{L}$ meets both c_1 and c'.

Lemma 7.11 (routes reach balls)**.** *Let $X \in \mathfrak{P}$, $c' \notin C(X)$ and $t \in \mathcal{P}$. Every point v of copy c' with $d((t, X), (c', v)) \le 7$ lies in*

$$B(t, 4) \;\cup \bigcup_{c_1 \in N(X, c')} B(\lambda_{c'} \lambda_{c_1}^{-1}(t), 4) \;\cup \bigcup_{(c,d) \in W_2} B(\lambda_c \lambda_d^{-1}(t), 2),$$

where W_2 is the set of ordered pairs of distinct copies of the following three kinds: (c', c_2) with c_2 sharing a class of $\mathfrak{P}$ with some $c_1 \in C(X)$ and a class of $\mathfrak{L}$ with c'; (c', c_1) with $c_1 \in C(X)$ and some c_2 sharing a class of $\mathfrak{L}$ with c_1 and with c'; (c_2, c_1) with $c_1 \in C(X)$ and c_2 sharing a class of $\mathfrak{L}$ with c_1 and a class of $\mathfrak{P}$ with c'.

Proof. A path from $h = (t, X)$ to (c', v) goes to point t of a copy $c_1 \in C(X)$, then alternates walks inside copies (of lengths $a_1, \ldots, a_r \ge 0$) with $r - 1$ passes copy → hub → copy, ending in c'. A hub meets each copy at most once, so consecutive copies differ; $c_1 \ne c'$, so $r \ge 2$. The length is $1 + \sum a_i + 2(r - 1) \le 7$, so $\sum a_i \le 8 - 2r$ and $r \le 4$. A pass through a $\mathfrak{P}$-hub happens at a point and keeps it. A pass through an $\mathfrak{L}$-hub (m, Y) happens at a line and goes from $\lambda_c L_0(m)$ in c to $\lambda_d L_0(m)$ in d: it applies $\lambda_d \lambda_c^{-1}$. Collineations preserve distances, so by induction the end lies in

$B(g(t), \sum a_i)$, g the product of the maps applied. The walk starts and ends at points; a segment before an $\mathfrak{L}$-pass ends at a line, so the segments around $\mathfrak{L}$-passes are odd and $\sum a_i$ is even.

- $r = 2$ ($\sum a_i \leq 4$): a $\mathfrak{P}$-pass gives $g = \mathrm{id}$; an $\mathfrak{L}$-pass needs a class of $\mathfrak{L}$ meeting c_1 and c', i.e. $c_1 \in N(X, c')$, and gives $g = \lambda_{c'}\lambda_{c_1}^{-1}$.
- $r = 3$ ($\sum a_i \leq 2$): $\mathfrak{P}$ then $\mathfrak{P}$ gives id. $\mathfrak{P}$ then $\mathfrak{L}$ forces $a_1 = 0$, $a_2 = a_3 = 1$ and $g = \lambda_{c'}\lambda_{c_2}^{-1}$ (first kind). $\mathfrak{L}$ then $\mathfrak{P}$ forces $a_1 = a_2 = 1$, $a_3 = 0$ and $g = \lambda_{c_2}\lambda_{c_1}^{-1}$ (third kind). $\mathfrak{L}$ then $\mathfrak{L}$ forces $a_1 = a_3 = 1$, $a_2 = 0$: both passes use the same row, so $g = \lambda_{c'}\lambda_{c_1}^{-1}$ (second kind).
- $r = 4$: all $a_i = 0$, so the walk never reaches a line, all passes are $\mathfrak{P}$-passes, and $v = t$.

□

Relabelling every copy c by λ_c^{-1} gives $L_c = L_0$ for all c and $P_c = \lambda_c^{-1}P_0$, so the same statement holds with points and lines exchanged and λ_c replaced by λ_c^{-1}, for $X \in \mathfrak{L}$. For a pair (X, c'), X in either bar, let $r_4 = |N(X, c')|$, and let m_2 be the number of pairs in W_2 that are not of the form (c', c_1) with $c_1 \in N(X, c')$.

Corollary 7.12. *Every hub of X has at least $q^5 - r_4q^4 - m_2q^2$ vertices of c' at distance at least 9 (points if $X \in \mathfrak{P}$, lines if $X \in \mathfrak{L}$). If $r_4 = \alpha$, then $m_2 \leq (4\alpha - 1) + 2\alpha \cdot 2(2\alpha - 1) = 8\alpha^2 - 1$.*

Proof. Take $X \in \mathfrak{P}$. By Lemma 7.10, $B(t, 4)$ misses $\mathrm{opp}(t)$, each other ball of radius 4 meets it in at most q^4 points and each ball of radius 2 in at most q^2. The remaining points of $\mathrm{opp}(t)$ in c' are at distance more than 7 from (t, X), and at odd distance (all points of all copies have one colour), so at distance at least 9. For m_2: the first two kinds are pairs (c', b) with $b \neq c'$, at most $5\alpha - 1 - r_4 = 4\alpha - 1$ of them outside the radius-4 words; the third kind has at most 2α choices of c_1 and, since c_1 is met by two classes of $\mathfrak{L}$ that meet 2α copies each, at most $2(2\alpha - 1)$ choices of c_2. □

Call (X, c'), with X in either bar and $c' \notin C(X)$, *starved* if $r_4 = \alpha$.

Lemma 7.13 (starved pairs)**.** *For every block pattern there are starved pairs with X unbarred and starved pairs with X barred: for $X = k$ unbarred, $c' = 5B' + k + 3$ for a block B' of type a (if all blocks are of type b, $c' = 5B' + k$); for $X = \bar{k}$ barred, $c' = 5B' + k + 3$ for a block B' of type b (if all blocks are of type a, $c' = 5B' + k + 1$).*

Proof. The classes meeting the copy at position i are: in an a-block, unbarred $i \pm 1$ and barred $i + 2, i + 3$; in a b-block, unbarred $i + 1, i + 2$ and barred $i \pm 1$. For $X = k$ (so $\mathfrak{L}$ is barred), c' at position $k + 3$ of an a-block has barred classes $\{k, k + 1\}$. In an a-block, $C(k)$ has positions $k - 1$ (barred $\{k + 1, k + 2\}$: shares a class with c') and $k + 1$ (barred $\{k + 3, k + 4\}$: does not). In a b-block, $C(k)$ has positions $k + 3$ (barred $\{k + 2, k + 4\}$: no) and $k + 4$ (barred $\{k, k + 3\}$: shares). If all blocks are b, c' at position k has barred $\{k - 1, k + 1\}$; $C(k)$ has $k + 3$ (barred $\{k + 2, k + 4\}$: shares $k + 4 = k - 1$) and $k + 4$ (barred $\{k, k + 3\}$: no). For $X = \bar{k}$ ($\mathfrak{L}$ unbarred), c' at position $k + 3$ of a b-block has unbarred $\{k + 4, k\}$. In an a-block $C(\bar{k})$ has $k + 2$ (unbarred $\{k + 1, k + 3\}$: no) and $k + 3$ (unbarred $\{k + 2, k + 4\}$: shares); in a b-block, $k - 1$ (unbarred $\{k, k + 1\}$: shares) and $k + 1$ (unbarred $\{k + 2, k + 3\}$: no). If all blocks are a, c' at position $k + 1$ has unbarred $\{k, k + 2\}$; $C(\bar{k})$ has $k + 2$ (unbarred $\{k + 1, k + 3\}$: no) and $k + 3$ (unbarred $\{k + 2, k + 4\}$: shares). In each case exactly one copy of $C(X)$ per block shares a class of the other bar with c', and $c' \notin C(X)$. □

Fact 7.14 (computer-checked: route words)**.** For $\alpha = 5, 6, 7, 8$ and every block pattern, the routes of Lemma 7.11 were enumerated for every pair (X, c'): every pair has $r_4 \geq \alpha$; for $\alpha = 5$ ($q = 7, 8$) every starved pair, of either bar and in every pattern, has $m_2 \leq 140$; for $\alpha = 6$ ($q = 9$) every starved pair has $m_2 \leq 201$; and in every pattern some starved pair has $m_2 \leq 134$ ($\alpha = 5$), 197 ($\alpha = 6$), 262 ($\alpha = 7$), 347 ($\alpha = 8$). This was checked by two programs written separately.

Fact 7.14 is a finite enumeration over block patterns and wiring words; it does not depend on q. Should we replace it by a hand count, or include the enumeration as an appendix table?

Theorem 7.15 (geometric labellings, $q \geq 8$)**.** *Let $q \geq 8$. For every block pattern, every choice of inner classes and every geometric labelling, $\Omega\mathrm{H}_q$ has diameter at least 9. More precisely, in the normal form, for every starved pair (X, c') every hub (t, X) has at least $q^5 - \alpha q^4 - m_2 q^2 > 0$ vertices of c' at distance at least 9. For $q \geq 10$ the proof is by hand; for $q = 8, 9$ it uses Fact 7.14.*

Proof. By Lemma 7.9 we may assume the normal form, in reading 2 or its dual. Take a starved pair (Lemma 7.13) and apply Corollary 7.12 with $r_4 = \alpha$. For $q \geq 10$, $m_2 \leq 8\alpha^2 - 1$ and the bound is positive when $q^2(q-\alpha) \geq 8\alpha^2$; with $\alpha \leq (q+3)/2$ it suffices that $q^2(q-3) \geq 4(q+3)^2$, i.e. $q^3 - 7q^2 - 24q - 36 \geq 0$, which holds at $q = 10$ (value 24) and for larger q. For $q = 8$ ($\alpha = 5$), Fact 7.14 gives $m_2 \leq 140$ for every starved pair, so the bound is at least $32\,768 - 20\,480 - 8960 = 3328$; for $q = 9$ ($\alpha = 6$), $m_2 \leq 201$ and the bound is at least $59\,049 - 39\,366 - 16\,281 = 3402$. □

The case $q = 7$. At $q = 7$ the bound of Corollary 7.12 is negative ($16\,807 - 5 \cdot 2401 - 140 \cdot 49 = -2058$), so the overlaps of the five radius-4 balls must be used. Here q is odd and 7 is not a power of 3, so the collineations are elements of $G_2(7) \leq \mathrm{SO}(7,7)$ and the reading is reading 2 or its dual. Fix a starved pair (X, c') of the point bar, write $g_1, \ldots, g_5$ for the elements $\lambda_{c'}\lambda_{c_i}^{-1}$, $c_i \in N(X, c')$, and $y_i = g_i(t)$. By Lemma 2.4(c), $B(y, 4) \cap \mathcal{P} = y^\perp \cap Q(6, q)$ for a point y, so with $C_i := y_i^\perp \cap \mathrm{opp}(t)$ the points of c' within distance 7 of (t, X) that are opposite t lie in $C_1 \cup \cdots \cup C_5$ and in at most $m_2 \leq 140$ balls of radius 2.

Fact 7.16 (eigenvalues in G_2)**.** For $h \in G_2(q)$ acting on V, the eigenvalues of h (in the algebraic closure, with multiplicity) are $1, x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}$ with $x_1 x_2 x_3 = 1$. (The semisimple part of h is conjugate over $\bar{\mathbb{F}}_q$ into the maximal torus $\mathrm{diag}(a, b, c, 1, a^{-1}, b^{-1}, c^{-1})$, $abc = 1$, of the subgroup $\mathrm{SL}(3)$ of Fact 2.3.)

Lemma 7.17 (overlaps, q odd)**.** *Let y, y', t be distinct points of $Q(6, q)$.*

(a) If $y \perp y'$ and t lies on the line yy', then $y^\perp \cap y'^\perp \cap \mathrm{opp}(t) = \emptyset$. Otherwise $|y^\perp \cap y'^\perp \cap \mathrm{opp}(t)| \geq q^3 - q$.

(b) If y, y', y'' are points not on one line, $|y^\perp \cap y'^\perp \cap y''^\perp \cap Q(6, q)| \leq (q+1)^2$.

(c) If y, y', y'' lie on one totally singular line ℓ not through t, all the pairwise and triple intersections of $y^\perp \cap \mathrm{opp}(t)$, $y'^\perp \cap \mathrm{opp}(t)$, $y''^\perp \cap \mathrm{opp}(t)$ equal $E_\ell := \ell^\perp \cap \mathrm{opp}(t)$, and $|E_\ell| \geq q^3 - q$.

Proof. If $y \perp y'$, the line yy' is totally singular; if t is on it, $y^\perp \cap y'^\perp = (yy')^\perp \subseteq t^\perp$. If y, y' are not perpendicular, the line yy' meets $Q(6, q)$ only in y, y', so t is not on it. In both remaining cases $y^\perp \cap y'^\perp$ is the perp of a line, a 5-dimensional space: a parabolic $Q(4, q)$ (if $y \not\perp y'$) or a cone with vertex the line yy' over a conic (if $y \perp y'$); both have $q^3 + q^2 + q + 1$ points. Its points not opposite t are those of $\Pi^\perp$, $\Pi = \langle y, y', t\rangle$ a plane. For three independent singular vectors spanning Π the Gram matrix of b has zero diagonal, so its rank is 3, 2 or 0 (a symmetric matrix with zero diagonal and rank 1 would be vv^T with all $v_i^2 = 0$). The radical of $\Pi^\perp$ is $\Pi \cap \Pi^\perp$, of dimension 0, 1 or 3. In the first case $\Pi^\perp$ (dimension 4) carries a non-degenerate form with $(q+1)^2$ or q^2+1 singular points; in the second it is a cone over a conic, with $q^2 + q + 1$ points; in the third Π is totally singular and $\Pi^\perp = \Pi \oplus \langle w\rangle$ with $Q(w) \neq 0$ ($Q(6, q)$ has no totally singular 4-space), so its singular points are those of Π, $q^2 + q + 1$. So $|\Pi^\perp \cap Q(6, q)| \leq (q+1)^2$, which gives (a) and, with $\Pi = \langle y, y', y''\rangle$, (b). For (c), $y^\perp \cap y'^\perp = \ell^\perp \subseteq y''^\perp$, and $|E_\ell| \geq q^3 - q$ by (a). □

Lemma 7.18 (coincidences)**.** *If two of $t, y_1, \ldots, y_5$ coincide, the hub (t, X) has at least $q^5 - 4q^4 - 140q^2 = 343$ vertices of c' at distance at least 9 (using $m_2 \leq 140$ from Fact 7.14).*

Proof. At most four distinct centres other than t remain; apply Corollary 7.12. □

Now let $t, y_1, \dots, y_5$ be distinct. Call y_i, y_j *aligned* if they lie on one totally singular line through t (then $C_i \cap C_j = \emptyset$ by Lemma 7.17(a)); the aligned classes partition $\{y_1, \dots, y_5\}$. Call a set of at least three y_i lying on one line of $\mathrm{PG}(6,q)$ not through t, and maximal with this property, a *rich line*. Three collinear singular points span a totally singular line, so every collinear triple of y's is inside an aligned class or a rich line; a rich line meets each aligned class at most once, and two rich lines share at most one point.

Lemma 7.19 (configuration types at $q = 7$)**.** *Let A_2 be the number of aligned pairs, T_3 the number of triples meeting three different aligned classes, and $R_3 = \sum \binom{m}{3}$ over the rich lines (m points each). Then*

$$|C_1 \cup \cdots \cup C_5| \le U := 5q^4 - (10 - A_2 - R_3)(q^3 - q) + (T_3 - R_3)(q+1)^2,$$

except when one rich line holds all five points, where $U := 5q^4 - 4(q^3 - q)$. Classified by the sizes of the aligned classes and of the rich lines there are exactly 17 kinds of configuration. Table 7 lists them for $q = 7$ with $F = 16\,807 - U - 6860$ and the number p of pairs among $\{t, y_1, \dots, y_5\}$ that every configuration of the kind has perpendicular: $p = \sum(s + \binom{s}{2})$ over aligned classes of size $s \ge 2$, plus $\sum \binom{m}{2}$ over rich lines. In the 9 kinds with $F \le 0$, $p \ge 6$.

TABLE 7. The configuration types of Lemma 7.19 at $q = 7$.

aligned classes	rich lines	A_2	T_3	R_3	U	F	p
5	–	10	0	0	12 005	−2058	15
4, 1	–	6	0	0	10 661	−714	10
3, 2	–	4	0	0	9989	−42	9
3, 1, 1	–	3	3	0	9845	102	6
3, 1, 1	3	3	3	1	10 117	−170	9
2, 2, 1	–	2	4	0	9573	374	6
2, 2, 1	3	2	4	1	9845	102	9
2, 2, 1	3, 3	2	4	2	10 117	−170	12
2, 1, 1, 1	–	1	7	0	9429	518	3
2, 1, 1, 1	3	1	7	1	9701	246	6
2, 1, 1, 1	3, 3	1	7	2	9973	−26	9
2, 1, 1, 1	4	1	7	4	10 517	−570	9
1, 1, 1, 1, 1	–	0	10	0	9285	662	0
1, 1, 1, 1, 1	3	0	10	1	9557	390	3
1, 1, 1, 1, 1	3, 3	0	10	2	9829	118	6
1, 1, 1, 1, 1	4	0	10	4	10 373	−426	6
1, 1, 1, 1, 1	5	0	10	10	10 661	−714	10

Proof. $|C_i| = q^4$ (Lemma 7.10, $y_i \ne t$). Bonferroni: $|\bigcup C_i| \le S_1 - S_2 + S_3$, S_k the sum of the k-fold intersections. By Lemma 7.17: an aligned pair contributes 0 to S_2, and every triple containing one contributes 0 to S_3; a pair inside a rich line ℓ contributes $|E_\ell|$, and so does a triple inside it; every other pair contributes at least $q^3 - q$, and every other triple (it is not collinear) at most $(q+1)^2$. A rich line with $m \in \{3, 4\}$ points contributes $-(\binom{m}{2} - \binom{m}{3})|E_\ell| = -2|E_\ell| \le -2(q^3 - q)$. Collecting terms gives U. If all five points lie on one rich line, all pairwise intersections equal E_ℓ, so $|\bigcup C_i| = 5q^4 - 4|E_\ell|$ exactly. The kinds: two 3-subsets of a 5-set sharing at most one point share exactly one, a third 3-subset would share two points with one of them, and a 4- or 5-point rich line leaves no room for another; so the rich-line systems are none, one 3-line, two 3-lines

through a common point, one 4-line, one 5-line. A rich line takes at most one point from each aligned class, which leaves exactly the 17 rows (partitions 5, 41, 32 admit no rich line; 311: none or one 3-line; 221: none, one, or two 3-lines through the singleton; 2111: none, 3, 3+3, 4; 11111: none, 3, 3+3, 4, 5). The numbers A_2, T_3, R_3 and p depend only on the row. □

Lemma 7.20 (few points are moved to perpendicular points)**.** *Let q be odd with $3 \mid q-1$ and let $h \in G_2(q)$ fix no point of $Q(6,q)$. Then* $\mathrm{Perp}(h) := \{s \in \mathcal{P} : h(s) \perp s\}$ *has at most $(q^5-1)/(q-1) + q^{3.5} + q^3 + (q-2)q^{1.5}$ points; at $q=7$, at most 4144 of the 19 608 points of $H(7)$.*

Proof. Count $N_h := \#\{v \in V : Q(v) = 0,\ F(v) = 0\}$ with $F(v) := b(v, hv)$; then $|\mathrm{Perp}(h)| = (N_h - 1)/(q-1)$. F is a quadratic form with polar form $b(v,(h+h^{-1})w)$, since $b(w,hv) = b(h^{-1}w, v)$. With ψ a non-trivial additive character of $\mathbb{F}$, $N_h = q^{-2}\sum_{\lambda,\mu\in\mathbb{F}}\sum_{v\in V}\psi(\lambda Q(v) + \mu F(v))$. The terms with $\mu = 0$ give $q^{-2}\cdot q\cdot q^6 = q^5$ (Q has q^6 zeros). For $\mu \neq 0$, $\lambda Q + \mu F = \mu(F - \gamma Q)$ with $\gamma = -\lambda/\mu$, and for fixed μ, γ runs over $\mathbb{F}$. The form $R_\gamma = F - \gamma Q$ has polar form $b(v,(h+h^{-1}-\gamma)w)$, so its rank is $7 - d_\gamma$ with $d_\gamma = \dim\ker(h^2 - \gamma h + 1)$. For a quadratic form of rank r on $\mathbb{F}^7$ (q odd), $|\sum_v \psi(R(v))| = q^{7-r/2}$. Hence $|N_h - q^5| \le (q-1)\sum_\gamma q^{1.5+d_\gamma/2}$ and $|\mathrm{Perp}(h)| \le (q^5-1)/(q-1) + q^{1.5}\sum_\gamma q^{d_\gamma/2}$.

For $\gamma \neq \gamma'$ the polynomials $x^2 - \gamma x + 1$ and $x^2 - \gamma' x + 1$ are coprime, so the kernels are independent and $\sum_\gamma d_\gamma \le 7$. Each $d_\gamma \le 4$: (i) if $x^2 - \gamma x + 1 = (x-\nu)(x-\nu^{-1})$ with $\nu \in \mathbb{F}$, $\nu \neq \pm 1$, the kernel is $\ker(h-\nu)\oplus\ker(h-\nu^{-1})$, and an eigenvector v with $hv = \nu v$ has $Q(v) = \nu^2 Q(v)$, so $Q(v) = 0$ and $\langle v\rangle$ is a fixed point; so $d_\gamma = 0$. (ii) $\gamma = \pm 2$: $\ker(h \mp 1)$ contains no singular vector, so it has dimension at most 2 (a quadratic form in at least 3 variables over $\mathbb{F}$ has a non-trivial zero), and $\dim\ker(h\mp 1)^2 \le 4$. (iii) If $x^2 - \gamma x + 1$ is irreducible with roots $\nu, \nu^q = \nu^{-1}$ in $\mathbb{F}_{q^2}$, then $d_\gamma = 2\dim\ker(h-\nu) \le 2m(\nu)$, $m(\nu)$ the multiplicity of ν as an eigenvalue. By Fact 7.16, since $\nu \neq \nu^{-1}$, ν occurs at most once for each index, so $m(\nu) \ge 3$ means $\nu = x_1^{\varepsilon_1} = x_2^{\varepsilon_2} = x_3^{\varepsilon_3}$, hence $\nu^{\varepsilon_1+\varepsilon_2+\varepsilon_3} = 1$ with $\varepsilon_1+\varepsilon_2+\varepsilon_3 \in \{\pm 1, \pm 3\}$; $\nu \neq 1$ forces $\nu^3 = 1$, and since $3 \mid q-1$ the cube roots of unity lie in $\mathbb{F}$, contradicting irreducibility. So $m(\nu) \le 2$ and $d_\gamma \le 4$. The function $d \mapsto q^{d/2}$ is increasing and convex; for integers $0 \le d_\gamma \le 4$ with $\sum d_\gamma \le 7$ the maximum of $\sum q^{d_\gamma/2}$ is at $(4,3,0,\ldots,0)$: $q^2 + q^{1.5} + (q-2)$. At $q=7$: $2801 + 907.5 + 343 + 92.6 = 4144.1$. □

Theorem 7.21 (geometric labellings, $q = 7$)**.** *For $q = 7$, every block pattern, every choice of inner classes and every geometric labelling, $\Omega\mathrm{H}_7$ has diameter at least 9. The proof uses Fact 7.14 ($m_2 \le 140$).*

Proof. By Lemma 7.9(a),(c) we have reading 2 or its dual in normal form; take a starved pair (X, c') of the point bar and suppose that every hub of X is within distance 8 of every point of c'. Consider the 15 elements g_i ($1 \le i \le 5$) and $g_i^{-1}g_j$ ($i < j$) of $G_2(7)$. If one of them fixes a point t, then two of $t, y_1(t), \ldots, y_5(t)$ coincide, and Lemma 7.18 gives a far vertex. So none of them has a fixed point. Then for every point t the six points $t, y_1, \ldots, y_5$ are distinct, and by Lemma 7.19 and Corollary 7.12 the configuration at t is of one of the 9 kinds with $F \le 0$, so at least 6 of the 15 pairs among $t, y_1, \ldots, y_5$ are perpendicular. The pair (t, y_i) is perpendicular if and only if $t \in \mathrm{Perp}(g_i)$, and (y_i, y_j) if and only if $t \in \mathrm{Perp}(g_i^{-1}g_j)$. Summing over the 19 608 points t: $\sum_h |\mathrm{Perp}(h)| \ge 6 \cdot 19\,608 = 117\,648$ over the 15 elements h, while Lemma 7.20 gives $\sum_h |\mathrm{Perp}(h)| \le 15 \cdot 4144 = 62\,160$. This is a contradiction. □

Corollary 7.22. *For every table value of $\Omega\mathrm{H}_q$, $(10,8)$, $(11,8)$, $(12,8)$, $(14,8)$, $(15,8)$ and $(16,8)$, no geometric labelling of the copies gives diameter 8, under any reading of Figs. 2.21a/b, any choice of inner classes and any mix of block types. For $(14,8)$, $(15,8)$ and $(16,8)$ the proof is entirely by hand. For $q = 7, 8, 11, 13$ (not powers of 3), reading 1 with both block types has no*

geometric labelling at all (Lemma 7.9(a)), so there the thesis labelling is covered by Theorem 7.5 only.

7.5. **What remains open.**

(1) *Arbitrary labellings* ($q \geq 4$). Whether some non-geometric labelling of Gómez's wiring gives diameter 8 at $q = 7, 8, 9, 11, 13$ is open. Counting does not exclude it. In every graph where this was checked by computer, all pairs at distance more than 8 were pairs (hub, vertex of a copy the hub misses). With labellings found by computer, ΩH_2 (1890 vertices) and ΩH_3 (14 560 vertices, 6-regular) have diameter 8 (computer-checked); these are far below the table at $(5, 8)$ and $(6, 8)$, and we know no description of their labellings. No labelling with diameter 8 was found at $q = 4$ or $q = 5$.

(2) *Labellings per (hub class, copy).* If the two classes of one bar meeting a copy may use different bijections, the model of Section 7.4 does not apply, even when every transfer is a collineation, and the counting method fails for every q.

(3) *Other wirings* of the copies.

(4) *The geometric class at $q \leq 5$* (not table values).

What can be claimed. The six table entries are the orders of Gómez's graphs ΩH_q; the thesis claims diameter at most 8 without proof; as defined there these graphs have diameter at least 9 for each of these q, under every reading of the figures and every pairing of point labels with line labels; geometric relabellings do not repair them. So these cells have no checkable graph of diameter 8. We do not claim that no graph of these orders with maximum degree $q + 3$ and diameter 8 exists. The $(13, 8)$ entry $5\,314\,680 = 15 \cdot |H(11)|$, also credited to [17], is not in the thesis, and nothing here bears on it; the thesis also lists ΩH_{11}d with $N(15, 8) \geq 14\,978\,080$, which is not defined in §§2.7–2.9 and which the table does not use.

The thesis value ΩH_5 at $(8, 8)$ (195 300) is below the 2013 value 201 480, so it is not a table entry; $(8, 8)$ is improved by a Haar graph in Section 8. Our own graphs at $(11, 8)$ and $(15, 8)$ (Table 8) beat the table values whatever their status; at $(10, 8)$, $(12, 8)$, $(14, 8)$, $(16, 8)$ we have no graph above the table values.

8. Haar graphs: diameter 5 and diameters 8 to 10

This section is group-theoretic, not geometric. It gives the criterion that makes the diameter of a Haar graph a finite covering statement, decidable by two breadth-first searches, and lists Haar graphs that improve the 2013 table at $D = 5$ and $D = 8, 9, 10$. Each of these is a single explicit graph; none comes from an infinite family, and their diameters are computer-checked (Appendix C).

8.1. **Definition and the covering criterion.**

Definition 8.1. Let G be a finite group with identity e and $S \subseteq G$ with $|S| = \Delta$. The *Haar graph* $H(G, S)$ has vertex set $G \times \{A, B\}$, and (g, A) is adjacent to (sg, B) for every $g \in G$ and $s \in S$. So

$$N(g, A) = \{(sg, B) : s \in S\}, \qquad N(h, B) = \{(s^{-1}h, A) : s \in S\}.$$

The name is from Hladnik, Marušič and Pisanski [29] (cyclic groups; Estélyi and Pisanski [20] for arbitrary groups). $H(G, S)$ is the regular G-cover of the dipole with Δ parallel edges carrying the voltages $s \in S$; voltage lifts of small base graphs are the standard tool behind many entries of the bipartite table (Loz and Širáň [30]; [31, §3.4.5]). Joining (g, A) to (gs, B) instead gives an isomorphic graph (via $(g, X) \mapsto (g^{-1}, X)$ and $S \mapsto S^{-1}$). Calling the A-vertices points and the B-vertices lines, $H(G, S)$ is the incidence graph of the development of S^{-1}: the line (h, B) is incident with the points $(s^{-1}h, A)$, $s \in S$. For example $H(\mathbb{Z}_7, \{0, 1, 3\})$ is the incidence graph of the Fano plane.

Lemma 8.2. *Let $|S| = \Delta \geq 1$. $H(G, S)$ is bipartite, simple and Δ-regular on $2|G|$ vertices; each right translation $\rho_h(g, X) = (gh, X)$ is an automorphism, and these act regularly on each side.*

Proof. Edges join the two sides. The neighbours (sg, B), $s \in S$, of (g, A) are distinct, and so are the $(s^{-1}h, A)$. If $(g, A) \sim (sg, B)$ then $(gh, A) \sim (sgh, B)$, and $\rho_h(e, X) = (h, X)$. □

For $X, Y \subseteq G$ write $XY = \{xy : x \in X, y \in Y\}$, $X^{-1} = \{x^{-1} : x \in X\}$, $X^0 = \{e\}$ and $X^j = X \cdot X^{j-1}$.

Proposition 8.3 (distances from the two base vertices)**.** *Let $S \neq \emptyset$, $x \in G$ and $j \geq 0$. Then $d((e, A), (x, A)) \leq 2j$ if and only if $x \in (S^{-1}S)^j$; $d((e, A), (x, B)) \leq 2j + 1$ if and only if $x \in S(S^{-1}S)^j$; $d((e, B), (x, B)) \leq 2j$ if and only if $x \in (SS^{-1})^j$; and $d((e, B), (x, A)) \leq 2j + 1$ if and only if $x \in S^{-1}(SS^{-1})^j$.*

Proof. A walk of length l from (e, A) passes through $(g_0, A), (g_1, B), (g_2, A), \ldots$ with $g_0 = e$, $g_i = t_i g_{i-1}$ for odd i and $g_i = t_i^{-1} g_{i-1}$ for even i, where $t_i \in S$ is the letter used at step i; every choice of letters gives a walk. So the ends of the walks of length exactly $2j$ are the (x, A) with $x \in (S^{-1}S)^j$, and those of length exactly $2j + 1$ are the (x, B) with $x \in S(S^{-1}S)^j$. A shortest path followed by back-and-forth steps along one edge is a walk of any larger length of the same parity. From (e, B) the same argument gives the last two statements. □

Theorem 8.4 (covering criterion)**.** *Let $S \neq \emptyset$ and $D \geq 2$.*

(a) If $D = 2\kappa$, then $\operatorname{diam} H(G, S) \leq D$ if and only if $S(S^{-1}S)^{\kappa-1} = G$.

(b) If $D = 2\kappa + 1$, then $\operatorname{diam} H(G, S) \leq D$ if and only if $(S^{-1}S)^\kappa = G$ and $(SS^{-1})^\kappa = G$.

Moreover $\operatorname{diam} H(G, S) = \max(\operatorname{ecc}(e, A), \operatorname{ecc}(e, B))$.

Proof. The last statement holds because the right translations are automorphisms acting transitively on each side (Lemma 8.2): every vertex is mapped to (e, A) or (e, B) by an automorphism. (a) If diam $\leq 2\kappa$, then for $x \in G$ the distance $d((e, A), (x, B))$ is odd and at most 2κ, so at most $2(\kappa - 1) + 1$, and $x \in S(S^{-1}S)^{\kappa-1}$ by Proposition 8.3. Conversely, if $S(S^{-1}S)^{\kappa-1} = G$ then $d((g, A), (h, B)) = d((e, A), (hg^{-1}, B)) \leq 2\kappa - 1$ for all g, h. Fix $s \in S$. Since $(g', A) \sim (sg', B)$, $d((g, A), (g', A)) \leq 2\kappa$; since $(h', B) \sim (s^{-1}h', A)$, $d((h, B), (h', B)) \leq 2\kappa$. (b) If diam $\leq 2\kappa + 1$, same-side distances are even, hence at most 2κ, so both products are G. Conversely, if both hold, all same-side distances are at most 2κ, and $d((g, A), (h, B)) \leq 1 + d((sg, B), (h, B)) \leq 2\kappa + 1$. □

At $D = 5$ ($\kappa = 2$) the criterion reads: every element of G is a product $a^{-1}bc^{-1}d$ and also a product $ab^{-1}cd^{-1}$ with $a, b, c, d \in S$. A shortest path never backtracks, so the same count gives the per-side Moore bound $|G| \leq \sum_{j<D}(\Delta - 1)^j$ (that is, $2|G| \leq M_b(\Delta, D)$). For $\Delta \geq 3$ and $D \in \{8, 9, 10\}$ it is strict: with equality the reduced words of the binding parity and length at most $D - 1$ have distinct values, which forces girth at least $2D$, and a bipartite graph of diameter D, girth $2D$ and all degrees $\Delta \geq 3$ is the incidence graph of a thick generalized D-gon of order $(\Delta - 1, \Delta - 1)$, which does not exist for these D [23].

8.2. **Diameter 5.** The covering condition at $D = 5$ is a finite check: for each graph below, a representation $a^{-1}bc^{-1}d$ and $ab^{-1}cd^{-1}$ of every group element is computer-checked (and kernel-checked in Lean, Appendix B).

- $(7, 5)$, 1152 vertices: $G = S_4 \times \mathbb{Z}_{24}$ (order 576), with $(f, x)(h, y) = (f \circ h, x + y)$, and
  $S = \{(\mathrm{id}, 0), ((0\,1)(2\,3), 23), ((0\,3\,2), 0), ((0\,2), 18), ((0\,1\,2\,3), 18), ((1\,2\,3), 17), ((1\,2), 3)\}$,
  permutations of $\{0, 1, 2, 3\}$ in cycle notation.
- $(9, 5)$, 2662 vertices: G the Heisenberg group modulo 11, $(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab')$, and
  $S = \{(0, 0, 0), (8, 6, 10), (7, 9, 3), (10, 1, 8), (4, 4, 5), (5, 5, 8), (6, 5, 4), (1, 0, 9), (6, 10, 8)\}$.

These improve the 2013 values 1134 and 2496.

(9, 5): the repository now has a larger Haar graph, of $\mathbb{Z}_{197} \rtimes \mathbb{Z}_7$ with 2758 vertices (computer-checked and proved in Lean), and (8, 5) has a 2048-vertex lift that is not a Haar graph. Neither is in the refereed notes. Include?

8.3. **Diameters 8, 9 and 10.** The groups are $\mathrm{SL}(2,q)$, $\mathrm{PSL}(2,q)$, direct products $Q \times \mathbb{Z}_n$ and $Q_1 \times Q_2 \times \mathbb{Z}_n$ of these with a cyclic group, and metacyclic groups $\mathbb{Z}_m \rtimes_r \mathbb{Z}_l$, the pairs $(a,j) \in \mathbb{Z}_m \times \mathbb{Z}_l$ with $(a,j)(b,i) = (a + r^j b, j+i)$, where $r^l \equiv 1 \pmod m$. Table 8 lists the graphs; their connection sets are given as data (Appendix C).

Example 8.5 ($\Delta = 7$, $D = 8$, 95 760 vertices)**.** $G = \mathrm{PSL}(2,19) \times \mathbb{Z}_{14}$, of order $3420 \cdot 14 = 47\,880$, matrices modulo $\pm I$, and

$$S = \left\{(I,0),\ \left(\begin{smallmatrix}2 & 13\\ 14 & 6\end{smallmatrix},6\right),\ \left(\begin{smallmatrix}4 & 13\\ 7 & 4\end{smallmatrix},12\right),\ \left(\begin{smallmatrix}2 & 7\\ 7 & 6\end{smallmatrix},10\right),\ \left(\begin{smallmatrix}6 & 2\\ 8 & 6\end{smallmatrix},1\right),\ \left(\begin{smallmatrix}6 & 13\\ 15 & 1\end{smallmatrix},9\right),\ \left(\begin{smallmatrix}1 & 4\\ 12 & 11\end{smallmatrix},11\right)\right\}.$$

With $g = \left(\begin{smallmatrix}1 & 3\\ 6 & 0\end{smallmatrix}\right)$ (order 3 in $\mathrm{PSL}(2,19)$), the automorphism $\alpha(x,z) = (gxg^{-1}, 9z)$ of order 3 fixes the first element of S and permutes the others in two 3-cycles, $s_1 \mapsto s_2 \mapsto s_3 \mapsto s_1$ and $s_4 \mapsto s_5 \mapsto s_6 \mapsto s_4$. By Theorem 8.4(a), $\operatorname{diam} H(G,S) \le 8$ means that every element of G is a product $s_{i_7} s_{i_6}^{-1} s_{i_5} s_{i_4}^{-1} s_{i_3} s_{i_2}^{-1} s_{i_1}$; this covering is computer-checked (and the graph is proved in Lean). The order is 14.3% of the Moore bound $M_b(7,8) = 671\,846$; the 2013 value is 80 940.

In most of the graphs of Table 8, S is closed under an automorphism α of G of order 3 (or 5, or 6), as in Example 8.5; no check uses this symmetry.

Table 8: Haar graphs $H(G,S)$ at $D = 8, 9, 10$ (and one lift). "%": gain over the 2013 value. All computer-checked by at least two separately written programs, or by one program and a Lean proof.

cell	2013	this paper	%	group G
(3,8)	256	288	12.5	lift of a 4-vertex base over $\mathbb{Z}_3^2 \rtimes \mathbb{Z}_8$ (not a Haar graph)
(5,8)	9 234	9 760	5.7	$\mathbb{Z}_{61} \rtimes_8 \mathbb{Z}_{80}$
(6,8)	29 790	31 680	6.3	$\mathrm{PSL}(2,11) \times \mathbb{Z}_{24}$
(7,8)	80 940	95 760	18.3	$\mathrm{PSL}(2,19) \times \mathbb{Z}_{14}$
(8,8)	201 480	239 904	19.1	$\mathrm{PSL}(2,17) \times \mathbb{Z}_{49}$
(9,8)	449 480	540 360	20.2	$\mathrm{PSL}(2,19) \times \mathbb{Z}_{79}$
(11,8)	2 246 940[a]	2 258 520	0.5	$\mathrm{PSL}(2,59) \times \mathbb{Z}_{11}$
(13,8)	5 314 680	6 999 552	31.7	$\mathrm{SL}(2,7)^2 \times \mathbb{Z}_{31}$
(15,8)	14 172 480[a]	18 128 880	27.9	$\mathrm{SL}(2,109) \times \mathbb{Z}_7$
(5,9)	27 936	29 540	5.7	$\mathbb{Z}_{422} \rtimes_5 \mathbb{Z}_{35}$
(6,9)	117 360	127 296	8.5	$\mathrm{SL}(2,17) \times \mathbb{Z}_{13}$
(7,9)	400 160	462 840	15.7	$\mathrm{PSL}(2,29) \times \mathbb{Z}_{19}$
(8,9)	1 091 232	1 341 600	22.9	$\mathrm{SL}(2,25) \times \mathbb{Z}_{43}$
(9,9)	2 961 536	3 499 776	18.2	$\mathrm{PSL}(2,7) \times \mathrm{SL}(2,7) \times \mathbb{Z}_{31}$
(10,9)	7 057 400	8 003 856	13.4	$\mathrm{SL}(2,83) \times \mathbb{Z}_7$
(11,9)	15 200 448	16 792 080	10.5	$\mathrm{SL}(2,61) \times \mathbb{Z}_{37}$
(12,9)	30 001 152	34 026 720	13.4	$\mathrm{SL}(2,41) \times \mathbb{Z}_{247}$
(13,9)	50 990 610	67 742 688	32.9	$\mathrm{SL}(2,103) \times \mathbb{Z}_{31}$
(14,9)	95 087 738	116 068 656	22.1	$\mathrm{SL}(2,67) \times \mathbb{Z}_{193}$
(15,9)	168 016 334	208 029 120	23.8	$\mathrm{SL}(2,79) \times \mathbb{Z}_{211}$
(16,9)	288 939 118	337 394 400	16.8	$\mathrm{SL}(2,151) \times \mathbb{Z}_{49}$

cell	2013	this paper	%	group G
(4,10)	11 748	12 416	5.7	$\mathbb{Z}_{97} \rtimes_{19} \mathbb{Z}_{64}$
(5,10)	90 068	95 472	6.0	$\mathbb{Z}_{1989} \rtimes_{461} \mathbb{Z}_{24}$
(6,10)	452 032	524 160	16.0	$\mathrm{SL}(2,5) \times \mathrm{PSL}(2,7) \times \mathbb{Z}_{13}$
(7,10)	1 987 380	2 237 760	12.6	$\mathrm{PSL}(2,5) \times \mathrm{SL}(2,8) \times \mathbb{Z}_{37}$
(8,10)	6 927 210	7 818 096	12.9	$\mathrm{SL}(2,67) \times \mathbb{Z}_{13}$
(9,10)	20 017 260	23 975 280	19.8	$\mathrm{PSL}(2,71) \times \mathbb{Z}_{67}$
(10,10)	50 331 156	60 141 120	19.5	$\mathrm{SL}(2,79) \times \mathbb{Z}_{61}$
(11,10)	130 592 354[b]	148 729 680	13.9	$\mathrm{PSL}(2,89) \times \mathbb{Z}_{211}$
(12,10)	300 383 050	325 832 976	8.5	$\mathrm{SL}(2,107) \times \mathbb{Z}_{133}$
(13,10)	617 330 936	677 714 400	9.8	$\mathrm{SL}(2,199) \times \mathbb{Z}_{43}$
(14,10)	1 213 477 190	1 326 627 120	9.3	$\mathrm{SL}(2,139) \times \mathbb{Z}_{247}$
(15,10)	2 300 326 510	2 467 618 800	7.3	$\mathrm{SL}(2,149) \times \mathbb{Z}_{373}$
(16,10)	4 119 507 330	4 220 872 800	2.5	$\mathrm{SL}(2,151) \times \mathbb{Z}_{613}$

[a] A value of Gómez's $\Omega\mathrm{H}_q$ (Section 7); at $(15,8)$ the thesis lists the larger value 14 978 080 ($\Omega\mathrm{H}_{11}\mathrm{d}$), against which the gain is 21.0%. [b] The survey PDF prints "130 592 1354"; every revision of [11] has 130 592 354.

The $(3,8)$ graph is a 72-fold regular cover of the cubic bipartite multigraph on $2+2$ vertices with edges 0–2 twice, 0–3, 1–2, 1–3 twice, with voltages in $\mathbb{Z}_3^2 \rtimes_\phi \mathbb{Z}_8$, where the generator of $\mathbb{Z}_8$ acts by $\phi = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 0 \end{smallmatrix}\right)$. It is not a Haar graph: its automorphism group is not transitive on a colour class (computer-checked: two distance profiles in each class).

Values at $D = 8$–10 raised after the refereed Haar note (computer-checked by the coordinator and proved in Lean on the lean-all branch): $(9,8)$ 546 720 ($\mathrm{SL}(2,16) \times \mathbb{Z}_{67}$), $(5,9)$ 29 756 ($\mathbb{Z}_{346} \rtimes \mathbb{Z}_{43}$), $(7,10)$ 2 298 240 ($\mathrm{SL}(2,7) \times \mathrm{PSL}(2,19)$), $(8,10)$ 8 043 840 ($\mathrm{PSL}(2,7) \times \mathrm{PSL}(2,19) \times \mathbb{Z}_7$); also $(3,10)$ 810 (a lift over $\mathbb{Z}_3 \times \mathrm{He}_3$). Use these in Table 8?

Cells we do not improve at $D = 8, 9, 10$: $(4,8)$ 2184, $(10,8)$ 1 176 480, $(12,8)$ 4 650 100, $(14,8)$ 14 172 480, $(16,8)$ 36 201 060, $(3,9)$ 506, $(4,9)$ 4970 and $(3,10)$ 800 (the last improved to 810 by a lift; see the note above). The values at $(10,8)$, $(12,8)$, $(14,8)$, $(16,8)$ are those of Section 7.

8.4. **Prior work.** The type of construction is not new. Haar graphs of $\mathrm{SL}(2,q)$ appear in the cage literature: according to the dynamic cage survey [22], the smallest known cubic graph of girth 30, on $1\,143\,408 = 2|\mathrm{SL}(2,83)|$ vertices, is a voltage lift over $\mathrm{SL}(2,83)$ of the dipole with three parallel edges [21], that is, a Haar graph $H(\mathrm{SL}(2,83), S)$ with $|S| = 3$; and Bray, Parker and Rowley [9] use $\mathrm{PSL}(2,q) \times \mathbb{Z}_3$. Haar graphs of cyclic groups appear at $D = 3$ in [25], and dipole lifts over metacyclic groups are probably behind Loz's values [30]. In the sources we searched we found no use of Haar graphs of $\mathrm{SL}(2,q)$ or $\mathrm{PSL}(2,q)$, alone or times a cyclic group, for large bipartite graphs of diameter 8, 9 or 10. Theorem 8.4 is routine and we claim no novelty for it.

9. The updated table

Table 9 lists, for every cell with $\Delta \le 16$ and $D \le 10$ treated in this paper, the 2013 value and ours. Status: P = member of a family with a complete proof in this paper (for the $D = 7$ records: the general theorems, with explicit sets that are computer-checked); C = the graph is computer-checked (Appendix C); L = the graph itself is proved in Lean; T = the general theorem is proved in Lean (so the value is a Lean-checked lower bound); L* = a Lean proof exists on a branch of the repository but has not yet been re-checked independently (Appendix B). "reg.": the order of the largest regular graph given here, when the record graph is not regular.

Table 9: Updated values. A dagger marks a 2013 value that is the order of one of Gómez's graphs $\Omega \mathrm{H}_q$, which have diameter at least 9 as defined (Section 7).

(Δ, D)	2013	this paper	reg.	where	status
(11,3)	190	192		Thm. 3.2	P C L T
(13,3)	270 (274)	276		Thm. 3.2	P C L T
(15,3)	370 (374)	376		Thm. 3.2	P C L T
(7,4)	346	359	350	Ex. 4.6, Cor. 4.8	P C L
(11,4)	1 734	1 782		Cor. 4.8	P C L T
(13,4)	3 064	3 146		Cor. 4.8	P C L T
(15,4)	4 946	5 070		Cor. 4.8	P C L T
(16,4)	5 134	5 440		Cor. 4.12	P C L T
(7,5)	1 134	1 152		Sec. 8	C L
(9,5)	2 496	2 662		Sec. 8	C L
(7,6)	8 992	9 592	9 348	Thm. 5.8, 5.18	P C L T
(11,6)	142 464	148 944	147 500	Thm. 5.8, 5.18	P C T
(13,6)	374 452	388 972	386 328	Thm. 5.8, 5.18	P C T
(15,6)	842 048	870 440	866 068	Thm. 5.8, 5.18	P C T
(16,6)	884 062	940 846	927 724	Thm. 5.8, 5.18	P C T
(7,7)	23 436	23 598		Cor. 6.24	P C L*
(9,7)	117 648	118 238		Cor. 6.24	P C L*
(10,7)	224 694	225 152		Cor. 6.24	P C L*
(11,7)	398 580	413 180		Cor. 6.25	P C L*
(12,7)	664 300	693 500		Cor. 6.25	P C L*
(13,7)	1 062 936	1 064 472		Cor. 6.24	P C L*
(14,7)	1 771 560	1 789 588		Cor. 6.26	P C L*
(15,7)	2 480 184	2 501 491		Cor. 6.26	P C L*
(16,7)	4 022 340	4 052 108		Cor. 6.26	P C L*
(3,8)	256	288		Sec. 8	C L
(5,8)	9 234	9 760		Table 8	C L*
(6,8)	29 790	31 680		Table 8	C L*
(7,8)	80 940	95 760		Ex. 8.5	C L
(8,8)	201 480	239 904		Table 8	C L
(9,8)	449 480	540 360		Table 8	C L
(10,8)	1 176 480†	–		Sec. 7	
(11,8)	2 246 940†	2 258 520		Table 8	C L*
(12,8)	4 650 100†	–		Sec. 7	
(13,8)	5 314 680	6 999 552		Table 8	C L*
(14,8)	14 172 480†	–		Sec. 7	
(15,8)	14 172 480†	18 128 880		Table 8	C L*
(16,8)	36 201 060†	–		Sec. 7	
(5,9)	27 936	29 540		Table 8	C
(6,9)	117 360	127 296		Table 8	C L
(7,9)	400 160	462 840		Table 8	C L
(8,9)–(16,9)		Table 8		Table 8	C L*
(4,10)	11 748	12 416		Table 8	C L
(5,10)	90 068	95 472		Table 8	C L
(6,10)	452 032	524 160		Table 8	C L

(Δ, D)	2013	this paper	reg.	where	status
(7,10)–(16,10)		Table 8		Table 8	C L*

Table 9: (i) how to show the four $\Omega\mathrm{H}_q$ cells we do not improve ($(10,8)$, $(12,8)$, $(14,8)$, $(16,8)$): keep the 2013 value with a mark, or leave the cell empty? (ii) The 2013 table has 112 cells with $\Delta \le 16$, $D \le 10$; should the paper print the whole table (with the cells we do not touch and the Moore cells)? (iii) The CombinatoricsWiki was last read for these cells in October 2026; re-check it before submission.

Appendix A. The $(7,4)$ example in coordinates

$Q(4,5)$ is the quadric $X_0^2+X_1X_2+X_3X_4=0$ of $\mathrm{PG}(4,5)$, with 156 points and 156 lines (points are written with first non-zero coordinate 1). The ovoid is $\mathcal{O} = Q(4,5) \cap \{X_4 = 2X_3\}$ (-2 is a non-square modulo 5), an elliptic quadric with 26 points. The structure of Example 4.6 consists of

- the incidence graph of $Q(4,5)$ (312 vertices, all of degree 6);
- one clone o' of each $o \in \mathcal{O}$, joined to the 6 lines on o (26 vertices; every line gains exactly one clone, as $\mathcal{O}$ is an ovoid);
- 21 shadow lines, one for each line below, joined to its six points, except that a starred point is replaced by its clone (the stand-in).

```
 1  (0:0:0:0:1) (0:1:0:0:0)  (0:1:0:0:1) (0:1:0:0:2)  (0:1:0:0:3)  (0:1:0:0:4)
 2  (0:0:0:1:0) (0:1:0:0:0)* (0:1:0:1:0) (0:1:0:2:0)  (0:1:0:3:0)  (0:1:0:4:0)
 3  (0:0:1:0:0) (1:0:0:2:2)  (1:0:1:2:2) (1:0:2:2:2)  (1:0:3:2:2)  (1:0:4:2:2)
 4  (0:0:1:0:1) (1:2:0:3:3)  (1:2:1:3:4) (1:2:2:3:0)  (1:2:3:3:1)  (1:2:4:3:2)
 5  (0:0:1:0:2) (1:1:0:2:2)  (1:1:1:2:4) (1:1:2:2:1)  (1:1:3:2:3)  (1:1:4:2:0)
 6  (0:0:1:0:4) (1:3:0:3:3)  (1:3:1:3:2) (1:3:2:3:1)  (1:3:3:3:0)  (1:3:4:3:4)
 7  (0:0:1:1:0) (1:4:0:4:1)  (1:4:1:0:1) (1:4:2:1:1)  (1:4:3:2:1)  (1:4:4:3:1)
 8  (0:0:1:2:0) (1:4:0:3:3)  (1:4:1:0:3) (1:4:2:2:3)  (1:4:3:4:3)  (1:4:4:1:3)
 9  (0:0:1:3:0) (1:2:0:4:1)  (1:2:1:2:1) (1:2:2:0:1)  (1:2:3:3:1)* (1:2:4:1:1)
10  (0:0:1:4:0) (1:4:0:1:4)  (1:4:1:0:4) (1:4:2:4:4)  (1:4:3:3:4)  (1:4:4:2:4)
11  (0:1:1:1:4) (1:0:3:4:1)  (1:1:4:0:0) (1:2:0:1:4)  (1:3:1:2:3)  (1:4:2:3:2)
12  (0:1:1:2:2) (1:0:0:1:4)  (1:1:1:3:1) (1:2:2:0:3)  (1:3:3:2:0)  (1:4:4:4:2)
13  (0:1:1:3:3) (1:0:2:3:3)  (1:1:3:1:1) (1:2:4:4:4)  (1:3:0:2:2)  (1:4:1:0:0)
14  (0:1:1:4:1) (1:0:2:4:1)  (1:1:3:3:2) (1:2:4:2:3)  (1:3:0:1:4)  (1:4:1:0:0)*
15  (0:1:2:1:3) (1:0:3:3:3)  (1:1:0:4:1) (1:2:2:0:4)  (1:3:4:1:2)  (1:4:1:2:0)
16  (0:1:2:2:4) (1:0:3:1:4)  (1:1:0:3:3) (1:2:2:0:2)  (1:3:4:2:1)  (1:4:1:4:0)
17  (0:1:3:2:1) (1:0:1:3:3)  (1:1:4:0:4) (1:2:2:2:0)  (1:3:0:4:1)  (1:4:3:1:2)
18  (0:1:3:3:4) (1:0:4:1:4)  (1:1:2:4:3) (1:2:0:2:2)  (1:3:3:0:1)  (1:4:1:3:0)
19  (0:1:4:1:1) (1:0:4:3:3)  (1:1:3:4:4) (1:2:2:0:0)  (1:3:1:1:1)  (1:4:0:2:2)
20  (0:1:4:2:3) (1:0:0:3:3)  (1:1:4:0:1) (1:2:3:2:4)  (1:3:2:4:2)  (1:4:1:1:0)
21  (0:1:4:4:4) (1:0:0:4:1)  (1:1:4:3:0) (1:2:3:2:4)* (1:3:2:1:3)  (1:4:1:0:2)
```

What to check by hand (Theorem 4.2(b) with $s = t = 5$, $\Delta = 7$): each listed line lies on the quadric; the 21 lines are pairwise disjoint except at the four ovoid points $(0:1:0:0:0)$, $(1:2:3:3:1)$, $(1:4:1:0:0)$, $(1:2:3:2:4)$, each on exactly two of them (lines 1 and 2, 4 and 9, 13 and 14, 20 and 21), and on the second of the two the point is replaced by its clone. So every old point and every clone serves at most one shadow line. Degrees: old lines $6+1=7$; old points 6 or 7; clones 6 or 7; shadow lines 6. The order is $156+156+26+21=359$, and the diameter is 4 by Theorem 4.2(a).

Appendix B. Formal verification

Many statements of this paper are proved in Lean 4 (version 4.34.1) with Mathlib, in the library `lean/BipartiteDD` of the repository. For every theorem listed below, `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry`, `admit`, new `axiom` or `native_decide`, and finite checks use `decide` or `decide +kernel`, which the kernel checks. A graph statement has the form "there is a graph on `Fin` N that is 2-colourable, has every vertex of degree at most Δ (or exactly Δ), and has diameter at most D", so the vertex count is exactly N.

Decide how much of this appendix to keep, and whether to cite the Lean files by name or only describe what is covered.

What is covered.

- Section 2.3: Theorem 2.6(a)–(c) for any finite graph under (L0), (L1), (C_D), with the degree bound $\Delta + m - 1$ and the exact count; Theorem 2.6(d); the hat lemma in both directions, for generalized polygons given as graphs of diameter n and girth $2n$ with all degrees at least 2; Corollary 2.8 ("if").
- Section 3: the regular version of Theorem 3.2 for every finite projective plane of order $q \geq 4$ with a frame (not necessarily Desarguesian), with the incidence rules of Definition 3.1 in coordinate-free form; the graphs of order 192, 276 and 376.
- Section 4: Theorem 4.2 with arbitrary stand-ins, and Corollary 4.3, for every finite GQ (more precisely for the hypothesis "lines have $s + 1$ points, points are on $t + 1$ lines, two points are on at most one line, and for an anti-flag some point of the line is collinear with the point"); the model of AS(q) of Section 4.2 satisfies this hypothesis, Theorem 4.7, and Corollary 4.8, for every finite field of odd characteristic; Proposition 4.10(b), the spread $\{yK\}$ and Corollary 4.12 (and its regular form) for every odd prime power q, in the cyclic model; the graphs of order 359, 1782, 3146, 5070 and 5440.
- Section 5: $H(q)$ in the coordinates of Section 2 is a generalized hexagon (every finite field); Theorem 5.5 and Lemmas 5.4, 5.7 and 5.6(d); Theorem 5.3 for any whisker set; Theorem 5.8 with the exact vertex counts; Theorem 5.18 (completion non-constructive, as on paper); the even case for $q \geq 4$, and $q = 2$, $m = 2$ with an explicit whisker set; the graph of order 9592.
- Section 7: Theorem 7.3 in the general form stated; Theorem 7.5 (for every finite field with $q \geq 7$, in a form that allows any mix of blocks and any colour convention per copy; the case $q = 8$ needs a ten-line wrapper written by the verifier); Theorem 7.15 for $q \geq 8$, reading 2 and its dual, every block pattern and every labelling whose transfers are automorphisms of the incidence graph (the reduction of other readings for $q = 3^h$, Lemma 7.9(b), and the case $q = 7$ are not formalised, and the explicit numbers of far vertices are not exported).
- Section 8: Theorem 8.4 (both directions, every group) and the per-side Moore bound; the bipartite Moore bound $|V| \leq M_b(\Delta, D)$; the graphs at $(7, 5)$, $(9, 5)$, and those marked `L` in Table 9.

What is not covered. The identification of the coordinate models with the classical AS(q), $W(3, q)$ and $H(q)$ (the proofs use the models directly, and the classical facts are cited); that the 359 graph decomposes as in Example 4.6 (the graph is checked, not its description); Theorem 7.21; and the $D = 7$ theorems of Section 6 as general statements (the nine record graphs are checked, see below).

Verification of the Lean work. The library behind Sections 2–5 was rebuilt module by module from a fresh clone on 25 October 2026, with the axiom lists regenerated. The additions behind Theorem 2.6(d), the Moore bounds, Theorem 8.4 (converse), Proposition 4.10 for every odd q, and Theorems 7.3, 7.5 and 7.15 were each rebuilt by a separate verifier from a clean checkout,

with regenerated axiom lists compared byte for byte, a search for forbidden tokens, and a reading of every main statement against the paper.

The proofs marked L^* in Table 9 (the nine $D = 7$ records, the four $D = 6$ records as instances of Theorem 5.8, and most Haar graphs at $D = 8$–10) are on a separate branch of the repository and have not yet been re-checked by a separate verifier. For several of the largest graphs the generated Lean files are too large to store; they are regenerated by deterministic scripts whose outputs are fixed by recorded hashes. In four $D = 7$ proofs a different hub graph, whose properties are also kernel-checked, replaces the one of the record graph.

Before submission: have the L^* proofs re-checked independently, or drop the column.

Appendix C. Computer checks

Every graph of Tables 1 and 8, and every "computer-checked" statement, was checked as follows. All programs and logs are in the repository `https://github.com/dogaozden/degree-diameter`.

- *All-pairs check.* A C program reads an edge list, checks that the graph is simple, bipartite and connected and has maximum degree at most Δ, and runs a bit-parallel breadth-first search from every vertex, with no symmetry assumed. Every graph in this paper with at most 4.1 million vertices (all graphs of Sections 3–6) was checked this way, after being rebuilt from its definition or data file. The $D = 7$ graphs were built from their data files by the builder that produced them; the check of the resulting edge list is independent of that builder, and three of them ($(7, 7)$, $(9, 7)$, $(10, 7)$) were also checked with a second breadth-first-search program.
- *Families.* The fourth hat: built three ways (from (C1)–(C6), from the point rows and from the line rows of Table 2), for every prime power $q \le 13$ and every admissible (a, c); degrees, both covering properties and exact diameter of G and R; a separate program rebuilt it for $q = 4, 5, 7, 8, 9, 11, 13$ and compared every row of Table 2 with the computed neighbourhoods. Theorem 4.7: line sums and the graph for $q = 3, 5, 7, 9$. Proposition 4.10: exactly the pairs $\{a, q - a\}$ give 2-ovoids, for $q = 3, 5, 7, 9, 11, 13, 25, 27$; Corollary 4.12 for $q = 3, 5$. Section 5: the graphs for $q = 5, 7, 9, 11, 13$ (and $q = 13$, $m = 3$), exactness of W on all free points for $q = 3, 5, 7$ and $q = 4, 8, 16$, the identities of Lemmas 5.6, 5.7, 5.12, 5.13 re-checked symbolically, and the five regular graphs of Theorem 5.18.
- *Diameter 7, small q.* Every theorem of Section 6 was tested on $H(q)$ for $q = 2, 3, 4, 5$, built from (2.1), with positive tests and negative controls (adding one turning point to a segment set, or breaking (H3) at one point, gives diameter 8).
- *Diameter 8.* The hexagons $H(q)$, $q \le 13$, three times by separate programs; the exact diameters and eccentricities stated after Corollary 7.6; Fact 7.14 by two separate programs; the rows of Table 7 by three programs.
- *Haar graphs.* By Theorem 8.4, two breadth-first searches, from (e, A) and (e, B), decide the diameter. Each graph of Table 8 was accepted by at least two separately written programs (each building the group from scratch, including the finite field and the matrix group), or by one program and a Lean proof; twelve of them (up to 540 360 vertices) also passed the all-pairs check. Negative controls (one generator altered, or D lowered by one) are rejected.

Should this appendix name the programs and log files (for instance `research/lifts/coordinator_check/apsp_mt.c`, `research/frontier/build_haar2.py`, `research/coordinator_check/check_2026-10-25.txt`)? The data files that define the $D = 7$ sets and hub graphs and the Haar connection sets are in `research/frontier/sols/`.

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